
DISSERTATION OF GRADUATION

Presented at

Higher Institute of Computer Science of Mahdia

Carried out at

Sumitomo Electric Bordnetze



In view of obtaining the diploma of

Bachelor's Degree in Business Computing : Business Intelligence

Prepared by:

Belghouthi Ichrak

Optimizing Employee Transportation System via Business Intelligence and
Generative-AI

Defended on 16/06/2025
before the jury composed of:

President	: Ines Zribi
Reviewer	: Ben Kraiem Maha
Academic Supervisor	: Soufien Boutaib
Academic Co-Supervisor	: Maha Elarbi
Industrial Supervisor	: Moetaz Fatin

Academic Year: 2024-2025

DEDICATION

First and foremost, I want to thank God, the Most Gracious and the Most Merciful. Without His guidance, strength, and blessings, none of this would be possible. Each step, every hurdle I overcame, and every mile I celebrated would have felt a much heavier without His divine presence in my life.

To my mother "**Cherifa Slimi**", the most important person in my world, this effort is dedicated to you. Your unwavering support, selfless sacrifice, and prayers have sustained my journey. You remained steadfast as a mother and believed in me when I didn't believe in myself. Your prayers have also been my shield. Thank you for being my light in dark times.

To my dear father "**Kamel Belghouthi**", your strength, wisdom, and quiet encouragement have reminded me to keep moving forward. You've taught me the value of diligence and integrity, and instilled within me the confidence I needed, by believing in my potential. Your trust in me gave me courage and conviction to pursue whatever the dreams I had. This accomplishment is yours as much as it is mine.

To my lovely sister "**Islem Belghouthi**", you have always been such a joy and comfort. Thank you for your supportive words, for being my shoulder in stressful times, for being my best friend. I love you very much.

Lastly, I would like to dedicate this to me - to myself, for not giving up. Thank you for being strong in the sleepless nights, for moving through doubt, and for tackling every challenge with resilience and determination. This report reflects my drive and I am proud of my progress.

GRATITUDE

First, I would like to acknowledge my God, for the strength, patience, and perseverance given to finish this project.

Moreover, I would like to express my sincere gratitude to my academic supervisors, **Mr. Sofien Boutaib** and **Mrs. Maha Elarbi**, for their support, guidance, and constant availability. Their know-how, well-deserved advice and feedback, were priceless throughout every aspect of this project. Their professionalism, discipline, and motivation for students deeply inspired me. I am very thankful for their constant motivation, especially during challenging situations, and for the trust they had in me and my work.

I also sincerely thank my professional supervisor **Mr. Moetaz Fatin**, who has facilitated my internship and guided me with his support, availability, and constructive feedback. I appreciate the chance to have learned and have the trust placed in me throughout my internship.

Lastly, I thank all **the jury members** who will assess my work. I appreciate you all being here, and your impressions are valued. Thank you for contributing to this valuable stage of my academic journey.

General Introduction	1
1 General Context of the Project	3
1.1 Introduction	3
1.2 Host Organization Presentation	3
1.2.1 Organizational Structure of SEBN-TN2	4
1.2.2 Services provided by the company	4
1.3 Project presentation	5
1.3.1 General context of the project	5
1.3.2 Analysis of the Existing System	5
1.3.3 Criticism of the Current System	5
1.3.4 Problematic	6
1.3.5 Proposed Solution	6
1.4 Project Management Methodologies Study	7
1.4.1 The GIMSI Method	7
1.4.1.1 GIMSI objectives	8
1.4.1.2 Main Phases of GIMSI	8
1.4.1.3 Advantages of GIMSI	9
1.4.2 SCRUM Methodology	9
1.4.2.1 Key Principles of Scrum	9
1.4.2.2 Advantages of Scrum	10
1.4.3 Adopted Methodology	11
1.5 Conclusion:	11
2 Sprint 0: Preparation phase	12
2.1 Introduction	12
2.2 Requirements specification	12

2.2.1	Identification of system actors	12
2.2.2	Functional requirements	13
2.2.3	Non-Functional requirements	13
2.2.4	Requirements modeling	14
2.3	Project Management with Scrum	16
2.3.1	Identification of the Scrum Team	16
2.3.2	Product backlog	16
2.3.3	Sprints Planning	18
2.3.4	Project Planning	19
2.4	Technical architecture of the decision-making solution	20
2.5	Overall design of the data warehouse	21
2.5.1	Data Collection	22
2.5.2	Key Performance Indicators (KPIs) Selection	22
2.5.3	Data Warehouse Design Approaches	23
2.5.3.1	Inmon Approach	24
2.5.3.2	Kimball Approach	24
2.5.3.3	Adopted Approach	25
2.6	Multidimensional Modeling	25
2.6.1	Conceptual Model Analysis	26
2.6.2	Choice of the Conceptual Model:	28
2.7	Work Environment	29
2.7.1	Hardware environment	29
2.7.2	Software environment	30
2.8	Conclusion	32
3	Sprint 1: Design and Development of Dashboards	34
3.1	Introduction	34
3.2	Requirements Definition	34
3.3	Sprint 1 Backlog	36
3.4	Data integration phase (ETL)	36
3.4.1	General Overview of the ETL Process	36
3.4.2	Data Extraction Phase (Extract)	37
3.4.3	Data Transformation Phase (Transform)	40
3.4.4	Data Loading Phase (Load)	44
3.4.5	Creating the dimension tables and the fact table	45
3.5	Dashboard Wireframe Design	46
3.5.1	Overview dashboard wireframe	47
3.5.2	Line Analysis dashboard wireframe	47
3.5.3	Financial Analysis dashboard wireframe	48

3.5.4	Delay Analysis dashboard wireframe	49
3.6	Data Presentation Phase	49
3.6.1	Overview Dashboard	50
3.6.2	Line Analysis Dashboard	53
3.6.3	Financial Analysis Dashboard	54
3.6.4	Delay Analysis Dashboard	56
3.7	Conclusion	59
4	Sprint 2: Chatbot-Driven Dashboard Generation and Insight Analysis	60
4.1	Introduction	60
4.2	Requirements Definition	60
4.3	Sprint 2 Backlog	61
4.4	Sprint 2 Analysis	61
4.4.1	Sprint 2 Use Case Diagram	61
4.4.2	Text description of use case "Generate AI report"	62
4.5	Definition of generative Artificial Intelligence	63
4.5.1	Traditional AI vs Generative AI	63
4.5.2	The Usefulness of Generative Artificial Intelligence in Business In- telligence	64
4.5.3	Generative AI Types	64
4.5.4	Selected Type	65
4.6	Large Language Model	65
4.6.1	LLM Models Comparison	66
4.6.2	Selected model	67
4.6.3	Working Mechanism of the Mistral-Saba-24b Model	67
4.7	Integration of Groq API	67
4.7.1	API Key and Authentication	68
4.7.2	Groq Cloud Usage	68
4.8	Architecture diagram of the intelligent agent	69
4.9	Realization	71
4.10	Conclusion	74
5	Sprint 3:Transport management	75
5.1	Introduction	75
5.2	Sprint 3 Backlog	75
5.3	Sprint 3 Analysis	77
5.3.1	Sprint 3 Use Case Diagram	77
5.3.2	Detailed use case study	78
5.3.2.1	Use case diagram for the “Authenticate” functionality . . .	78
5.3.2.2	Use case diagram for the “Display Fill Rate” functionality	79

5.3.2.3	Use case diagram for the “Manage employees” functionality	80
5.3.3	Text descriptions of use case scenarios	81
5.3.3.1	Text description of use case "Authenticate"	81
5.3.3.2	Text description of use case "Display Fill Rate"	82
5.3.3.3	Text description of use case "Add employee"	82
5.4	Conception	83
5.4.1	Sequence Diagrams	83
5.4.1.1	Authenticate sequence diagram	84
5.4.1.2	Display Fill Rate sequence diagram	84
5.4.1.3	Add employee sequence diagram	85
5.4.2	Class Diagram	86
5.5	Realization	87
5.5.0.1	Authentication interface	87
5.5.0.2	Home interface	88
5.5.0.3	Fill rate interface	89
5.5.0.4	Employees interface	90
5.6	Conclusion	91
6	Sprint 4: Real-Time Employee Authentication and Transport Request System	92
6.1	Introduction	92
6.2	Sprint 4 Backlog	92
6.3	Sprint 4 Analysis	93
6.3.1	Sprint 4 Use Case Diagram	93
6.3.2	Detailed use case study	94
6.3.2.1	Use case diagram for the “Scan Badge” functionality . . .	94
6.3.2.2	Use case diagram for the “Send complaint” functionality .	94
6.3.2.3	Use case diagram for the “Show bus position” functionality	94
6.3.3	Text descriptions of use case scenarios	95
6.3.3.1	Text description of use case "Scan Badge"	95
6.3.3.2	Text description of use case "Send complaint"	96
6.3.3.3	Text description of use case "Show bus position"	96
6.4	Conception	97
6.4.1	Sequence Diagrams	97
6.4.1.1	Scan Badge sequence diagram	97
6.4.1.2	Send complaint sequence diagram	98
6.4.1.3	Show bus position sequence diagram	99
6.4.2	Class Diagram	100
6.5	Realization	101

6.5.1	Scan badge Interface:	101
6.5.2	Send complaints Interface	104
6.5.3	Track Bus Interface	106
6.6	Conclusion	107
General Conclusion		108
Bibliography		110

LIST OF FIGURES

1.1	SEBN-TN2's Logo.	3
1.2	Organizational chart of SEBN-TN2.	4
1.3	Main Phases of GIMSI.	8
1.4	SCRUM PROCESS.	9
1.5	SCRUM Roles.	10
2.1	Global Use Case Diagram.	15
2.2	Gantt chart.	20
2.3	Technical architecture of the system	21
2.4	Inmon Approach architecture.	24
2.5	Kimball Approach architecture.	25
2.6	Star Schema Model.	27
2.7	Snowflake Schema Model.	27
2.8	Galaxy Schema Model.	28
2.9	the Conceptual Model of the DW.	29
2.10	VS Code Logo.	30
2.11	MySQL Logo.	30
2.12	phpMyAdmin Logo.	31
2.13	XAMPP Logo.	31
2.14	Talend Logo.	31
2.15	Power BI Logo.	32
2.16	Draw.io Logo.	32
2.17	Overleaf's Logo.	32
3.1	The ETL Process.	37
3.2	"arrival_time" table dimensions.	38
3.3	"employee" table dimensions.	38
3.4	"lignes" table dimensions.	38

3.5	"arrets" table dimensions.	38
3.6	"bus" table dimensions.	39
3.7	MySQL Database Connection.	39
3.8	tDBInput.	40
3.9	SQL Query to Identify Orphan Route IDs in 'arrets' Table.	41
3.10	Orphan Route IDs in 'arrets' Table.	41
3.11	tMap component configuration with lookup.	41
3.12	SQL Query to detect "prestataire" redundancy.	42
3.13	Redundancy Count of "Prestataire"	42
3.14	Normalization using tUniqRow.	42
3.15	Sample of non-descriptive numeric "shift" IDs.	43
3.16	tMap component with conditional expressions.	43
3.17	SQL Query to detect "Buses" redundancy.	44
3.18	Sample lines with more than one bus.	44
3.19	tMap component to create the "ligne_bus" table.	44
3.20	tDBOutput.	45
3.21	Job Creation.	45
3.22	Populating the Fact Table.	46
3.23	Wireframe of the overview dashboard.	47
3.24	Wireframe for the line analysis dashboard.	48
3.25	The financial analysis dashboard wireframe.	48
3.26	The delay analysis dashboard wireframe.	49
3.27	Navigation Menu.	50
3.28	General Overview Dashboard.	51
3.29	Line Analysis Dashboard.	53
3.30	Financial Analysis Dashboard.	55
3.31	Delay Analysis Dashboard.	57
3.32	filtering menu.	58
4.1	Sprint 3 Use Case Diagram.	62
4.2	Setting up Groq LLM with API key in project.	68
4.3	API activity and resource consumption.	69
4.4	Architecture diagram of the intelligent agent.	70
4.5	Main Interface of the Dashboard Generation Platform.	71
4.6	Natural language query submitted by the user: "Analyse des employés". . .	72
4.7	Analytical visualization of the number of employees per shift and per bus line.	72
4.8	Analytical visualization of the total employee contribution and the yearly trend in worker numbers.	73

4.9	Automatic analysis and strategic recommendation.	73
5.1	Sprint 3 Use Case Diagram.	78
5.2	Use case diagram for the “Authenticate” functionality.	79
5.3	Use case diagram for the “Display Fill Rate” functionality.	80
5.4	Use case diagram for the “Manage employees” functionality.	80
5.5	Sequence diagram of the “Authenticate” functionality.	84
5.6	Sequence diagram of the “Display Fill Rate” functionality.	85
5.7	Sequence diagram of the “Add employee” functionality.	86
5.8	Class Diagram of Sprint 3.	87
5.9	the HR employee authentication interface.	88
5.10	The Home interface of the web platform.	88
5.11	Fill rate interface.	89
5.12	Modal of the scanned employees.	90
5.13	Manage employees interface.	90
5.14	Add employee form.	91
6.1	Sprint 4 Use Case Diagram.	93
6.2	Use case diagram for the “Scan Badge” functionality.	94
6.3	Use case diagram for the “send complaint” functionality.	94
6.4	Use case diagram for the “Show bus position” functionality.	95
6.5	Sequence diagram of the “Scan Badge” functionality.	98
6.6	Sequence diagram of the “Send complaint” functionality.	99
6.7	Sequence diagram of the “Show bus position” functionality.	100
6.8	Class Diagram of sprint 4.	101
6.9	Scan Result Interface – Authorized Badge.	102
6.10	Scan Result Interface – Unauthorized Badge.	103
6.11	Scan Result Interface – Warning message.	104
6.12	Send complaints Interface.	105
6.13	The history of submitted complaints.	106
6.14	Track Bus Interface.	107

LIST OF TABLES

2.1	Product Backlog.	17
3.1	Sprint 1 Backlog.	36
3.2	Analysis of the visuals of the “Overview” dashboard.	51
3.3	Analysis of the visuals of the “Line Analysis ” dashboard.	54
3.4	Analysis of the visuals of the “Financial Analysis” dashboard.	55
3.5	Analysis of the visuals of the “Delay Analysis” dashboard.	57
4.1	Sprint 2 Backlog.	61
4.2	Text description of use case "Generate AI report".	62
4.3	Traditional AI vs Generative AI.	64
4.4	Comparison of Language Models.	66
5.1	Sprint 3 Backlog.	75
5.2	Text description of use case "Authenticate".	81
5.3	Text description of use case "Display Fill Rate".	82
5.4	Text description of use case "Add employee".	83
6.1	Sprint 4 Backlog.	92
6.2	Text description of use case "Scan Badge".	95
6.3	Text description of use case "Send complaint".	96
6.4	Text description of use case "Show bus position".	96

List of Acronyms

SEBN-TN2: Sumitomo Electric Bordnetze EL FEJJA

SaaS: Software as a Service

HR: Human Resources

iOS: iPhone Operating System

BI: Business Intelligence

ETL: Extract, Transform, Load

KPI: Key Performance Indicator

DW: Data Warehouse

RDBMS: Relational Database Management System

SQL: Structured Query Language

DB: Database

UI: User Interface

SPA: Single Page Application

MVC: Model–View–Controller

REST API: Representational State Transfer Application Programming Interface

AI: Artificial Intelligence

GenAI: Generative Artificial Intelligence

GAN: Generative Adversarial Network

GPT: Generative Pre-trained Transformer

LLM: Large Language Model

API: Application Programming Interface

GENERAL INTRODUCTION

In today's fast-paced digital environment, businesses are re-evaluating how to do business differently. To optimize operations and improve decision-making processes, more organizations are looking for unique ways for data analytics, artificial intelligence, and web technology to enter organizations' operations. The connection of data and technology into organizational workflows is a key driver of efficiency, competitiveness, and overall value. Transportation and human resources management are a major area affected by digital change, in particular large companies whose workforce mobility poses a significant operational and financial challenge.

Data is now a key resource for organizations to drive improved performance. If raw data can be transformed into actionable intelligence with the effective use of tools and methods, organizations can use insights gathered from the intelligence to influence strategic decisions. Business Intelligence (BI), using interactive dashboards and analytics with real-time comparison, is a useful model for how businesses are leveraging data. Not only that, but Generative AI now opens up creating intelligent personalized interfaces that make the user experience more flexible and functional.

This is the context in which my end-of-studies internship took place. Hosted within SEBN-TN2 which is concerned with the optimization of its employees' transport, the aim of this project was to design and create a complete, intelligent, and expandable solution that uses modern web development, Business Intelligence tools, and AI; for monitoring, managing, and improving employee transportation processes.

This project was not only suitable for my degree which was obtained with a specialization in Business Intelligence, but also allowed me to exercise, deepen and expand my technical skills in a professional context. The solution that was realized includes: data visualization dashboards, generative AI supported dashboard generation, and the realisation of a complete web application with full-stack development for HR and employee

To better describe the trajectory, methodology and impact of this project the report is composed of six chapters representing the six distinct stages of the project;

First Chapter "General Context of the Project": This chapter conveys the context and motivation for the project, it will introduce the host company, the needs they have, the identified issues with relation to transport management and the objectives established for the internship, and present the overall methodology.

Second Chapter, "Sprint 0 – preparation phase": This section identify the preliminary preparation and planning work completed at the beginning of the internship, it will ask the surveys as the identified functional and technical specifications and the choices of tools, technologies and architectural choices that were need to be made to appropriately complete the project.

Thrd Chapter, "Sprint 1 – Dashboard Design and Development": In this chapter, I explain how I designed and built interactive and dynamic dashboards using Business Intelligence (BI) tools. These dashboards track key transportation indicators, such as bus usage and punctuality, as well as cost analysis, providing HR professionals with an effective, easy-to-understand, and quick-to-understand decision-making tool.

Fourth Chapter, "Sprint 2 – Chatbot-Driven Dashboard Generation and Insight Analysis": In this chapter, I highlight the innovative aspect of this project: the use of generative AI to dynamically generate personalized dashboards based on user requests.

Fifth Chapter, "Sprint 3 – Transport Management": This chapter describes the creation of a responsive web platform for the HR department, enabling them to more efficiently manage employee transportation.

Sixth Chapter, "Sprint 4 – Real-Time Employee Authentication and Transport Request System": This chapter presents a real-time badge scanning system that authenticates employees instantly, ensuring secure access. Additionally, a web application was developed to allow employees to submit transport requests via a simple and secure method.

And finally, I will provide a general conclusion which serves to summarize all the work which has been done throughout this internship.

CHAPTER 1

GENERAL CONTEXT OF THE PROJECT

1.1 Introduction

This chapter outlines the general framework of the project by first introducing the host company. Then it evaluates the existing solutions, outlines the weaknesses of these solutions, and finally proposes suggestions to enhance service quality. Lastly, it highlights the methodology selected to achieve the objectives of the project.

1.2 Host Organization Presentation

Sumitomo Electric Bordnetze El Fejja S.A.R.L. is a subsidiary of Sumitomo Electric Industries, specializes in the manufacturing of electrical harnesses for the automotive industry. Located in Néopark El Fejja, Mornaguia, Manouba, Tunisia, the company was established in 2019 and it has over 1,500 employees.



Figure 1.1: SEBN-TN2's Logo [1].

1.2.1 Organizational Structure of SEBN-TN2

To ensure uninterrupted operations, SEBN-TN2 operates on a three-shift system, . The teams are structured as follows:

- **Morning Shift:** 6:00 AM – 2:00 PM
- **Afternoon Shift:** 2:00 PM – 10:00 PM
- **Night Shift:** 10:00 PM – 6:00 AM

In addition, there is a fixed administrative schedule:

- **Administrative Shift:** 8:00 AM – 4:30 PM

Production managers and quality control teams are responsible for each shift, and everything is inspected for efficiency and product quality. By maintaining this structure, SEBN-TN2 ensures consistent output and delivers value to the automotive industry.

The departments are represented in the following organizational chart:

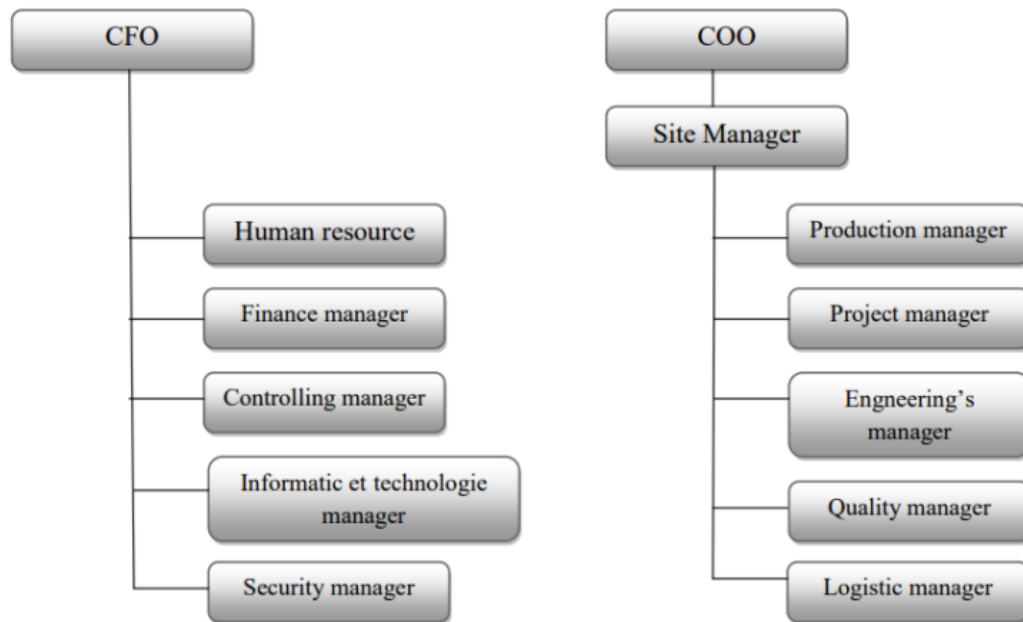


Figure 1.2: Organizational chart of SEBN-TN2.

1.2.2 Services provided by the company

SEBN-TN2 provides a variety of services exclusively offered to customers in the automotive industry. The company produces wiring harnesses for vehicles, facilitating power and signal transmission. The company also develops customer-specific wiring systems (from design to series production) and specializes in new approaches for electric and hybrid vehicles, including high-voltage harnesses to support the wholesale electrification of the automotive sector.

1.3 Project presentation

This part introduces the project in an organized format by outlining its context, analyzing the current situation, identifying the key problem, and proposing an appropriate and effective solution.

1.3.1 General context of the project

This project was completed within the context of the applied Bachelor degree in Business Information Technology, focusing in Business Intelligence, at the Mahdia Higher Institute of Computer Science and the Mahdia Faculty of Economics and Management. Conducted over a four-month term at SEBN-TN2, this provided the opportunity to apply theory to business practice with regards to sophisticated data analysis and decision-making. The focus of this initiative was to develop an innovative solution for managing employee transportation, providing better supervision, and lowering costs.

1.3.2 Analysis of the Existing System

SEBN-TN2 provides transport to all its employees based on different time slots (morning, afternoon, night, and administrative time). The service is currently in the manual mode with rudimentary tools, like Excel sheets or written records, to plan routes and monitor attendance. Bus route planning, schedule management, and personnel deployment are handled randomly by managers, without the aid of digital decision-making tools or automated monitoring. That approach gives weak coordination and transparency into the actual utilization of assets (buses, routes, schedules). The information collected is disparate and not very helpful, which makes it difficult to perform intensive analysis or optimization actions.

1.3.3 Criticism of the Current System

An analysis of the existing system is a critical part of any change or improvement process, as it provides a clear understanding of existing operations, documents inefficiencies, and indicates areas of optimization opportunity. Breaking down employee transport management into minute details, we will be able to identify the causes of issues at the very root level and ensure that any recommended solution meets actual needs and challenges effectively. This preliminary analysis forms the basis for the planning of a more cost-effective, efficient, and reliable transport system. The analysis of the existing system revealed several flaws in the current management of the transport of employees:

- **Ineffective Decision-Making:** Lack of consistent and central monitoring makes it difficult to correctly assess transport performance. Without a unified dashboard,

weaknesses are difficult to identify or efficiently optimize route planning.

- **Lack of Control Over Bus Utilization:** The lack of a proper verification system has resulted in some employees taking buses they are not authorized to take. This kind of control is leading to the improper use of transportation and the integrity of the system.
- **Lack of Flexibility:** The current tools do not have the ability to properly manage changes or unplanned events in schedules. This requires managers to make route adjustments manually, which are time-consuming and can also lead to mistakes.
- **Enhanced Operating Risk:** The lack of real-time bus tracking or employee tracking leads to problems with delays, diversion or people not boarding on time not being detected, which adds to the operating risk.

In summary, SEBN-TN2's employee transportation system relies on fragmented, manual processes, leading to inefficient management and limited visibility of resource utilization. The absence of a centralized database and automation not only increases operational costs, but also makes monitoring much more difficult and timely responses to problems impossible. The lack of effective tools and an organized operational overview indicates that SEBN-TN2 needs a solution—a unique value proposition such as the one planned in this project to improve employee transportation management.

1.3.4 Problematic

In this aspect, SEBN-TN2 also has numerous daunting problems in managing the issues of employees' transportation. The lack of automated, intelligent, and centralized systems significantly impede the company from possessing a smooth, efficient, and responsive company. Human error, unavailability of reliable real-time data, and the absence of tracking devices discredit not only the profitability of the business, but also employee convenience. It is therefore imperative to reconsider how transportation works by relying on sophisticated technologies in tracking, validation and analysis of travel in an attempt to streamline routes, save unnecessary costs, and increase improved management of flow.

1.3.5 Proposed Solution

In order to rectify the concerns with SEBN-TN2's employee transportation, this project proposes the utilization of a centralized, scalable system for intelligent transportation management. The goal of this concept is to provide a modern and integrated platform that allows Human Resources departments to oversee everyday bus operations in an effective manner while providing the employees with an intuitive mobile application to engage

with the service. By providing real-time monitoring, centralized data information, and automatic reporting features, the solution will enhance the overall operations, reduce overall wasted costs, and improve quality of service and decision-making. To address these challenges, we propose a complete smart transportation management system that includes the following key components:

- **Dynamic Dashboards:** A set of real-time dashboards for HR employees and decision-makers to track bus usage, employee attendance, delays, and optimize costs.
- **Chatbot:** Integrated chatbot by which the HR employees can view personalized dashboards, access applicable analysis, and receive personalized recommendations in an interactive and frictionless way.
- **HR Platform for Transportation Management:** A centralized web platform for HR to assign buses, view statistics, generate reports, and manage complaints.
- **Implementation of a Badge Scanning System:** Employees scan their badge when boarding, automatically recording presence and verifying access rights.
- **Employee Mobile Application:** A user-friendly app for transport requests, real-time tracking, and communication with HR.

This solution aims to automate employee transport tracking, reduce costs, improve satisfaction, and provide a powerful decision-support system data visualization and intelligent features.

1.4 Project Management Methodologies Study

Project management is a critical discipline to organize, plan, coordinate, and monitor progress of a project in order to achieve some specified goals within defined time and cost frameworks. A project management methodology refers to an organized set of rules, practices, and processes that aim to guide the successful execution of a project. It provides a systematic framework that ensures enhanced control over scope, time, cost, quality, and risk. Various project management methods exist and understanding them is crucial, as it helps in selecting the most suitable method in accordance with the project context and requirement. Each method has its pros, cons, and process that can easily affect the success of a project. The aim of this section is to present various methods as GIMSI and Scrum.

1.4.1 The GIMSI Method

The GIMSI approach is a project management method for information systems projects. Created in a French-speaking environment, it is an integrated approach that covers the

whole life cycle of an information system, from the identification of needs to the operational deployment.

1.4.1.1 GIMSI objectives

GIMSI is designed to:

- Organize the project management process methodically.
- Guarantee consistency among all project phases and deliverables.
- Bring all stakeholders together collaboratively.
- Favor knowledge capitalization and reuse.

1.4.1.2 Main Phases of GIMSI

The GIMSI methodology is composed of five main phases that are illustrated in the figure 1.3 and explained below:

1. **Preliminary Study:** Project justification, context analysis, and needs identification.
2. **Opportunity Study:** Feasibility study and potential solutions analysis.
3. **Detailed Study:** Functional requirements specification and target system modeling.
4. **Implementation:** Unit development and testing, integration.
5. **Deployment and Maintenance:** Change management support, go-live, and continuous improvement.

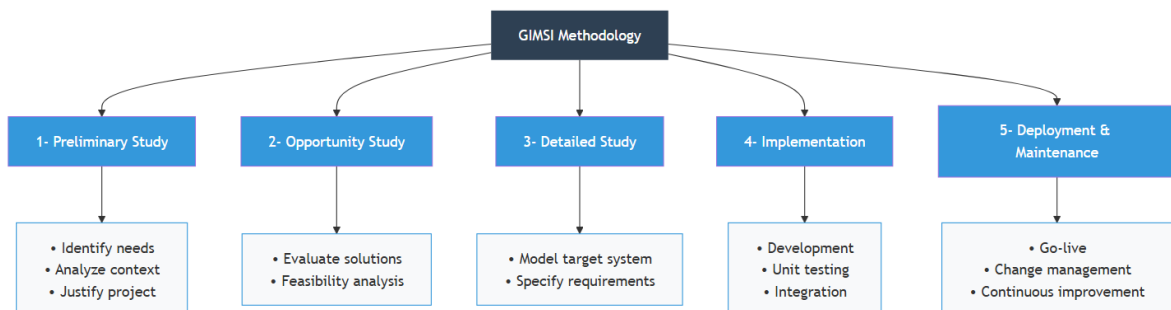


Figure 1.3: Main Phases of GIMSI.

1.4.1.3 Advantages of GIMSI

The GIMSI methodology offers numerous advantages, including:

- Good documentation at all stages for enhanced traceability.
- Well-structured and well adapted to complex information systems.
- Gives an overall overview of the to-be-built system.

1.4.2 SCRUM Methodology

Scrum is an adaptive project management framework that revolves around continuous change, teamwork, rapid delivery of working software, and customer satisfaction. Scrum is most suitable for projects with shifting requirements such as software development. Figure 1.4 shows the flow of the scrum process:

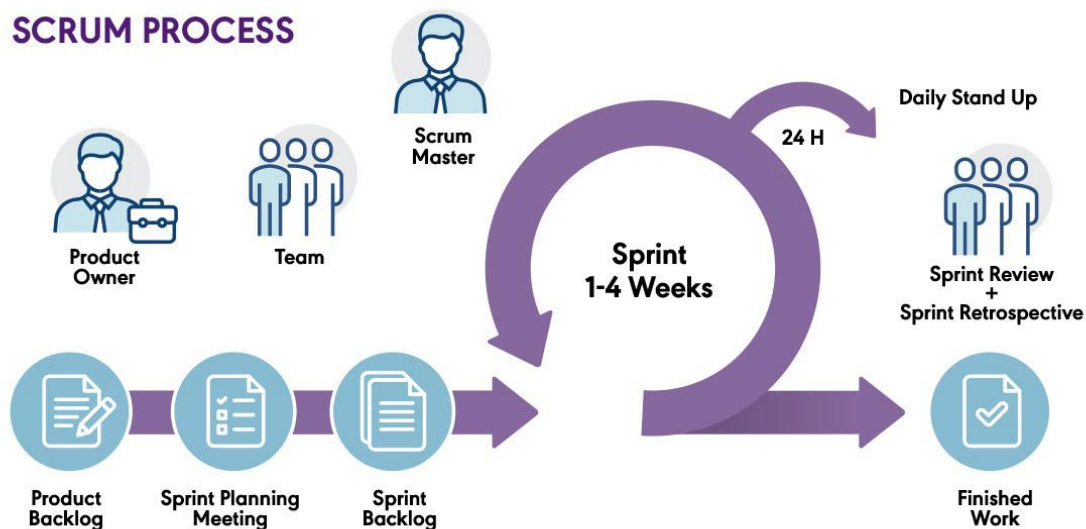


Figure 1.4: SCRUM PROCESS [2].

1.4.2.1 Key Principles of Scrum

The key principles of Scrum include:

- **Iterative and Incremental:** The project is divided into short periods referred to as sprints (typically 2 to 4 weeks).
- **Well-Defined Roles:** As shown in the figure 1.5 Scrum has three roles:
 - **Product Owner:** Responsible for the product vision and product backlog, prioritizing backlog items based on the business value. The Product Owner's

goal is to move the team to deliver the highest impact based on prioritized tasks.

- **Scrum Master:** Facilitates the process of Scrum, removes impediments for the team while forming and supporting the team and Product Owner, and encourages reflection and continuous improvement.
- **Development Team:** self-organizing, cross-functional team that delivers a product increment in each Sprint, and is responsible for determining how to do the work.

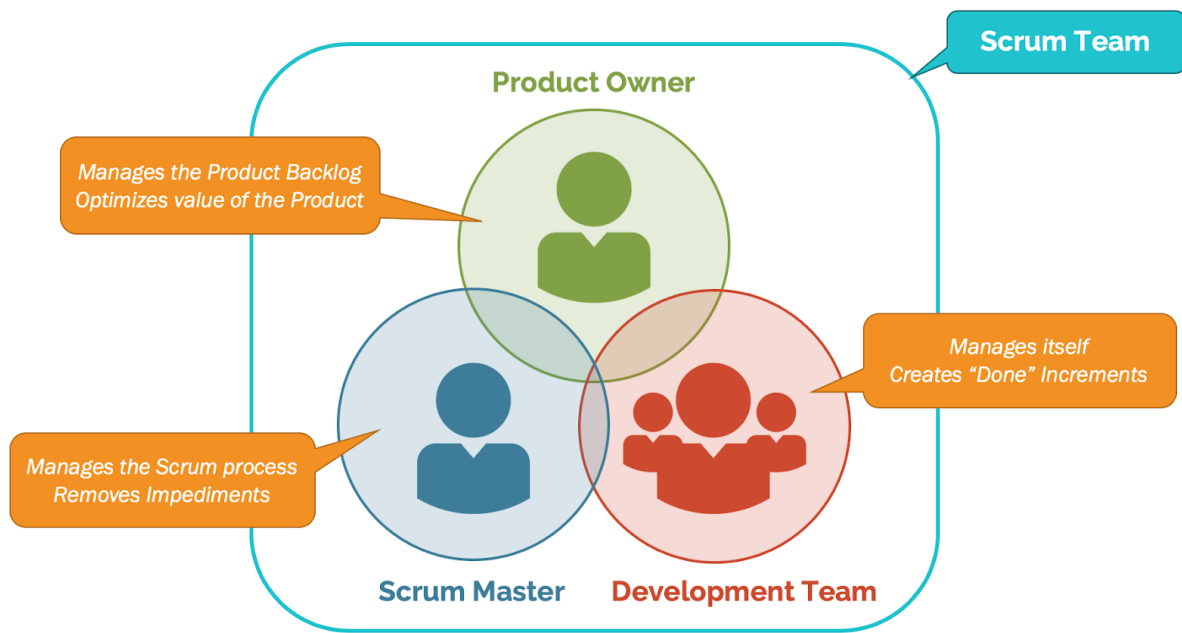


Figure 1.5: SCRUM Roles [3].

- **Continuous Delivery:** A potentially releasable product increment is delivered at the end of each sprint.
- **Scrum Ceremonies:** Daily Scrum, Sprint Planning, Sprint Review, and Sprint Retrospective.

1.4.2.2 Advantages of Scrum

The Advantages of Scrum include:

- High responsiveness to changes.
- Continuous client engagement.
- Continuous improvement using retrospectives.
- Complete visibility into project progress.

1.4.3 Adopted Methodology

For this project, Scrum was adopted due to its nature of flexibility and ability to handle altering requirements efficiently. The project requires an adaptable and iterative development process accommodating continuous user feedback and incremental value delivery. Scrum's iterative short cycles and regular ceremonies enable the team to adapt quickly to changes and continually improve.

1.5 Conclusion:

In this first introductory chapter, I introduced the context of the host organization as well as the framework of the project. After providing a general overview, I conducted an in-depth analysis of the current situation, which allowed me to gain a comprehensive understanding of the project. These aspects are considered the initial requirements that I will work to address in the upcoming chapters.

CHAPTER 2

SPRINT 0: PREPARATION PHASE

2.1 Introduction

The Sprint 0 phase is an important one since it sets the stage for the entire development. During the first phase, I identify stakeholders, their needs, and evaluate the functional and non-functional requirements of the system. After going through the Scrum process, I am able to define the backlog of the product and track what the next sprints will look like. By following an iterative approach, I can accomplish incremental improvements while seeing how the project is progressing along its road map.

2.2 Requirements specification

Requirements specification is a vital stage of any BI project. It is the process of identifying and documenting stakeholder requirements and expectations. A good specification confirms that the entire project is to be to business objectives and end-user needs.

2.2.1 Identification of system actors

An actor in a system is any user or external entity that uses the system to perform some tasks or receive services. Actors are required in system design as they define roles, responsibilities, access rights, and features related to users. The clear identification of actors results in a more efficient, secure, and personalized experience for each type of user. There are three roles present in our transport management solution: the **HR Manager**, the **HR Employee**, and the **Employee**.

- **The HR Manager:** Has all the privileges, including managing users, managing transport, real-time bus tracking, accessing dynamic dashboards, and using an intelligent chatbot to generate personalized visual insights and receive recommendations.

- **The HR Employee:** Is responsible for managing transport, real-time bus tracking, and viewing dashboards for operational performance.
- **The Employee:** Has the privileges to submit complaints, request line changes, make reservations, and track bus movements in real time, ensuring a seamless daily commute.

2.2.2 Functional requirements

Functional requirements correspond to the functionalities expected by users in order to effectively meet their business objectives. As part of optimizing employee transportation, SEBN TN2 wishes to implement a complete solution meeting the following functional requirements:

- Integration of a process of data extracting, transforming, and loading information regarding employees, routes, buses, stops, service providers, etc.
- Development of interactive dashboards for tracking punctuality, service provider efficiency, costs, etc.
- Developing an intelligent chatbot to assist in the automatic generation of customized visuals and give recommendations based on transport data.
- Developing a dedicated web portal for the HR department for transport management in integrated form, real-time bus tracking, and monitoring performance indicators.
- Creating a system for a badge scanning system for secure and automated control of bus entry.
- Implementation of a mobile application for workers to book seats, complain about problems, request changing the route and tracking buses.

2.2.3 Non-Functional requirements

In addition to the functional features, the system must also meet several non-functional requirements to ensure reliability, performance, and user-friendliness. It is essential that the platform complies with the following criteria:

1. **Availability:** The system must be available at almost all times.
2. **Security:** The system and data must be secured
3. **Usability:** The interface must be intuitive and user-friendly for HR staff and employees to navigate with minimal training.

4. **Accessibility:** The system should be accessible on both mobile and desktop, as well as compatible with Android and iOS.
5. **Scalability:** The system should be scalable as the number of buses and employees increase without losing performance.
6. **Maintainability:** The code should adhere to standard development practices and document to allow for maintenance updates and bug fixing.

2.2.4 Requirements modeling

Use case diagrams provide a modeling of the functional behaviour of a software system. It can also describe how it will be used by the stakeholder, and how it will also interact with the stakeholder. Our system is composed of the following use cases:

- Scan Badge.
- Reserve a seat.
- Send complaint.
- Request to change line.
- Show bus position.
- Display fill rate.
- View dashboard.
- Manage reservations.
- Show complaints.
- Manage employees.
- Manage Buses.
- Manage travel lines.
- Manage changing Requests.
- Manage users.
- Generate AI reports.

Figure 2.1 shows the general use case diagram, it provides a good view of the association between actors and different use cases.



Figure 2.1: Global Use Case Diagram.

As mentioned the system contains three identified actors: Employee, Human Resources (HR) Employee and Human Resources Manager. The Employee interacts with the system by scanning their badge. They reserve rides and can submit complaints if there is something wrong with their ride. The HR Employee looks at the system in a global view, he has dashboards viewable to show them their data for transportation, the control and reservation management, verification, they will receive any complaints submitted by

Employees, and they manage all of the Employees' transportation data management role. Both Employee and HR Employees have the ability to view buses in real time and must authenticate to use the system. Human resources managers are able to manage system users and can use the system to generate AI reports.

2.3 Project Management with Scrum

The Scrum process enables Business Intelligence project management to be more efficient due to the potential to easily adapt to change and user needs. It is therefore important to extensively define stakeholders and involve them in the development process. Secondly, organizing work according to a product backlog helps in prioritizing features to be developed in each sprint. Finally, using a consistent method in every cycle ensures steady and controlled project progress.

2.3.1 Identification of the Scrum Team

For my project, I have identified the team members, who are as follows:

- **The Product Owner:** Mr. Moetaz Fatin, Business Analyst and my professional supervisor, formalizes the feature to develop and the prioritization based on the business needs with reference to transportation activity from company SEBN-TN2.
- **The Scrum Master:** Mr. Sofiene Boutaeib, PhD in Computer Science from ISIMA, and my academic supervisor, who ensures the progress of my project.
- **The team members:** Ichrak Belghouthi, a BI developer, who is responsible for implementing the different features of the project and collaborate with the Product Owner for guidance and work closely with the Scrum Master for advice.

2.3.2 Product backlog

The product backlog is a prioritized list of features, improvements, bug fixes, and technical work that outlines the range of work for a project, particularly in an Agile and Scrum framework. The Product Owner manages the backlog and continually updates it based on feedback, new requirements, and business needs.

Table 2.1: Product Backlog.

ID	Functionality	US ID	User Story	Priority
1	Authentication	1	As a user, I want to authenticate myself.	High
2	Conceptual Modeling	2	As a HR employee, I want a conceptual model that is in relation to the requirements of the specifications document.	High
3	Dashboard Creation	3	As HR employee, I want complete dashboards bringing together key activity indicators to give a complete picture.	High
4	Create a chatbot	4	As an HR manager, I want to have access to a chatbot that can generate personalized visualizations based on my specific needs.	High
5	Manage users	5.1	As an HR Manager, I want to view the users list.	High
		5.2	As an HR Manager, I want to add a user.	High
		5.3	As an HR Manager, I want to delete a user.	High
6	Manage Employees	6.1	As an HR employee, I want to view the employees list.	High
		6.2	As an HR employee, I want to add an employee.	High
		6.3	As an HR employee, I want to modify an employee.	High
		6.4	As an HR employee, I want to delete an employee.	High
7	Manage Buses	7.1	As an HR employee, I want to view the Buses list.	High
		7.2	As an HR employee, I want to add a Bus.	High
		7.3	As an HR employee, I want to modify a Bus.	High
		7.4	As an HR employee, I want to delete a Bus.	High
8	Manage Lines	8.1	As an HR employee, I want to view the Lines list.	High
		8.2	As an HR employee, I want to add a Line.	High
		8.3	As an HR employee, I want to modify a Line.	High
		8.4	As an HR employee, I want to delete a Line.	High

Table 5.1: Product Backlog (Continued).

ID	Functionality	US ID	User Story	Priority
9	Manage Stops	9.1	As an HR employee, I want to view the Stops list.	High
		9.2	As an HR employee, I want to add a Stop.	High
		9.3	As an HR employee, I want to modify a Stop.	High
		9.4	As an HR employee, I want to delete a Stop.	High
10	View complaints	10	As an HR employee, I want to read complaints.	High
11	Show Bus Position	11	As an employee, I want to track buses in real time.	High
12	Display Fill Rate	12	As an employee, I want to view the fill rate for each bus line and shift.	High
13	Scan Badge	13	As an employee, I want to scan my badge.	High
14	Access to dashboards	14	As an HR employee, I want to see a well-structured Dashboard.	Medium
15	Manage Reservations	15.1	As an HR employee, I want to view the reservations list.	Medium
		15.2	As an HR employee, I want to accept a reservation.	Medium
		15.3	As an HR employee, I want to refuse a reservation.	Medium
16	Manage changing requests	16.1	As an HR employee, I want to view the changing requests list.	Medium
		16.2	As an HR employee, I want to accept a changing requests.	Medium
		16.3	As an HR employee, I want to refuse a changing requests.	Medium
17	Send complaints	17.1	As an employee, I want to send a complaint.	Medium
		17.2	As an employee, I want to see my complaint's history.	Medium
18	Request to change line	18	As an employee, I want to request to change my line.	Low
19	Reserve a Seat	19.1	As an employee, I want to reserve a bus.	Low
		19.2	As an employee, I want to see my reservations' history.	Low

2.3.3 Sprints Planning

Since the project was developed using the Scrum methodology, sprint planning was critical in establishing the project workflow and continuous progress. By separating the work into

focused and time-boxed sprints we were able to prioritize tasks, to be more adaptable, and to better track deliverables at key project milestones. we have defined the following sprints:

Sprint 1: Design and development of dashboards:

- Design and implementation of the ETL process for employee transport tracking.
- Design and development of interactive dashboards.

Sprint 2: Chatbot-Driven Dashboard Generation and Insight Analysis:

- Design and development of a chatbot to generate personalized visualizations based on user requests.

Sprint 3: Transport management:

- Authentication.
- Transport Management.
- Bus Tracking.
- Dashboard Access.

Sprint 4: Real-Time Employee Authentication and Transport Request System:

- Authentication.
- Request Handling.
- Bus Tracking.
- Badge Scanning Interface.

2.3.4 Project Planning

Gantt charts are sites for visual planning, and they break down a project into clear phases and sprints, allow you to assign tasks to each period, and allow you to monitor the progression over time. Figure 2.2 represents my project's Gantt chart:

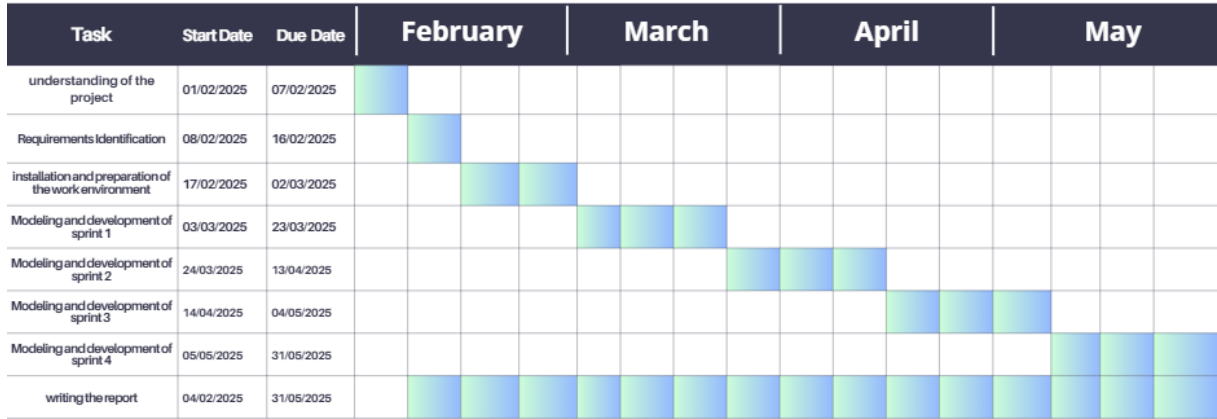


Figure 2.2: Gantt chart.

2.4 Technical architecture of the decision-making solution

My technical design adheres to the conventional ETL (Extract, Transform, Load) pipeline that corresponds to the three most significant phases of this project: Data Collection, Data Integration, and Data Delivery. The Figure 2.3 illustrates the different phases of our system and are as follows:

- Data Collection:** This is the phase where employee transport management data from diverse sources is gathered. In my case, data is extracted primarily from a central database that has data such as employee details, bus schedules, reservation, and history of tracking.
- Data Integration:** Once the data is collected, it is scrubbed and converted to make it homogeneous and of high quality. This means removing duplicates, fixing errors, reformatting fields (e.g., text-based dates into standard date formats), and generating calculated fields so that the dataset is richer. The goal is to create a formatted, homogeneous dataset that will be easy to analyze efficiently.
- Data Delivery:** At this phase, the data is delivered to the destination system, a centralized database to be analyzed and visualized. It prepares the data for further processing, such as generating dashboards, tracking employee transport behavior, and supporting decision-making through visualization tools.

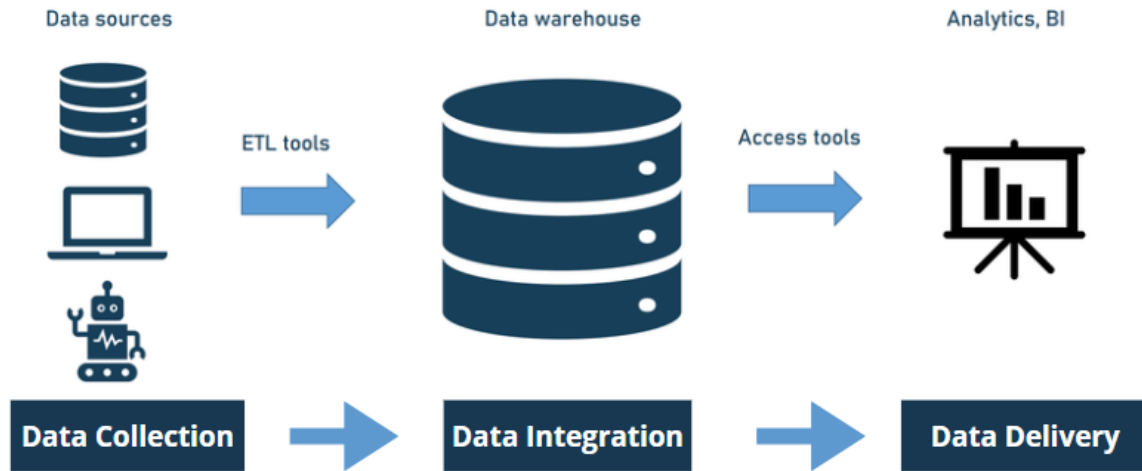


Figure 2.3: Technical architecture of the system [4].

Now, we will present key BI concepts required to build a successful analytical system. These concepts include the Data Warehouse, the Data Mart, and the ETL process. They serve as the foundation for collecting, organizing, and delivering valuable insights to support data-driven decision-making.

- **Data Warehouse:** A Data Warehouse is a consolidated repository of data from various sources in a centralized repository. It is tuned for reporting and analytical queries to allow organizations to track performance over time, identify trends, and make strategic decisions. In contrast to operational databases, a data warehouse is for quick access to historical and real-time data for the entire business.
- **Data Mart:** A Data Mart is a small, focused incarnation of a data warehouse that is utilized for the needs of one business unit or department, say sales, HR, or logistics. It provides quicker access to relevant data tailored to given use cases so that groups can more effectively examine and respond to what matters most to them.
- **ETL (Extract, Transform, Load):** The ETL process is responsible for extracting and aggregating data from a multitude of sources, transforming and cleansing it, and loading it into a central repository for analysis. It ensures that the data is presented to users in a format that is consistent, credible, and ready for visualization or decision support systems.

2.5 Overall design of the data warehouse

The overall structure of a data warehouse is the foundation of any business intelligence system. It dictates the organization, storage, and accessibility of information from varied

sources for analysis and decision-making. A well-designed data warehouse ensures data consistency, integrity, and performance by organizing information in such a way that supports effective querying and reporting. This part covers the general architectural specifications of a data warehouse, typically based on multidimensional models like the star schema or the snowflake schema, and how these facilitate analytical processing across business domains.

2.5.1 Data Collection

Data collection is a fundamental task in every BI initiative, as it forms the foundation upon which KPIs are defined and meaningful insight is developed. Collecting accurate, relevant, and complete data ensures that the indicators selected do reflect the reality of operations and are used to make sound decisions. For the scope of this project, the HR department of SEBN-TN2 provided a set of Excel files with complete information about the employee transportation system of the company. The data sets include records of bus lines, shifts, providers, stops, and employee assignments that were crucial to understanding the system structure and defining the KPIs to monitor.

2.5.2 Key Performance Indicators (KPIs) Selection

In all BI systems, KPIs are key measurements used to measure and contrast the performance of different aspects of an organization. They provide valuable information on which strategic choices can be made based on, are tied to business objectives, and reflect areas where enhancements can be created. In the employee transport management system for SEBN-TN2, a set of precisely defined KPIs has been selected and grouped into thematic modules in an attempt to monitor operations efficiently and offer continuous optimization. These KPIs are presented below:

- The "**General Overview**" module :
 - Total number of transported employees.
 - Total number of transportation lines.
 - Total number of available buses.
 - Number of employees per line and per shift.
 - Transportation cost per line.
 - Average bus capacity utilization rate.
 - Delay rate per shift and per line.
 - Total penalty cost per transportation provider.
 - Total amount of employee contributions.

- The "**Line Analysis**" module :
 - Total number of transportation lines.
 - Total number of transported employees.
 - Top 5 most saturated transportation lines.
 - Number of employees per line and shift.
 - Number of lines operated per provider.
 - Number of bus stops per line.
 - Bus occupancy rate per line.
- The "**Financial Analysis**" module:
 - Total employee transportation cost.
 - Average transportation cost per employee.
 - Transportation cost per line and per bus capacity.
 - Total cost incurred per provider.
 - the total costs per shift.
 - Cost of a single trip, broken down by line and bus capacity.
 - Total employee contribution to transportation expenses.
- The "**Delay and Punctuality Analysis**" module:
 - Overall delay percentage.
 - Overall punctuality percentage.
 - Delay rate by shift and line.
 - Average delay duration by shift and by provider.
 - Evolution of the delay rate per year.
 - Total penalty cost per service provider.
 - Delay rate per transportation provider.

2.5.3 Data Warehouse Design Approaches

Data warehouse design is a crucial step of setting up a robust BI system. Data warehouse is a centralized database storing historical, consolidated, subject-oriented data to support reporting and analytical operations. Two major methodologies have been developed over time for designing a data warehouse: the Inmon method and the Kimball method. Both offer varying approaches to structuring the data and mapping it with business needs.

2.5.3.1 Inmon Approach

The Inmon approach, created by Bill Inmon, is the top-down approach. It begins with the development of a centralized, normalized enterprise data warehouse (EDW) that data marts are then developed from as and when needed. Data integration and consistency are given prime importance with all data standardized to enterprise-wide standards before it is used for reporting [5].

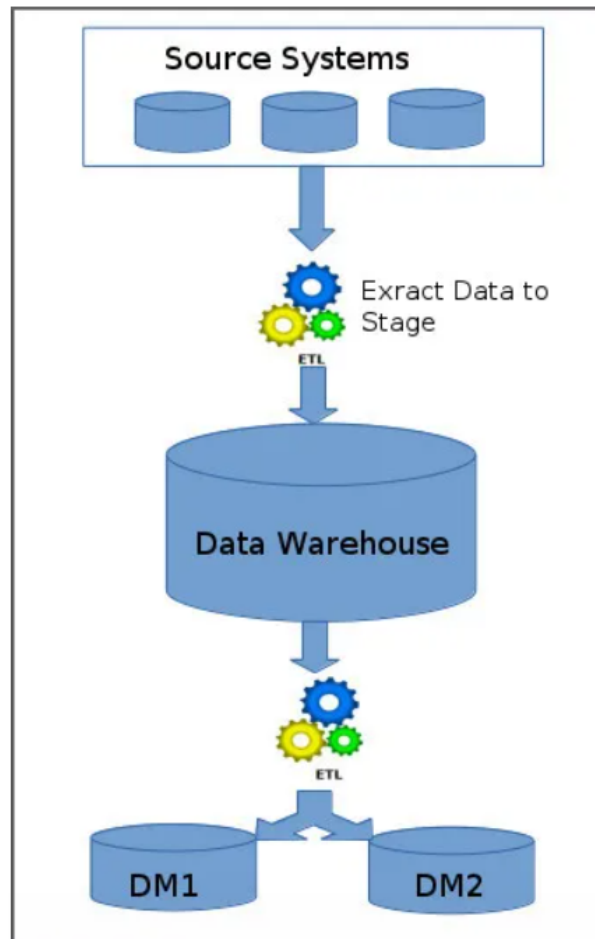


Figure 2.4: Inmon Approach architecture [6].

2.5.3.2 Kimball Approach

The Kimball approach, supported by Ralph Kimball, is also called the bottom-up approach to data warehouse design. It is about building data marts around specific business processes (e.g., transportation, sales, finance) and aggregating them later at a central data warehouse. The method relies heavily on the application of the dimensional modeling technique, where data is organized in fact and dimension tables, usually in a star or snowflake schema [7].

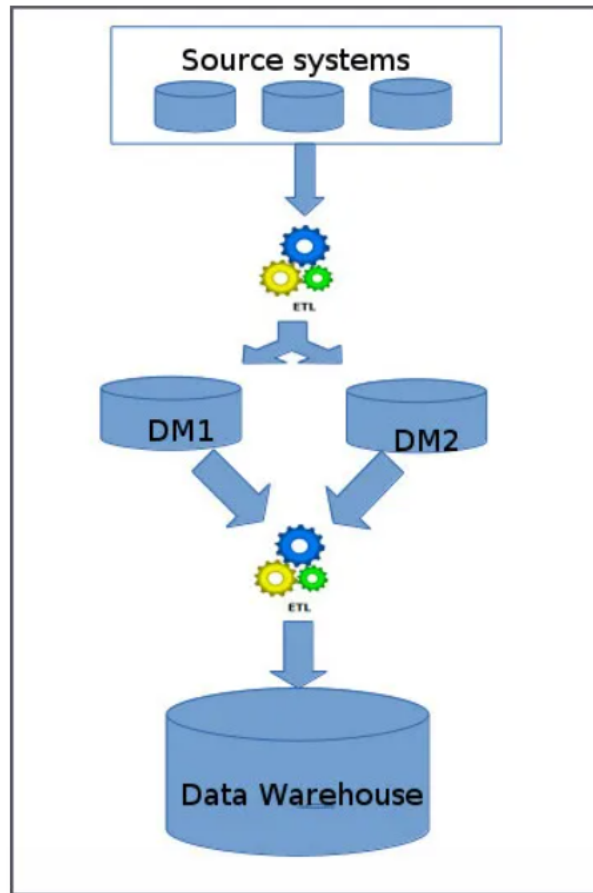


Figure 2.5: Kimball Approach architecture [8].

2.5.3.3 Adopted Approach

For this project, the Kimball methodology was used. The reasons for the usage were the need for fast access to insight, business-driven KPIs as high-priority, and modularity of the transport processes being analyzed. The bottom-up approach facilitated the rapid development of subject-specific data marts that aligned with the organization's short-term analytical needs. The simplicity of the dimensional model and performance gains also qualified it as the optimal solution for building interactive dashboards and delivering an end-user-friendly reporting experience to stakeholders such as HR and operations managers.

2.6 Multidimensional Modeling

Multidimensional modeling is a fundamental concept in BI and data warehousing. It organizes the data into facts and dimensions to facilitate good analysis and reporting. A fact table contains quantitative data, typically numeric measures or KPIs, which define business events or transactions. Dimension tables contain descriptive properties of the facts such that users can slice and dice the data around various angles such as time,

location, or product. In the following section, we present the tables utilized in our multidimensional model. They form the foundation of the analytical framework and support intensive and flexible exploration of the data.

- **Fact Table:**

- **Fact_suivi:** This is the fact table containing all information related to transportation tracking.

- **Dimension Tables:**

- **dim_employee:** Dimension table containing employee information.
- **dim_shifts:** Dimension table containing shift information.
- **dim_provider:** Dimension table containing provider information.
- **dim_stop:** Dimension table containing stop information.
- **dim_line:** Dimension table containing line information.
- **dim_bus:** Dimension table containing bus information.
- **dim_arrivalTime:** Dimension table containing arrival time information.

2.6.1 Conceptual Model Analysis

When working within the scope of data management and business intelligence, developing a conceptual model is in the structured organization of data that clearly illustrates the manner in which data integrates with a particular business goal or objective. A conceptual model is a logical and high-level representation of data and its relationship, providing a clear representation of data and relationships, prior to developing detailed design, and implementation directions. This section will cover three commonly used conceptual models in data architecture:

- **Star Schema Model:** The Star Schema is a model that consists of one central fact table, and the rest of the schema is dimension tables. The design is simple, which lends itself well to analytical queries and facilitating quick retrieval of data and analysis in data warehousing environments.

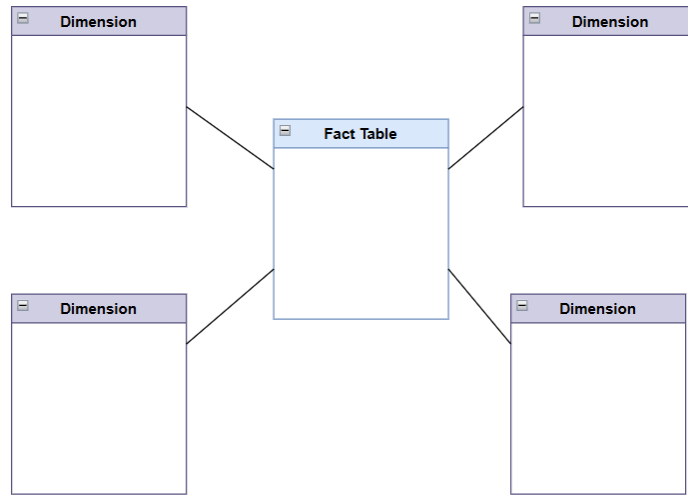


Figure 2.6: Star Schema Model.

- Snowflake Schema Model:** The Snowflake Schema is an extension of the Star Schema. Unlike the Star Schema, with respect to each of the dimension tables, Snowflake will normalize them into more than one related table. This should reduce redundancy and offer more data integrity with respect to the source data, but it may necessitate more complex queries.

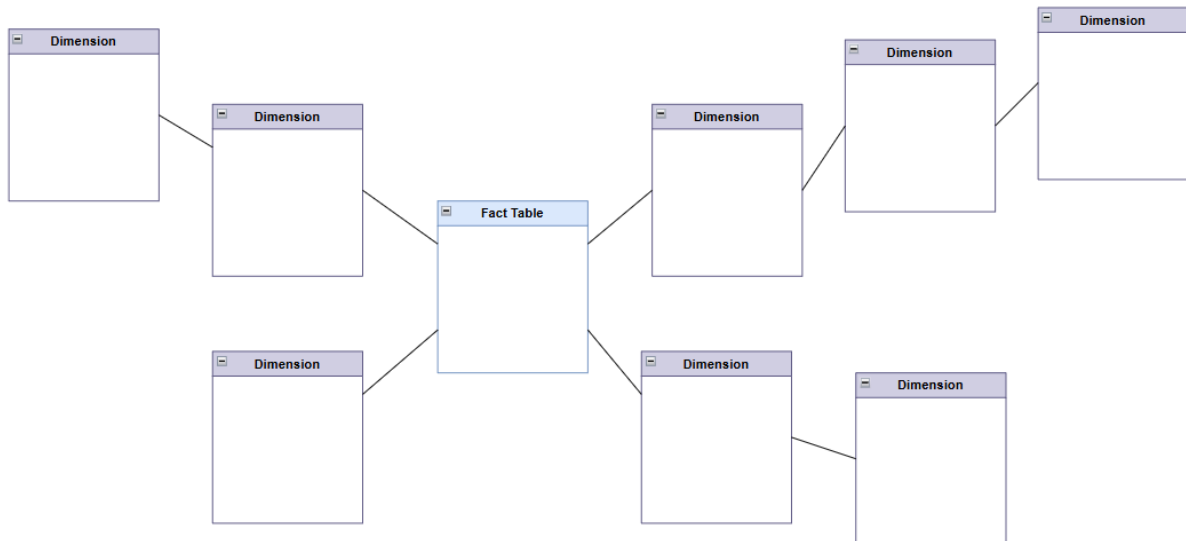


Figure 2.7: Snowflake Schema Model.

- Galaxy Schema Model (or Fact Constellation):** The Galaxy Schema consists of several fact tables, and each fact table uses shared dimension tables. This model provides good data analytic capabilities to satisfy more complex analysis. In other words, it allows for simultaneous analytical assessment of several different business processes or analytic perspectives.

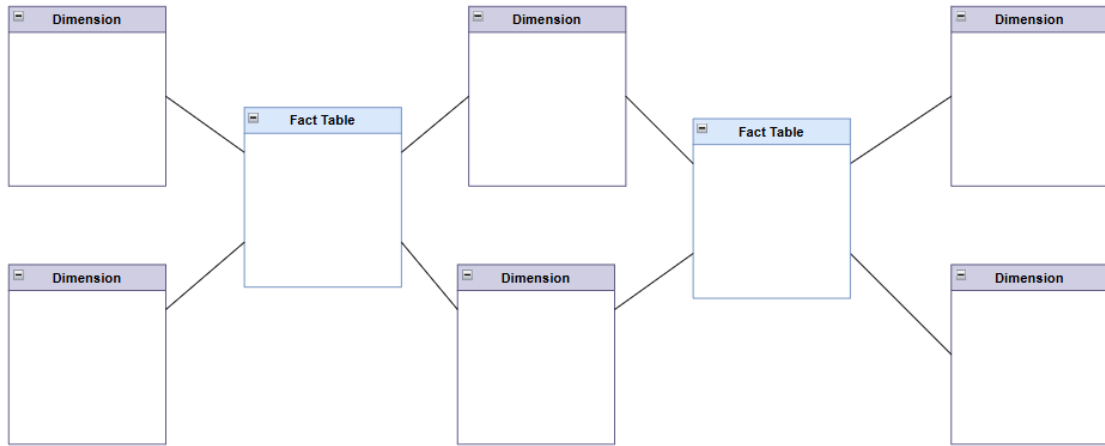


Figure 2.8: Galaxy Schema Model.

2.6.2 Choice of the Conceptual Model:

The decision was made for a snowflake schema to correctly model intricate relationships among some of the dimensions without sacrificing a clean, normalized design. The model joins all of the measures in the `fact_suivi` table and refers to several normalized dimension tables for an optimal balance between data consistency and analysis query performance. With the use of intermediary tables, like the `ligne_bus` bridge joining `dim_lignes` and `dim_bus`, we provide exact modeling of bus routes and assignments without duplication.

Model Description: We designed a customized snowflake schema to suit the company's requirements in our project. The fact table, `fact_suivi`, at the middle records each transport event, and dimensions `dim_employee`, `dim_shifts`, `dim_arrets`, `dim_lignes`, and `dim_bus` enable detailed analysis on various business dimensions. This hierarchical, normalized structure (shown in figure 2.9) enables fast and logical data extraction, and reporting and visualization are consistent and easy.

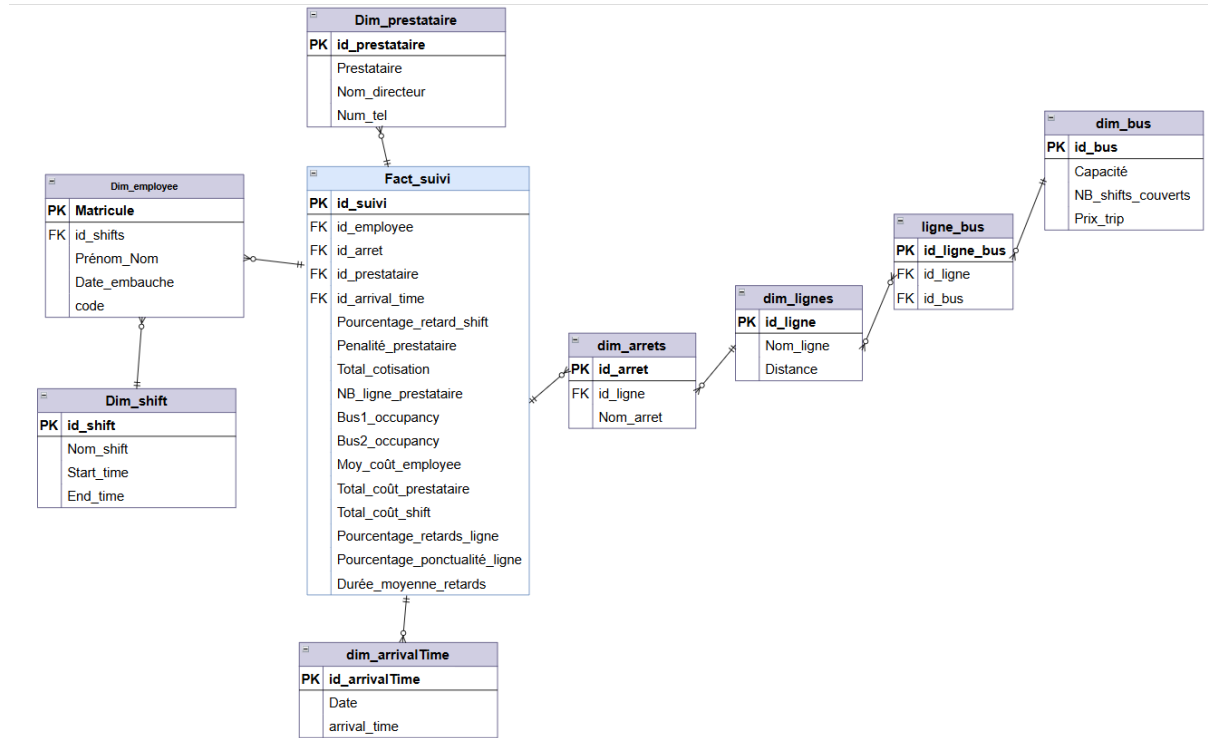


Figure 2.9: the Conceptual Model of the DW.

2.7 Work Environment

This section provides an overview of the work environment under which the project was developed during my internship. It has both hardware and software components used in all development and modeling processes. Understanding this environment is critical in understanding tools and resources available as well as constraints that would have influenced development choices.

2.7.1 Hardware environment

I used a personal laptop throughout the internship with the following details. These bits of equipment served all phases of development, ranging from coding and modeling to testing and deployment. The machine offered sufficient performance to sustain the varied tools and processes required by the project.

- **Type:** Laptop.
- **Brand:** HP.
- **Processor:** 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz.
- **RAM:** 8,00 Go.
- **Operating system:** Windows 11.

2.7.2 Software environment

This part describes the software tools and platforms used for modeling, developing, and executing the project. It describes the development environment, library, database systems, and frameworks that were core to the successful implementation of the application.

Visual Studio Code:

Visual Studio Code is a lightweight, cross-platform code editor for Windows, macOS, and Linux. Visual Studio Code natively supports JavaScript, TypeScript, and Node.js, and through its extensive extension ecosystem, developers can develop with almost any programming language and framework [9].



Figure 2.10: VS Code Logo [9].

MySQL:

MySQL is an RDBMS that operates under the dual licensing model—open-source (GPL) and commercial. It is among the widely used database products applied globally by individual users, primarily for web applications, as well as professionals. MySQL is a strong competitor to other top-rated RDBMS databases such as Oracle, PostgreSQL, and Microsoft SQL Server [10].



Figure 2.11: MySQL Logo [10].

phpMyAdmin:

phpMyAdmin is free and open-source software written in PHP that is used to handle MySQL databases through a web interface. phpMyAdmin offers support for MySQL as well as MariaDB, including wide-scale operations such as database, table, column, relationship, index, user, and permissions management. While it offers a straightforward graphical interface for everyday tasks, it also supports the direct execution of SQL queries for advanced tasks [11].



Figure 2.12: phpMyAdmin Logo [11].

XAMPP:

XAMPP is a software bundle offering to install a local web server, FTP, and email server. It stands for "X (cross-platform), Apache, MariaDB, PHP, and Perl." It is easy to install and adaptable and has enabled a wide range of users, including those with no technical knowledge, to use it. It supports most of the most widely used operating systems and is for local development and testing [12].



Figure 2.13: XAMPP Logo [12].

Talend:

Talend is a leader in Cloud and Big Data integration software. It allows organizations to make decisions in real time and become data-driven. With Talend, data is made more accessible, of higher quality, and can be transferred to the target systems at a higher speed [13].



Figure 2.14: Talend Logo [14].

Microsoft Power BI:

Microsoft Power BI is a business analytics service developed by Microsoft which enables users to create customized and interactive data visualizations. Its ease of use enables even non-technical users to create their own reports and dashboards easily. It consists of a collection of tools, services, and connectors that all together move separated data sources into rich, interactive, and visual insights. Power BI supports integration with several data sources, including Excel documents, SQL databases, and also hybrid data warehouses—hosted either locally or in the cloud [15].



Figure 2.15: Power BI Logo [15].

Draw.io:

draw.io, or Diagrams.net, is an open-source application that is employed to generate diagrams and flowcharts. It possesses both automatic layout capabilities and user-defined layout capabilities. It contains a huge collection of shapes and graphics elements and can be utilized to generate rich, visually appealing diagrams according to the need of the user [16].



Figure 2.16: Draw.io Logo [16].

Overleaf:

Overleaf is an internet platform that gives you all of the facilities to write and publish a document. It is great for professional typesetting, especially for academic and technical writing. The most useful thing about Overleaf is that it lends itself easily to collaborative work - you can work together on the same document at the same time [17].



Figure 2.17: Overleaf's Logo [17].

2.8 Conclusion

Sprint 0 was the foundation for successful initiation of our project. During this sprint, we targeted requirements gathering and modeling, establishing the technical and functional

scope, and setting up the project management framework upon the Scrum methodology. We also established the technical architecture, defined the KPIs, and created the overall data warehouse layout by knowing and choosing the most appropriate design strategies and modeling methods. In addition to this, we also established the instruments and technologies that will be used in supporting the development process and documented the hardware and software environment under which the project will be deployed. Having the basics in place, the project can now proceed to the subsequent phase. In Sprint 1, we will initiate the ETL process, focusing on data transformation and integration using Talend services. In parallel, we will also begin the development of the dashboards, laying the groundwork for the decision-making interface. This sprint marks the start of the live implementation of our business intelligence solution.

CHAPTER 3

SPRINT 1: DESIGN AND DEVELOPMENT OF DASHBOARDS

3.1 Introduction

This chapter outlines the main stages involved in creating dashboards. It begins with the data preparation and integration phase, followed by the dashboard design process, and concludes with the data presentation stage, aiming to ensure clear and effective visualization of the information.

3.2 Requirements Definition

This step is aimed at identifying, collecting and documenting the primary requirements for the information needed to design and develop the Dashboards based on the needs of the business and the users. After consultation with the Product Owner, he were to identify the key requirements, focused on being able to monitor in detail the transportation activity across the business. These criteria shaped the direction and evolution of this sprint:

- The first goal was to complete the full ETL (Extract, Transform, Load) process. Data had to follow this pipeline to ensure that it is verified, consistent, and ready for analysis.
- The second goal was to develop a decent structured dashboard that would allow for insights.

To satisfy these requirements, we have formed 4 modules which will monitor each particular aspect of transportation:

1. Module 1: General View of Employee Transportation Activities

This module provides a general view of the company's employee transportation activities, it enables to:

- Visualize the total number of employees.
- Visualize the total number of lines.
- Visualize the total number of buses.
- Visualize the number of employees per line and per shift.
- Visualize the total cost per line and the capacity of the buses.
- Visualize the delay rate of lines by shift
- Visualize the total penalties per provider.
- Visualize the total contributions from employees.

2. Module 2: Line Analysis

This module focuses on analyzing the lines, which are the set of stops where employees are collected, it enables to:

- Visualize the total number of lines.
- Visualize the total number of employees.
- Visualize the top five most saturated lines.
- Visualize the total number of employees per line and shift.
- Visualize the total number of lines per provider.
- Visualize the number of stops per line.
- Visualize the total bus occupancy per line.

3. Module 3: Financial Analysis

This module focuses on financial analysis, allowing for the tracking of costs, it enables to:

- Visualize the total transportation cost.
- Visualize the cost per employee.
- Visualize the total cost per line and bus capacity.
- Visualize the total costs per provider.
- Visualize the total costs per shift.
- Visualize the cost of a single trip per line and bus capacity.
- Visualize the total contributions.

4. Module 4: Delay Analysis

This module is dedicated to analyzing delays in the transportation system, it enables to:

- Visualize the percentage of delays.

- Visualize the percentage of punctuality.
- Visualize the delay rate of lines by shift.
- Visualize the average delay duration by shift and provider.
- Visualize Evolution of the delay rate per year.
- Visualize the total penalties per provider.
- Visualize the delay rate per service provider.

To filter the KPIs, we decided to use the following segments:

Year, Month, Day, Line, and Shift.

3.3 Sprint 1 Backlog

Sprint 1 backlog, table 3.1, was created to record and order the essential tasks to initiate the project and establish the framework of the employee transport management system.

Table 3.1: Sprint 1 Backlog.

ID	Functionality	US ID	User Story	Duration in days
1	Conceptual Modeling	1	As an HR employee, I want a conceptual model that is in relation to the requirements of the specifications document.	7
2	Dashboard Creation	2	As an HR employee, I want complete dashboards bringing together key activity indicators to give a complete picture.	14

3.4 Data integration phase (ETL)

Prior to analyzing business data in dashboards, an ETL process must be completed to produce clean, reliable, and structured data for decision-making.

3.4.1 General Overview of the ETL Process

ETL simply stands for Extract, Transform and Load, it is a process used to integrate information from several sources into a centralized system for a consistent and reliable view of information.

- **Extract:** The extraction phase consists of obtaining raw data from several source systems, such as databases, Excel files, web services, or application programming interfaces (APIs).

- **Transform:** In the transformation phase, the extracted data is cleaned, filtered and enriched. Specifically, duplicates may need to be removed, formats changed to a standard range, new metrics calculated, and connections identified. The transformation process converts raw data into relevant, meaningful and organized information to reflect the real contextual business situation.
- **Load:** In the end, the data will be loaded into a single and aggregated data warehouse or database. This repository is the single source of truth and optimized for data retrieval and dashboarding. After it has been loaded, the data will be prepared for visualization in reporting tools like Power BI to help with monitoring and decision making.

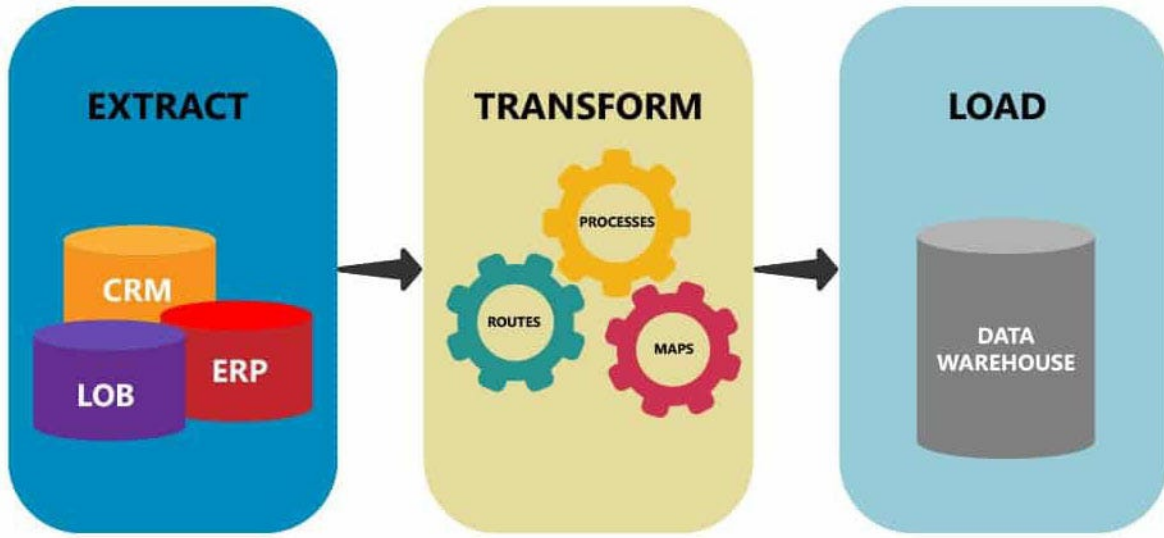


Figure 3.1: The ETL Process [18].

3.4.2 Data Extraction Phase (Extract)

The extraction is the first step in ETL and is critical for collecting the raw data that is needed for transformation and analysis.

After receiving the set of data from HR in the form of Excel files, the data was transferred from Excel into a MySQL relational database which was scalable and suitable for the ETL workflow and analytical tasks.

The database consisted of five tables, which contained the transportation system of the employees. These tables will be detailed below to better understand their structure and content, which will facilitate the subsequent transformation and analysis steps.

- **"arrival_time" Table:** The arrival_time table records bus arrival times for each route which allows detailed tracking of punctuality and schedule adherence. Our

data includes the timeframe from January 1st, 2022 to February 15th, 2025. The table contains a total of 109632 rows and 5 columns as illustrated in the figure 3.2.

table_name	column_count	row_count
arrival_time	5	109632

Figure 3.2: "arrival_time" table dimensions.

- **"employee" Table:** The "employee" table contains specific information about the employees of the company. It contains 6 columns and has a total of 1,572 rows as illustrated in the figure 3.3. Each row represents a unique employee, and the columns store key attributes such as employee ID, name, hire_date, and other relevant data necessary for analysis and tracking purposes.

table_name	column_count	row_count
employee	6	1572

Figure 3.3: "employee" table dimensions.

- **"lignes" Table:** The "lignes" table stores information about the various travel routes in the employee transportation system. It contains details such as the line ID, line name and the distance .It consists of 3 columns and a total of 24 rows, figure 3.4, with each row representing a distinct travel line.

table_name	column_count	row_count
lignes	3	24

Figure 3.4: "lignes" table dimensions.

- **"arrets" Table:** The table "arrets" has complete information about the stops that are associated with each travel line. Each record represents one of the stops where employees can get on or off the bus. it includes attributes such as the stop ID, stop name and associated line ID. The table consists of 5 columns and contains 96 rows, Figure 3.5.

table_name	column_count	row_count
arrets	5	96

Figure 3.5: "arrets" table dimensions.

- **"bus" Table:** The "bus" table contains useful information about the buses that are used to transport employees. Every record is associated with a specific bus and includes attributes such as the bus ID, capacity, and the service provider responsible for the vehicle. It consists of 5 columns and 27 rows as demonstrated in the figure 3.6.

table_name	column_count	row_count
bus	8	27

Figure 3.6: "bus" table dimensions.

We established the connection to the MySQL database using Talend's built-in components as shown in the figure 3.7 below. The connection has proven to be reliable and provides consistent access to the source tables throughout the integration pipeline.

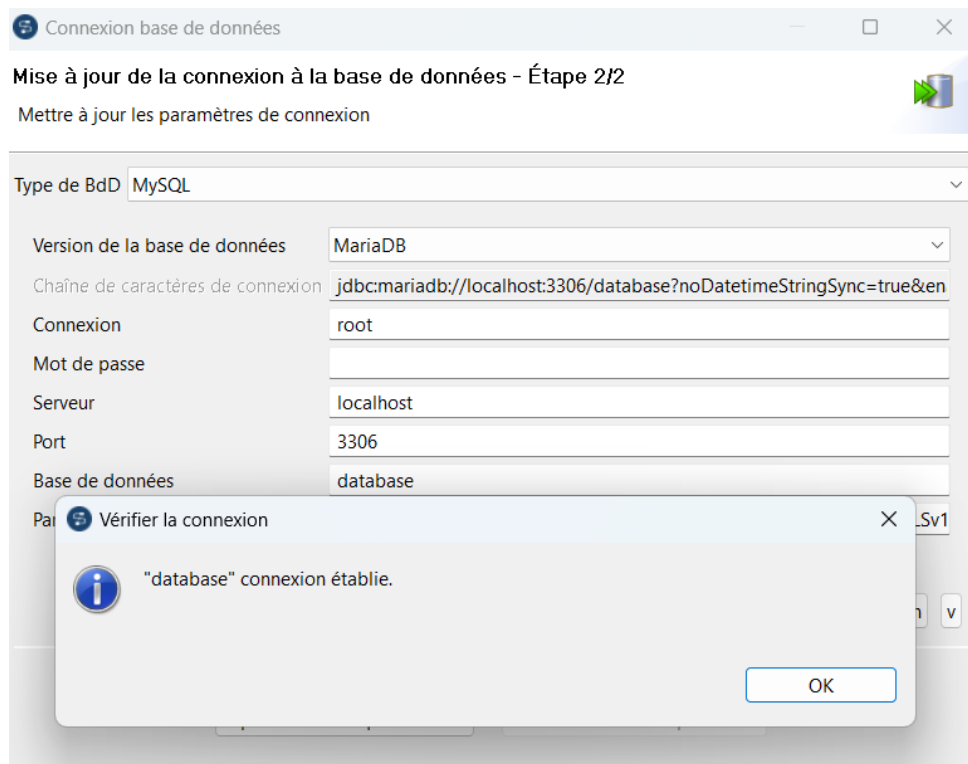


Figure 3.7: MySQL Database Connection.

For the extraction, the **tDBInput** component, figure 3.8, was utilized. This Talend component is used to connect to a database table or view and extract data through executing an SQL query. It is commonly the first component in an ETL job that extracts structured data from a relational source. Each of the source tables was connected to its own tDBInput component to ensure that all data could be accurately and completely extracted.

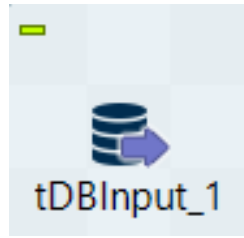


Figure 3.8: tDBInput.

3.4.3 Data Transformation Phase (Transform)

Before any analysis occurs, the data must first be transformed. This step is important to prepare the data for analysis by cleaning, standardizing, and structuring it to improve quality and consistency.

- **Identifying data quality issues:** During our initial review of the data source, we noted several quality issues. First, we found inconsistencies in some references, pointing to deleted records, leading to a loss of referential integrity and a risk of analytical errors. Second, some information appeared multiple times in many entries, creating unnecessary redundancies, thus increasing the risk of inconsistencies during updates. Third, some columns consisted exclusively of non-descriptive identifiers, making interpretation difficult without first applying transformations or contextual mappings. For non-descriptive identifiers, this can complicate the understanding of the data, but also prevent direct analysis. Also, a structural issue was identified with the depiction of hierarchical relationships. In particular, equating entities possessing a direct one-to-many relationship in a hierarchy could potentially lead to duplicate identifiers and redundant information in an attempt to depict all related elements. In an effort to overcome this, an associative mechanism was created to efficiently manage the association in order to deliver data normalization, cancel out redundancy, and facilitate appropriate and easy analysis. These issues clearly highlight the need for a thorough transformation phase that ensures data quality, ensures consistency and, therefore, helps improve the reliability of information regarding this dataset.
- **Referential Integrity Treatment:** During our review of the data, we identified a referential integrity issue: the "arrets" table had "id_ligne" values that did not exist in the "lignes" table. This happened because records from the "lignes" table were deleted without cascading updates to dependent records. figure 3.9 shows the SQL query used to identify these orphan values, and figure 3.10 displays the query results.

```

1 SELECT DISTINCT a.idLigne
2 FROM arrets a
3 WHERE a.idLigne NOT IN (
4     SELECT l.id
5     FROM lignes l
6 );

```

Figure 3.9: SQL Query to Identify Orphan Route IDs in 'arrets' Table.

idLigne

425

440

610

Figure 3.10: Orphan Route IDs in 'arrets' Table.

In order to correct this, we used the **tMap** component which had a **lookup** in Talend to check for the existence of each `id_ligne`. Records in the `arrets` table with no matches in `lignes` were removed in the transformation, ensuring tablewise referential security. Figure 3.11 shows the tMap configuration, where `row1` corresponds to the `arrets` table and `row2` to the `lignes` table.

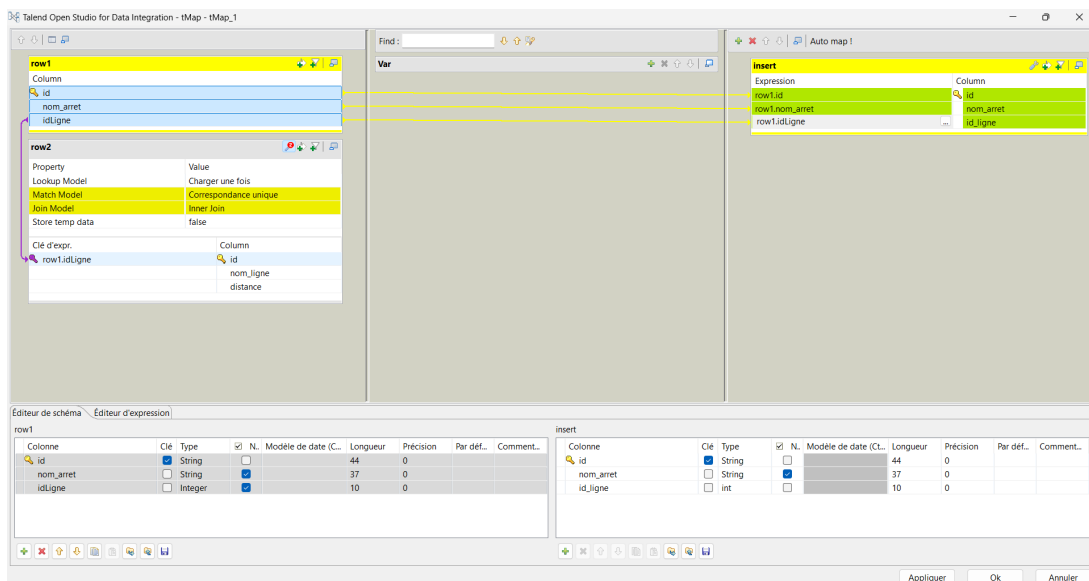


Figure 3.11: tMap component configuration with lookup.

- **Data Redundancy Treatment:** During the analysis of bus table also we noticed redundancy in data with regard to "prestataire", "num_tel", and "nom_directeur" being typical examples of repeated fields. This level of redundancy indicates a low level of normalization. Below, the query used to detect these redundancies and its result are presented in figures 3.12 and 3.13 respectively.

```

1 select Prestataire, COUNT(*) AS Redundancy
2 FROM bus
3 GROUP BY Prestataire;

```

Figure 3.12: SQL Query to detect "prestataire" redundancy.

Prestataire	Redundancy
BM	12
MBT	12
STRICT	3

Figure 3.13: Redundancy Count of "Prestataire" .

To remove redundancy, we used the "tUniqRow" component to get unique records for each service provider as illustrated in the figure 3.14.

The screenshot shows the 'tUniqRow_1' component configuration. On the left, there are tabs for 'Paramètres simples', 'Paramètres avancés', 'Paramètres dynamiques', 'Vue', and 'Documentation'. The 'Paramètres simples' tab is active. In the center, there's a 'Schéma' section with a 'Clé unique' dropdown. To the right, there's a table of columns with checkboxes for 'Attribut de clé' and 'Sensible à la casse'.

Column	Attribut de clé	Sensible à la casse
id	☐	☐
Capacite	☐	☐
NbreShiftCouvert	☐	☐
id_ligne	☐	☐
Prestataire	☒	☐
Nom_directeur	☐	☐
num_tel	☐	☐
num_trin	☐	☐

Figure 3.14: Normalization using tUniqRow.

- **Resolution of Non-Descriptive Shift Codes:** In the employee table, the shift field was encoded using numeric IDs, figure 3.15 , that were not self-explanatory which limited the readability of the data.



Figure 3.15: Sample of non-descriptive numeric "shift" IDs.

To clarify these values, we implemented a new shift dimension table with human-readable descriptions and time intervals. We used tMap combined with conditional expressions as illustrated in the figure 3.16.

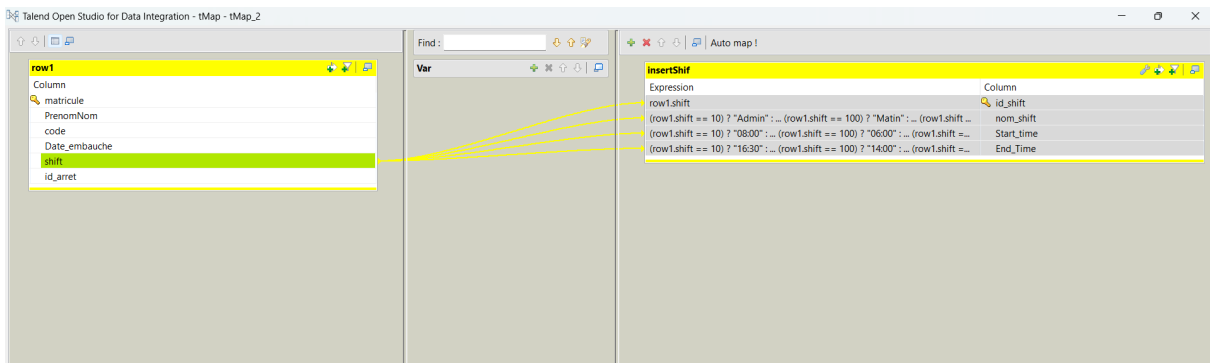


Figure 3.16: tMap component with conditional expressions.

- Normalization of Relationships:** Looking at the bus table, we identified a structural flaw concerning the redundancy of `id_ligne` in some entries. The redundancy revealed an unstated many-to-one association between buses and lines, and we identified a need for further normalization of the data model. To determine the issue, a SQL query was executed in an effort to determine the number of buses by line, and lines with more than one bus were queried. This query and its result are shown in figure 3.17 and figure 3.18, respectively.


```

1 SELECT id_ligne, COUNT(id) AS NB_Bus
2 FROM bus
3 GROUP BY id_ligne
4 HAVING COUNT(id) > 1;

```

Figure 3.17: SQL Query to detect "Buses" redundancy.

id_ligne	NB_Bus
430	2
435	2
900	2

Figure 3.18: Sample lines with more than one bus.

To rectify the problem and standardize the schema, we created a new table named `ligne_bus`. This table strengthens the relationship between buses and lines and contains three columns: generated id constructed by using `"L" + row1.id_ligne + "_B" + row1.id`, `id_ligne`, and `id_bus`. The building and configuration of this table were performed using the tMap component in Talend, as shown in figure 3.19.

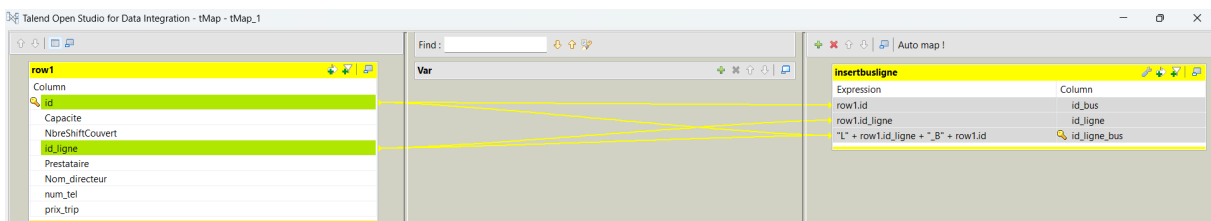


Figure 3.19: tMap component to create the "ligne_bus" table.

3.4.4 Data Loading Phase (Load)

After data transformation is completed, we now move to the next stage of loading the cleaned and structured data into the target database. With this step, we can ensure that the data is available to view and optimized for analytical or reporting products, which ultimately simplifies access for decision makers.

tDBOutput (figure 3.20): is a Talend component, which is used to write data into a database. Using this component you can insert, update or delete records into a database

table. This component is configured to work with different database types such as MySQL, PostgreSQL, and Oracle. It is necessary to verify that the transformed data is correctly loaded into the target system.

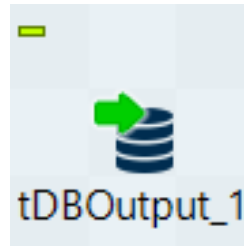


Figure 3.20: tDBOutput.

3.4.5 Creating the dimension tables and the fact table

This process started with the creation of an ETL **job**, figure 3.21, to automate the creation and population of the dimension and fact tables in the target database. The ETL job incorporates the **extraction, transformation, and loading** steps of the ETL workflow in one package so that all data is processed uniformly.

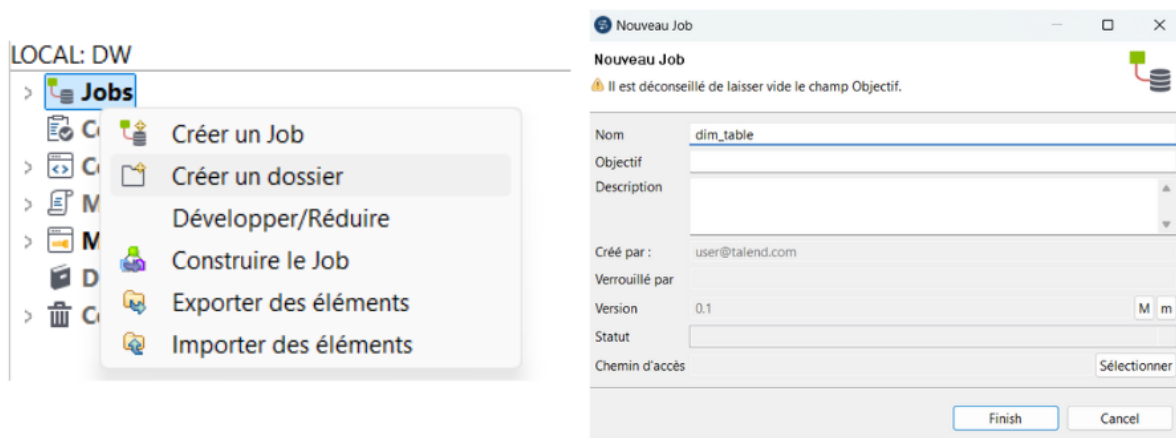


Figure 3.21: Job Creation.

We began the process by creating the required dimension tables for our analysis:

- **"dim_employee"**: Derived from the employee table, this dimension contains detailed information about the employees.
- **"dim_shifts"**: Derived from the employee table as well, this dimension decodes the shift IDs into meaningful shift descriptions.

- **"dim_arrets"**: Built from the arrets table, but included a lookup on the lignes table to ensure referential integrity and to add additional information (stops).
- **"dim_lignes"**: Contains detailed data related to the travel lines.
- **"dim_prestataire"**: Created to avoid redundancy and have a central source of information for the service providers taken from the bus table.
- **"dim_bus"**: Includes detailed information for the buses.
- **"dim_arrivalttime"**: Contains arrival times of buses per line across the entire range of the dates in our data set.
- **"ligne_bus"** Junction Table: To properly model the many-to-many relationship between the buses and lines, we created a junction table linking the two dimensions together.

Finally, we created the fact_suivi table, figure 3.22, maintaining the snowflake schema developed in the beginning of the project. This table consolidates all the transactional data in relation to employee transportation, and links all of the dimensions to enable a complete analysis of the transportation system.

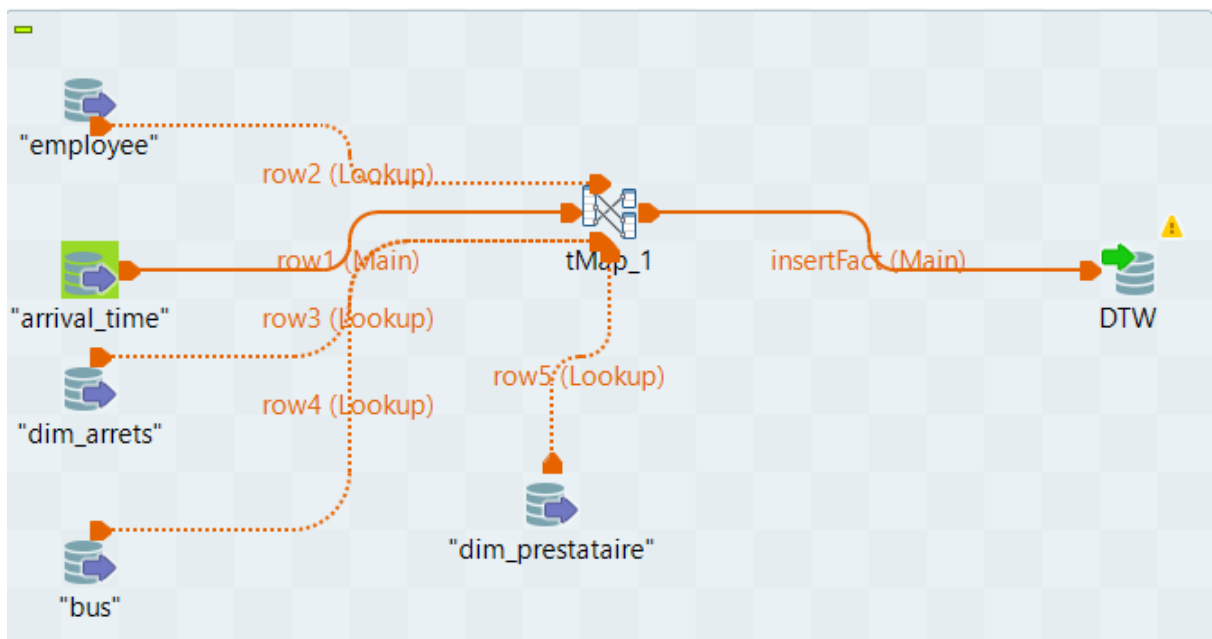


Figure 3.22: Populating the Fact Table.

3.5 Dashboard Wireframe Design

A basis for dashboard development is wireframe design that outlines how critical information and KPIs might look visually. I planned wireframes as low fidelity layouts about

the placement of charts, filters, navigation, etc., for each of the dashboard modules. The wireframe design phase is appropriate when the intention is to orient concepts around user experience and clarity, and I would not have to be focused on detailed visuals as they applied to usability. This is an important step to make sure the dashboards would be user friendly and user needs are acknowledged.

3.5.1 Overview dashboard wireframe

The overview dashboard wireframe presented in figure 3.23, offers a tidy and intuitive design that enables quick access to selected transportation activity metrics. At the top of the dashboard is the navigation button located next to the title identifying what the dashboard provides. Next to the title are informational cards that show the overall metrics, such as total number of employees, trips, and buses. In the middle of the dashboard, there are charts that help to show trends, distributions, and comparisons between different transportation characteristics, such as shifts, lines.

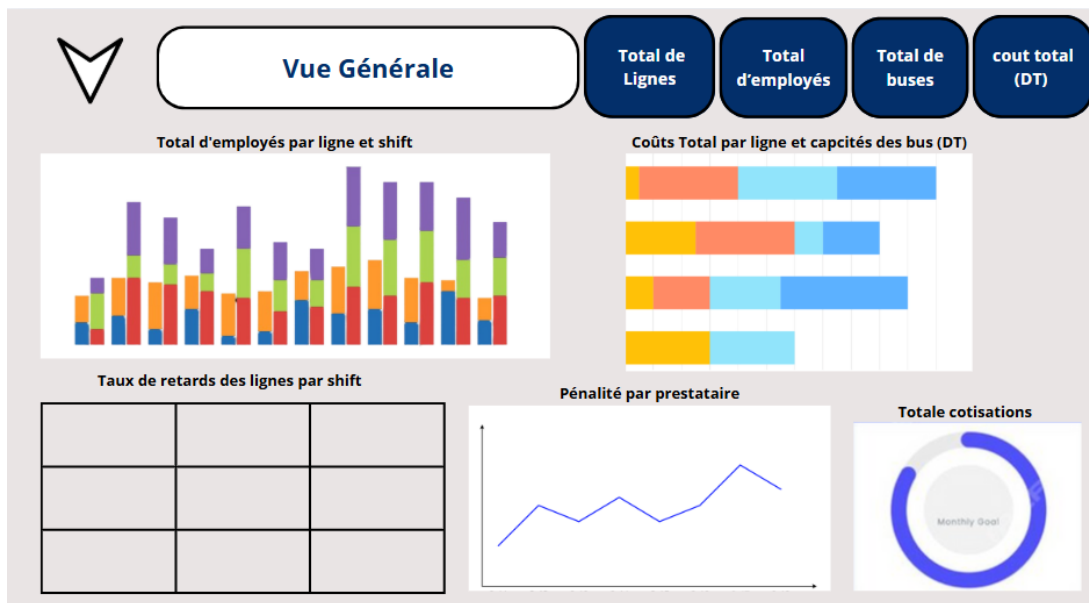


Figure 3.23: Wireframe of the overview dashboard.

3.5.2 Line Analysis dashboard wireframe

The wireframe for the line analysis dashboard, figure 3.24, is meant to provide an in-depth view of employee transportation lines. At the top, it has a navigation button, a distinguishable title, and summary cards with high-level statistics. It also has a dropdown to filter the data by shift. In the central area, the dashboard accesses various visualizations, such as a treemap, stacked column chart, decomposition tree, area chart, and pie chart, and provides a more highly detailed view of the transportation lines with some additional dimensions.

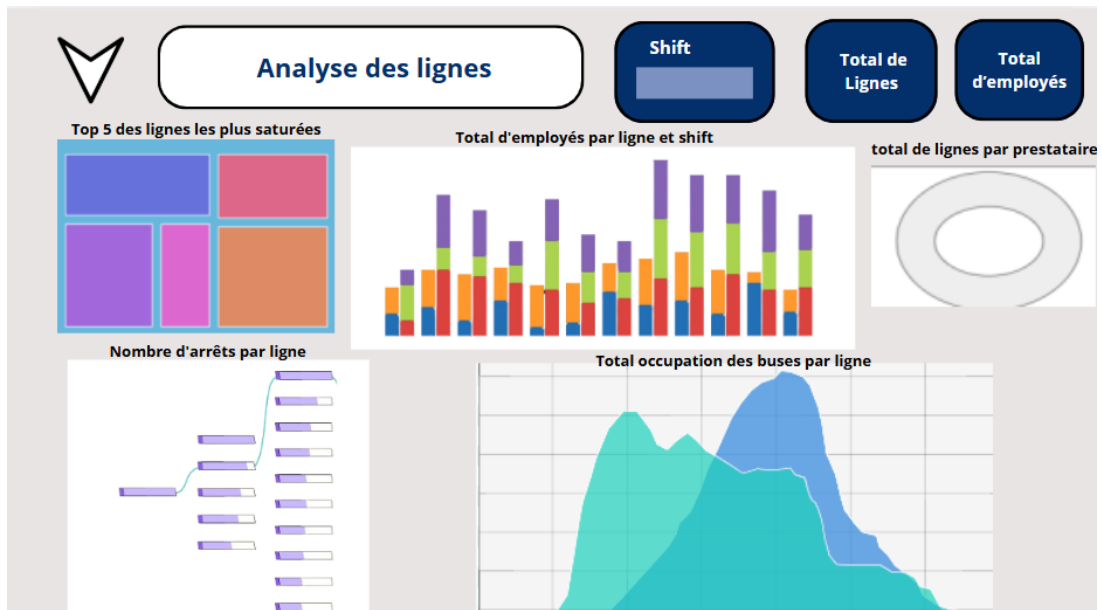


Figure 3.24: Wireframe for the line analysis dashboard.

3.5.3 Financial Analysis dashboard wireframe

The financial analysis dashboard wireframe, figure 3.25, is meant for viewing all transportation related expenses. It has a navigation button at the top and a clear title. There are summary cards for general financial measures, and a dropdown filter to filter the data by line. The middle section includes stacked bar chart, stacked column chart, scatter chart, and donut chart to analyze and view financial data in all perspectives.

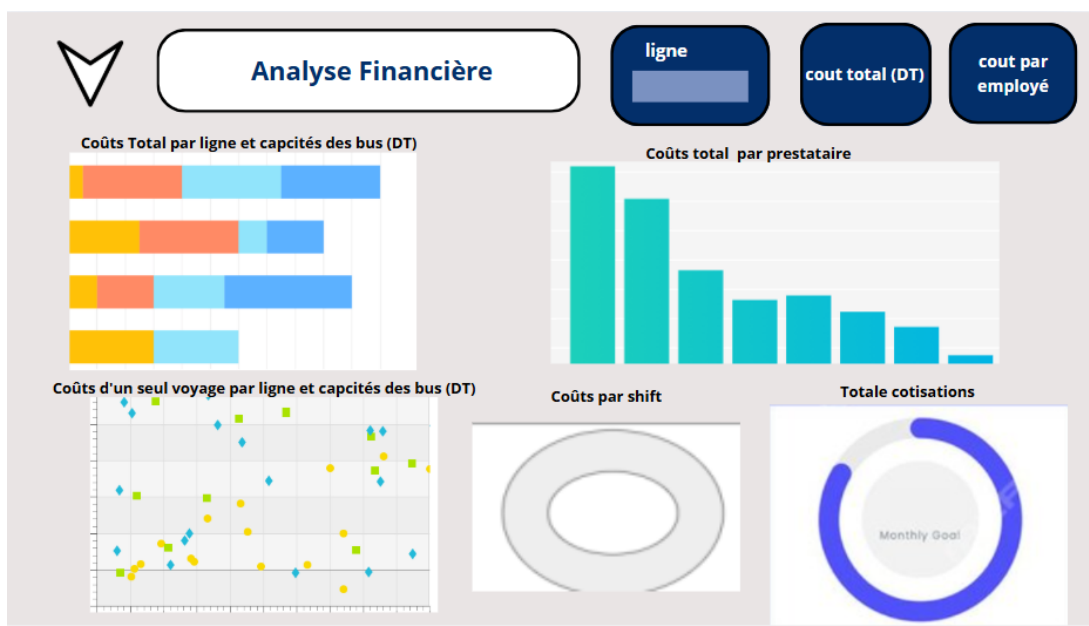


Figure 3.25: The financial analysis dashboard wireframe.

3.5.4 Delay Analysis dashboard wireframe

The delay analysis dashboard wireframe , figure 3.26 ,has been designed to measure and categorize on-time performance across transport lines. At the top of the page, there are the navigation button, the title, and the percentage cards indicating performance elements of major importance. The top navigation is complemented by a filter button to enhance interaction with the data. Next, in the middle of the page are visuals to illustrate delay rates (line chart), relative rates over a time series (clustered column chart), and utilization comparisons across different transports or shift periods (pie chart).

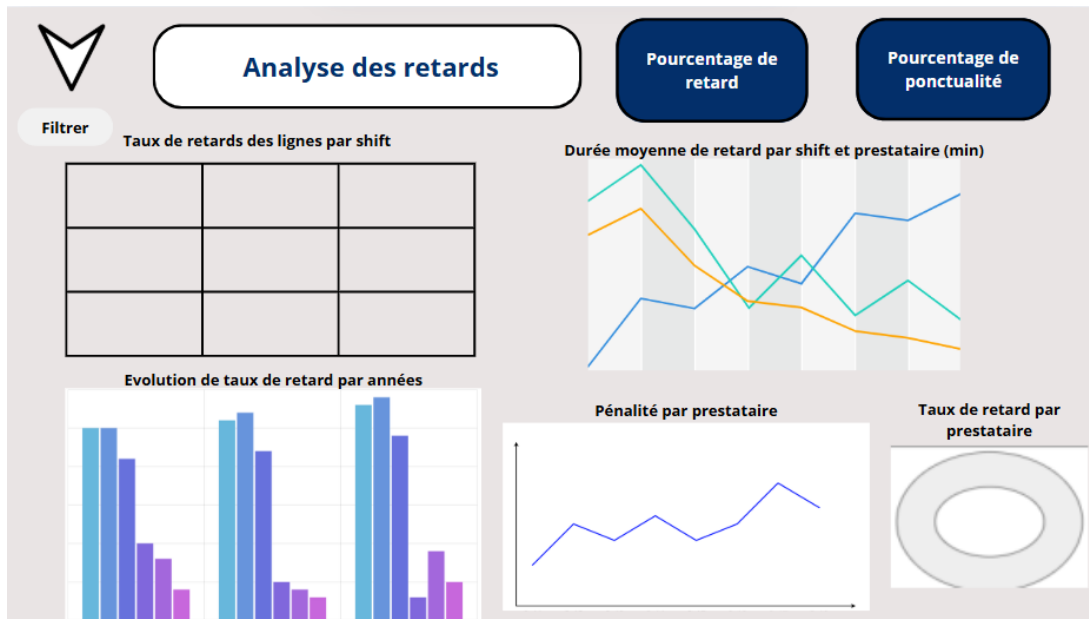


Figure 3.26: The delay analysis dashboard wireframe.

3.6 Data Presentation Phase

This step is the stage of the process where we take raw processed and structured data and, through interactive dashboards, change that into actionable insights. I worked with Power BI to create dashboards that are interactive, easy to use, and the visuals are dynamic to allow users to interact with and examine the indicators of transportation performance. The dashboards contained a combination of indicators to create a number of visuals including KPIs, charts, and filters to allow decision-makers to monitor costs, on-time travel and operational performance, at-a-glance and in real-time. To reiterate from the beginning of this chapter , I identified four modules:

1. **Module 1:** General View of Employee Transportation Activities
2. **Module 2:** Line Analysis
3. **Module 3:** Financial Analysis

4. Module 4: Delay Analysis

The dashboards were designed to be self-explanatory and easy to use, allowing stakeholders to explore key performance indicators through dynamic visuals. For example, bar charts, line graphs, pie charts, KPI cards, and slicers, which enable stakeholders to filter and drill down into subsets of data, were used for some visualizations. The navigation menu in the dashboard ,figure 3.27, allows users to seamlessly switch between different analysis modules. It is designed to provide easy access to all the key sections, enabling users to explore the data from different perspectives. This menu is consistent across all dashboards, ensuring a smooth user experience by maintaining a unified navigation structure throughout the platform.

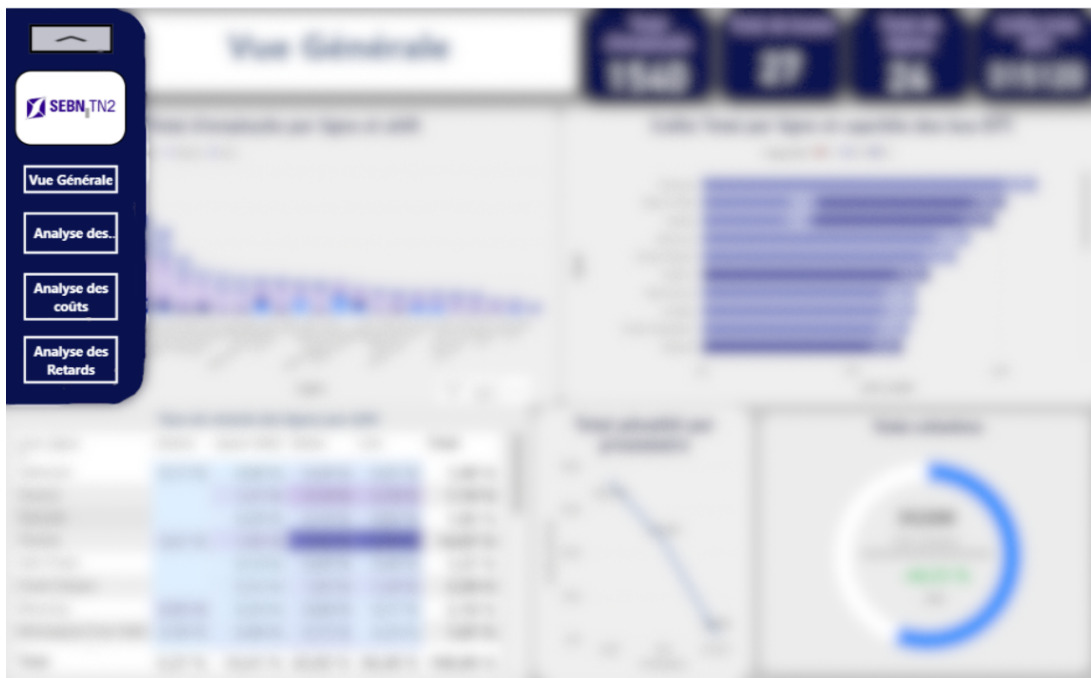


Figure 3.27: Navigation Menu.

3.6.1 Overview Dashboard

The **Overview Dashboard**, figure 3.28, provides an overview of the employee transportation system. It combines various key indicators regarding utilization, service capacity, costs, financial aspects, and punctuality. It provides decision-makers with key indicators on the overall health of the system (employee participation, number of routes and buses in operation, total costs, delay rates, etc.). It is intended to be a strategic entry point for monitoring transportation activities and identifying areas requiring further review or monitoring. The main visual elements of this dashboard are summarized and presented in the table below.

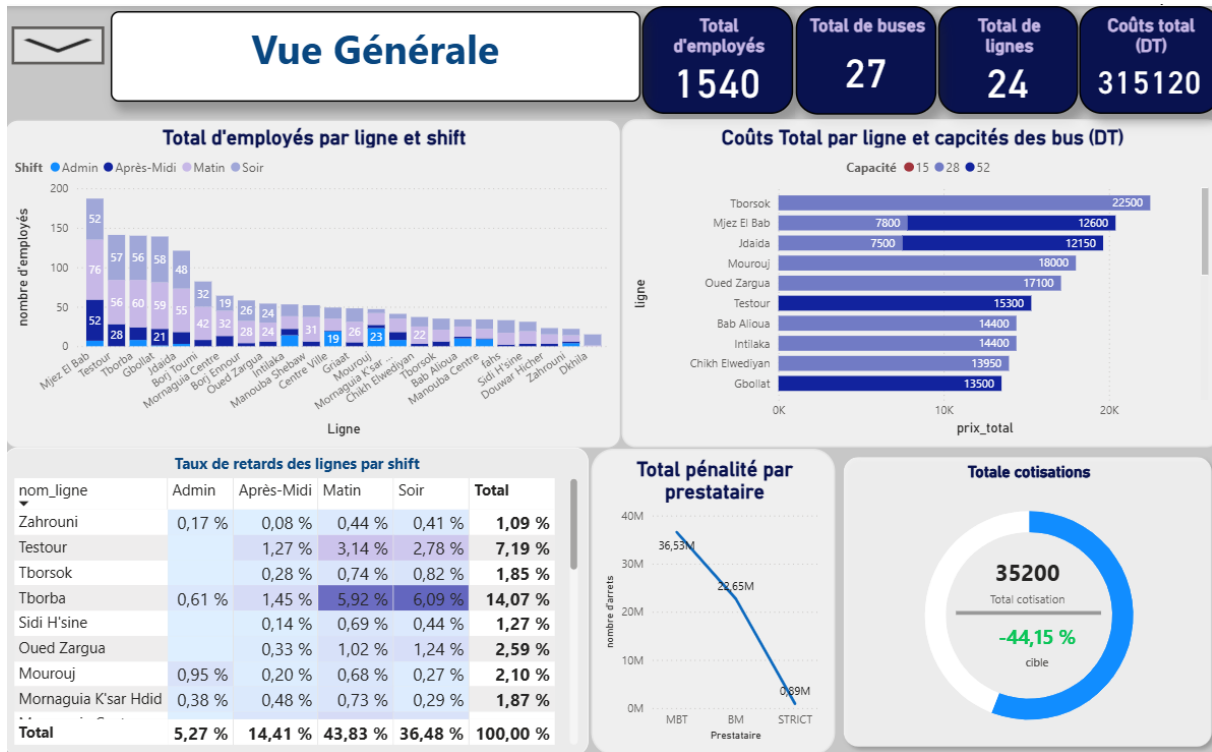


Figure 3.28: General Overview Dashboard.

The key visuals presented in this dashboard are explained in the table 3.2 below:

Table 3.2: Analysis of the visuals of the “Overview” dashboard.

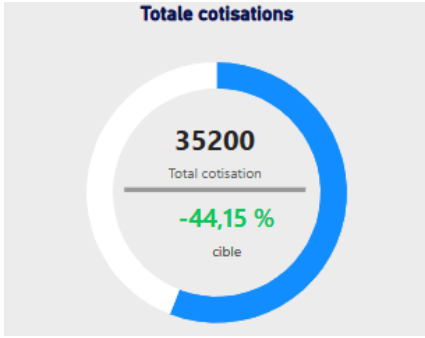
Visual	Analysis
	This KPI shows that the transportation services are being utilized by a total of 1,540 employees. This number indicates the general size of the system's user interface, and it will help us assess demand for transportation service over all routes.
	This KPI shows that 27 buses have been allocated by the providers of the company. It illustrates the general extent of the transportation fleet and is informative for measuring capacity, coverage, and provider allocation.
	The dashboard indicates that there is a KPI with the number of lines that are presently operational, which is 24. This illustrates the geographical extent of the transport network and segmentation.
	This card indicates that the total transport cost of the company is 315,120 TND. This value is reflective of the initial measurement to evaluate financial performance, and to optimize usage and performance-related costs.

Table 3.2: Analysis of the visuals of the “Overview” dashboard(continued).

Visual

</

Table 3.2: Analysis of the visuals of the “Overview” dashboard(continued).

Visual	Analysis
 The donut chart displays the total contribution (Total cotisation) as 35200. A green segment indicates a variance of -44.15% from the target (cible).	This is the total contribution made by the employees. The employees with more than one year of service contribute 20 TND/month, and new employees contribute 40 TND/month. The company target is to fund at least 20% of the total cost of transport through employee contributions, but the actual is -44.15% below the target, indicating the cost-sharing arrangement is not performing well.

3.6.2 Line Analysis Dashboard

The Route Analysis Dashboard, Figure 3.29, provides an in-depth understanding of the distribution and utilization of transit lines across the entire network. It integrates several indicators to facilitate decision-making regarding operator performance, staffing, and route planning. This dashboard helps identify the busiest lines, understand the distribution of employees by shift, and evaluate the distribution of work among operators. To assess coverage and efficiency, it also integrates indicators such as the number of stops and the occupancy rate per line. By allowing users to dynamically examine data based on the selected time slot, a shift filter enhances interactivity.

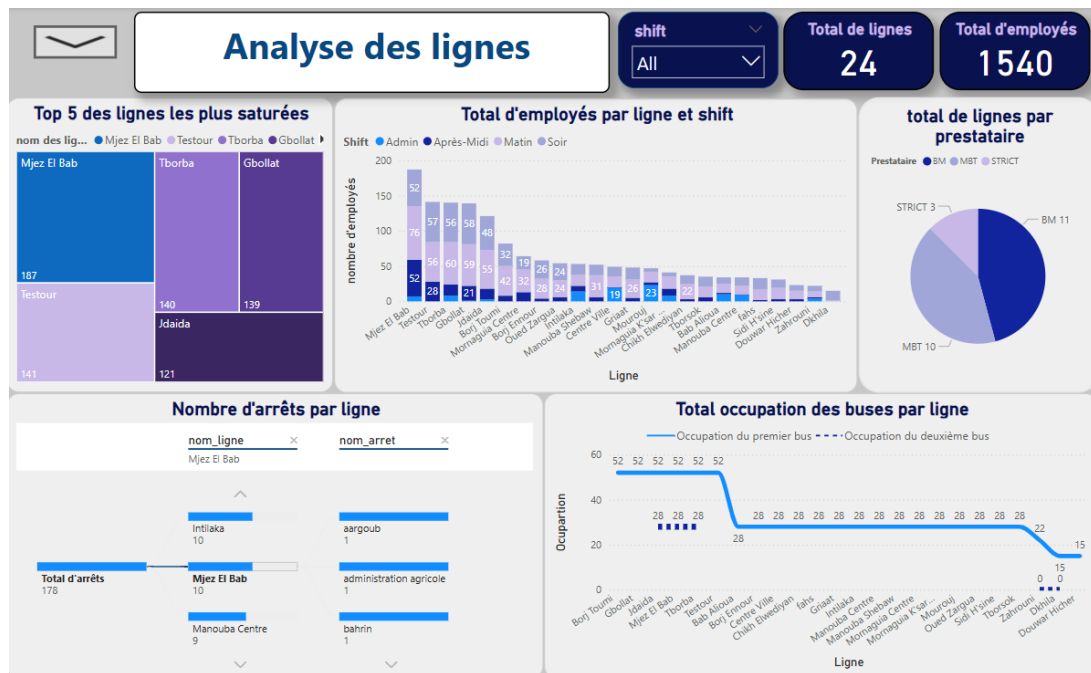


Figure 3.29: Line Analysis Dashboard.

[illegible]

3.6.3 Financial Analysis Dashboard

The purpose of the Financial Analysis Dashboard, figure 3.30, is to give a clear and concise picture of the total cost of employee mobility-related transportation. Its primary goals are to support data-driven financial decisions, detect inefficiencies, and make cost monitoring easier. This dashboard assists decision-makers in understanding how resources

are being allocated and whether financial goals are being met by visualizing important metrics pertaining to expenses, cost distribution, and employee contributions.

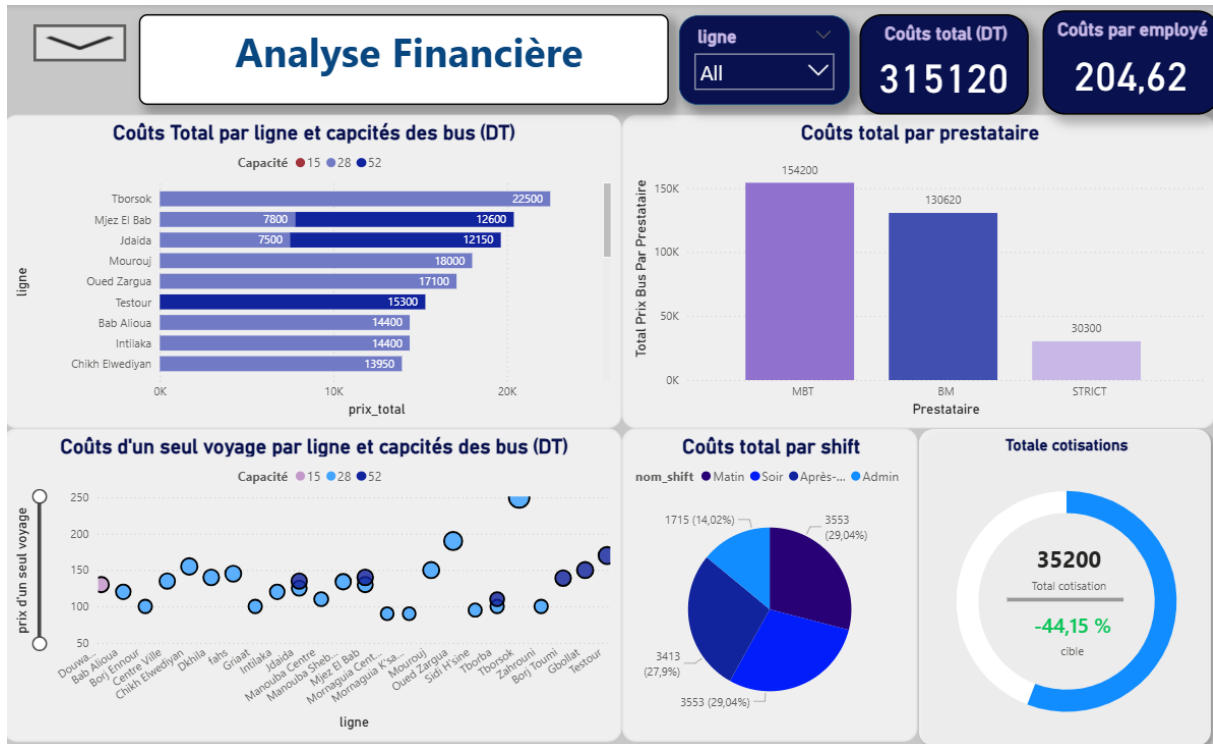


Figure 3.30: Financial Analysis Dashboard.

The table 3.4 below is an explanation of the dashboard's main visual elements.

Table 3.4: Analysis of the visuals of the "Financial Analysis" dashboard.

Visual	Analysis
	This measures that the company is paying on average 204.62 TND per employee for transportation costs. This represents a critical figure to measure per-capita efficiency as well as the setting of future budgeting levels or cost-cutting levels.
	This chart highlights sharp variations across service providers. MBT takes the lead with the highest total expense of 154,200 TND per month, followed by BM at 130,620 TND, and Strict at 30,300 TND. These variations may be attributed to the number of lines operated, distance, or service quality. This analysis may assist in decisions for contract renegotiation or provider performance measurement.

Visual

Coûts d'un seul voyage par ligne et capacités des bus (DT)

Capacité 15 28 52

prix d'un seul voyage

ligne

Analysis

This chart shows remarkable disparities in unit trip expenditures. "Tborsok", for instance, has the highest cost per trip even though it has the same bus capacity of 28 seats at 250 TND, while the lowest-cost trips are taken by the "Mornaguia Centre" and "Mornagui K'sar Hdid" lines at 90 TND per 28-seat bus. The principal reason for this difference appears to be proximity to the company from the terminal bus stop, and its effect on fuel and operating expenses. This is a helpful observation to make for capacity planning and route planning.

Visual

Coûts total par shift

nom_shift Matin Soir Après-midi Admin

1715 (14,02%)

3553 (29,04%)

3413 (27,9%)

3553 (29,04%)

Analysis

This pie chart shows that morning and evening shifts involve the highest total costs at 5,553 TND and 3,413 TND respectively, as opposed to 1,715 TND for the administrative shift.

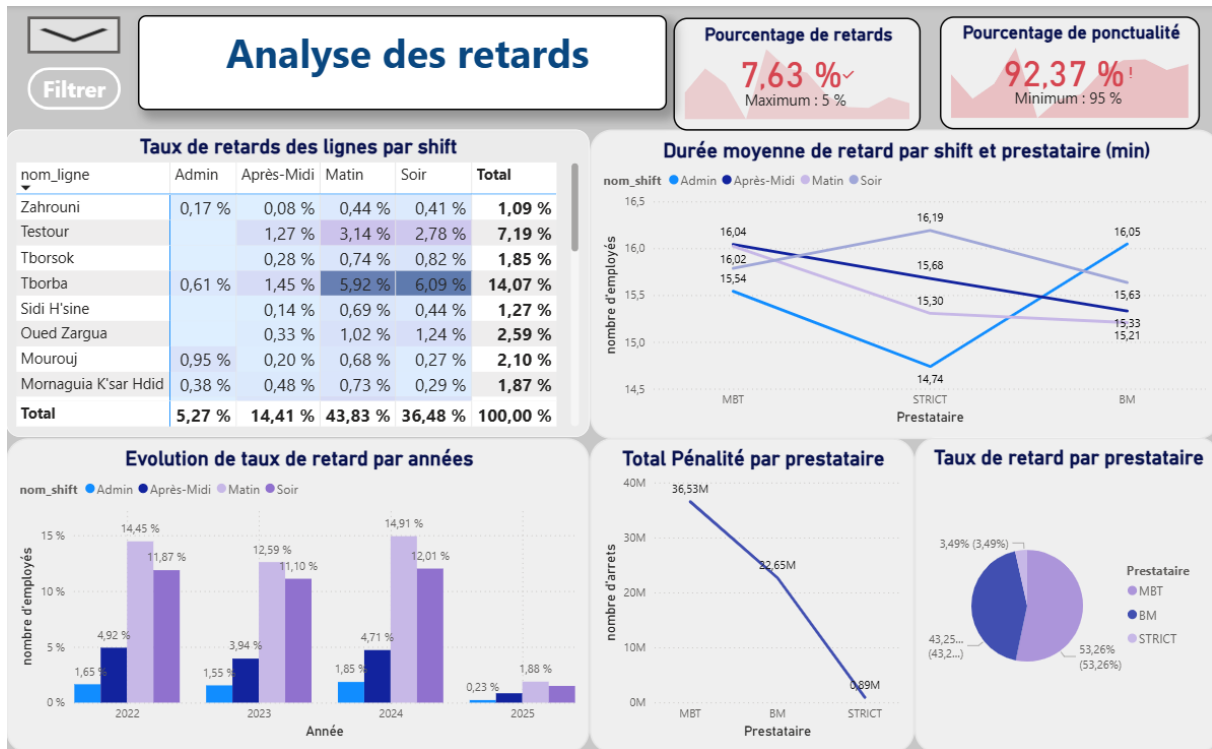


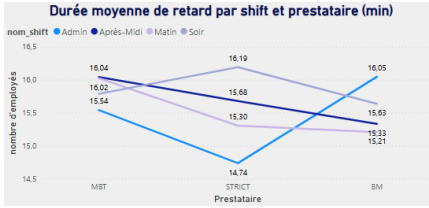
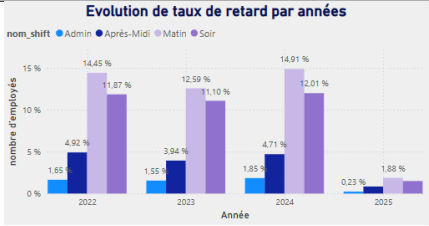
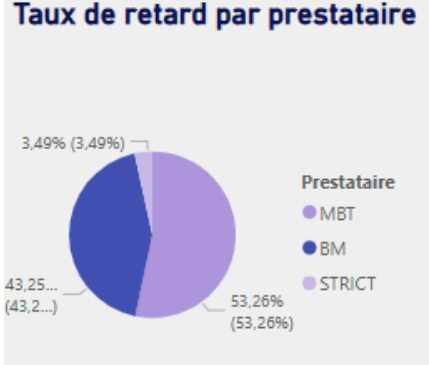
Figure 3.31: Delay Analysis Dashboard.

The table 3.5 below is an explanation of the dashboard's main visual elements.

Table 3.5: Analysis of the visuals of the “Delay Analysis” dashboard.

Visual	Analysis
	The current delay percentage is 7.63%, which exceeds the company's established margin of 5%. This was established as the standard to ensure the efficacy and reliability of staff transportation. Exceeding this threshold can lead to Reduced productivity, as staff members arrive late in workplaces.
	It has a 92.37% punctuality rate, failing to meet its minimum requirement of 95% to guarantee that almost all trips arrive on schedule. This shortfall may be an indication of weaknesses in operational performance and can lead to Negative impact on shift productivity and employee routines.

Table 3.5: Analysis of the visuals of the “Delay Analysis” dashboard(continued).

Visual	Analysis
	The chart suggests that delay times are relatively similar, all of 14.74 to 16.19 minutes, from the most accurate provider Strict. That this consistency—while being the most accurate provider in other data—is present shows that even small inefficiencies can add up.
	The column cluster chart of the delay rate evolution by shift and year displays obvious temporal trends for punctuality problems. Indeed, the morning shift always has the highest delay rate, at 14.91% in 2024, before dropping drastically to 1.88% in 2025.
	The delay rate pie chart by provider is a comparative view of performance among service providers. The delays are predominantly a result of MBT at 53.26%, followed by BM at 43.25%, and Strict at a significantly low rate of only 3.49%. This disparity indicates the issue of unreliability with MBT and BM, with a bearing on review of performance, audit of services, or even re-negotiation of contracts.

A **comprehensive filtering menu** , figure 3.32, allows users to refine the analysis based on multiple temporal and operational criteria: **Year, Month, Day, Shift, Line**.

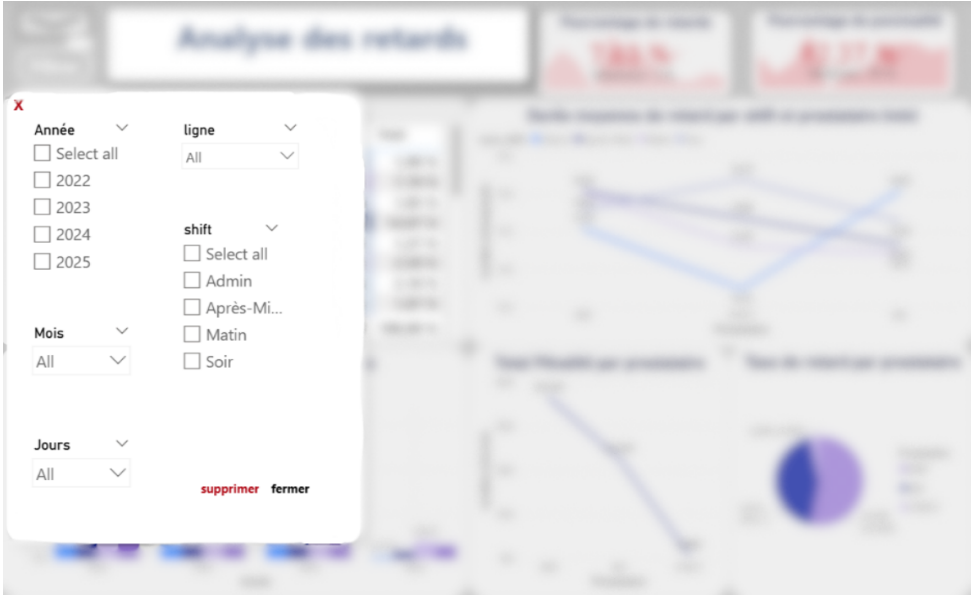


Figure 3.32: filtering menu.

3.7 Conclusion

At this sprint level, I implemented the ETL process using Talend to structure and centralize the data into a clean and reliable data warehouse. This foundation enabled the creation of dynamic dashboards divided into four key analysis modules: general overview, line analysis, financial analysis, and delay analysis. The next chapter will focus on the development of a chatbot designed to automatically generate dashboards using artificial intelligence, providing users with an intuitive and intelligent way to interact with and analyze transportation data.

CHAPTER 4

SPRINT 2: CHATBOT-DRIVEN DASHBOARD GENERATION AND INSIGHT ANALYSIS

4.1 Introduction

As a function of modernizing and automating the data analysis process for employee transportation, this sprint will develop an innovative feature: an intelligent chatbot that can provide direct user interaction, understand user's natural language questions and dynamically generate a personalized dashboard with visualizations and KPIs, automated analysis and recommendations.

4.2 Requirements Definition

Following the discussion with the Product Owner, we have established the requirements for Sprint 2 in line with the overall project vision. The most critical objective of this sprint is to implement a chatbot system that can understand and handle user queries in natural language. This functionality is critical for enhancing user experience due to its provision of an effortless as well as natural interface for communicating with the data system. Utilizing a simple-to-use dialogue interface, users will be able to ask questions in their own words, without necessarily needing technical skills or knowledge of the underlying data structures. The system, suitably responding, will not just generate dynamic and interactive visualizations that are focused on the user's intent, but also provide context-specific analysis and actionable recommendations. These are designed to help users better understand data, uncover significant findings, and make more informed decisions. With its focus on natural language processing, automatic generation of visualizations, and context-based suggestions, Sprint 2 aims to enhance the platform's usability along with its intelligence significantly, all in accordance with the project's overall objectives.

4.3 Sprint 2 Backlog

The Sprint 2 backlog outlines the set of prioritized tasks and user stories planned for this sprint. The table 4.1 below presents the backlog designed for Sprint 2.

Table 4.1: Sprint 2 Backlog.

ID	Functionality	US ID	User Story	Duration in days
1	Authentication	1	As a user, I want to authenticate myself.	2
2	Create a chatbot	2	As a user, I want to have access to a chatbot that can generate personalized visualizations based on my specific needs.	19

4.4 Sprint 2 Analysis

This section presents the analysis that was conducted as part of Sprint 2. This involves the use case diagram that specifies the major functionalities created as part of this sprint. The diagram provides a clear and structured view of the main user interactions and system responses related to the new features that were created. In addition to the graphical representation, this section provides a detailed textual description of one of the most significant use cases: "Generate AI Report." This use case focuses on the interaction flow between the user and the system, demonstrating how the chatbot accepts natural language queries and provides suitable visualizations, analysis, and recommendations. It assists in emphasizing the value addition by the AI-based solution and enhanced decision-making capability provided to the end user. The narrative description and the diagram collectively provide a complete overview of the functional scope covered in Sprint 2 and how the features developed assist in achieving the goals of the sprint.

4.4.1 Sprint 2 Use Case Diagram

The use case diagram presented in figure 5.1 reflects an overview of the services offered during Sprint 2 with a particular focus on system-Human Resources Manager interactions. The Human Resources Manager being a single actor begins by authenticating through the Authenticate use case in order to access the platform securely. At login, the user has access to utilize the central capability Generate AI Report, which enables natural language writing questions and getting insightful reports supported by visualizations, analysis, and recommendations. The user can also see a list of reports generated earlier via the Show Reports History use case and download any report using the Download Report feature. The following chart illustrates the functional deliverables in Sprint 2 and indicates how the chatbot interface contributes value to data interaction as well as decision-making.

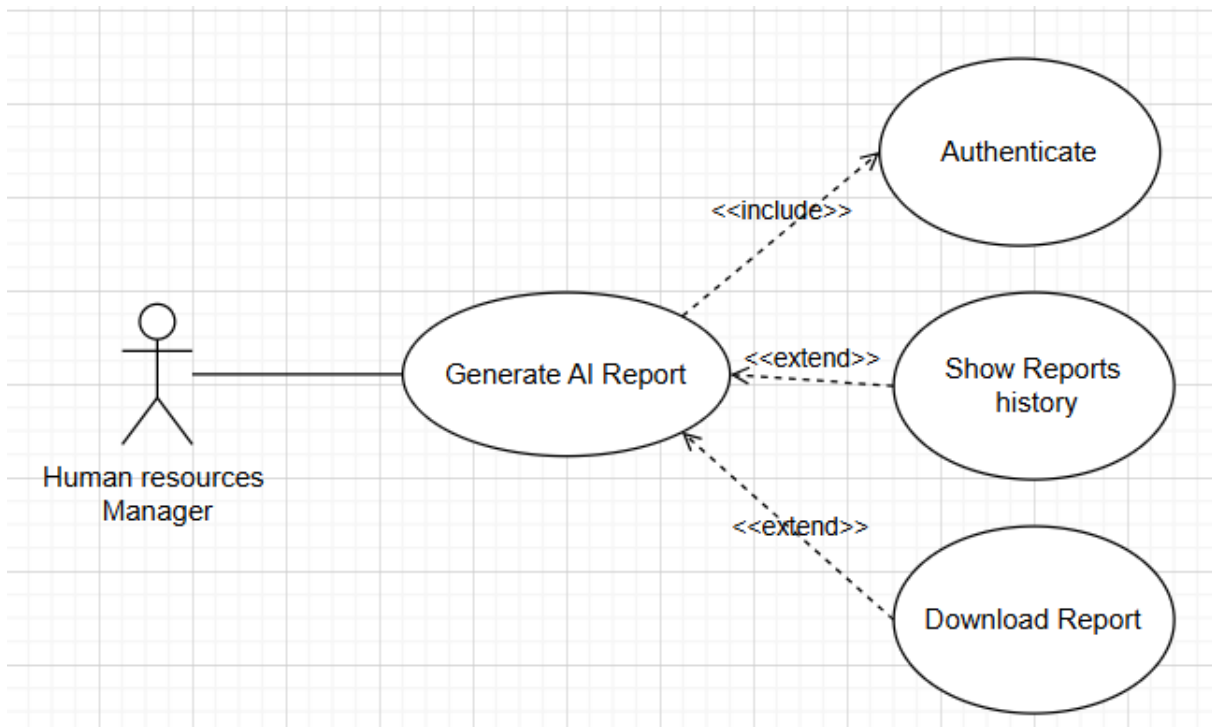


Figure 4.1: Sprint 3 Use Case Diagram.

4.4.2 Text description of use case "Generate AI report"

The use case "Generate AI Report" is the essential functionality that were designed in Sprint 2. It specifies how the Human Resources Manager employs the system in order to obtain AI-driven results through natural language queries. Table 4.2 presents the full textual description of this use case, including its primary actor, preconditions, postconditions, and nominal and alternative scenarios.

Table 4.2: Text description of use case "Generate AI report".

Use Case	Generate AI report
Main Actor	HR Manager
Precondition	the HR manager is authenticated
Postcondition	An ai report is generated
Nominal Scenario	1. The HR employee asks his question. 2. The System displays a structured Dashboard with Description.
Alternative scenario	the system displays nothing if the question is out of the context.

4.5 Definition of generative Artificial Intelligence

Generative AI is a specific sub-discipline of the broad artificial intelligence field that is designed to generate new and original content through learning from a vast pool of available information. In contrast to other AI, which has primarily been employed for tasks such as classification, prediction, or decision-making based merely on already available information, generative AI goes the extra step by generating content that has not been available before. Legacy systems are only able to identify patterns or provide suggestions, whereas generative AI systems are capable of creating new material entirely in any number of varying forms such as, but not limited to, pictures, writing, sound, video, code, and data visualizations. The technology is evolving continuously and will probably generate even more types of outputs that have not been conceived or even thought of yet. Generative AI is generally built on top of cutting-edge deep learning frameworks, and the most widely used one currently is the transformer model due to its superior performance and adaptability. These models are trained with large-scale, well-curated datasets that allow them to abstract the underlying patterns, structures, and relationships in the input data. As a result, generative AI can produce output not just coherent and contextually accurate, but also stylistically precise to human-authored content, whether in language or any other well-structured medium.

4.5.1 Traditional AI vs Generative AI

Table 4.3 is an integrated overview comparing Traditional AI and Generative AI. It highlights some of the most significant differences between these artificial intelligence technologies. The table openly enumerates their corresponding intended uses, showcasing how each methodology is crafted in an effort to address varying problems and yield varying kinds of outputs. It also illustrates the type of outputs each technology produces, showcasing how Traditional AI is primarily focused on classification and prediction, whereas Generative AI can produce brand new and original content. The table also explains how these technologies handle the data that they process and emphasizes major differences in data handling and learning styles. Finally, it defines common use cases and applications where each category of AI is optimally deployed, which helps to explain their uses and implications on a daily basis across various industries.

Table 4.3: Traditional AI vs Generative AI.

Aspect	Traditional AI	Generative AI
Purpose	Solves predefined problems	Creates new content (e.g., text, images, music)
Output	Predictions, classifications, decisions	New, original content (e.g., generated text, images)
Data Type	Structured data	Structured and unstructured data
Learning Process	Supervised or unsupervised learning	Deep learning, such as GANs or transformers
Applications	Predictive analytics, fraud detection, recommendation	Chatbots, content generation, deep-fakes, creative tools
Human Intervention	Requires predefined rules or training	Can generate content autonomously with minimal input
Interpretability	More interpretable and explained	Often harder to explain due to complexity

4.5.2 The Usefulness of Generative Artificial Intelligence in Business Intelligence

Generative artificial intelligence is having a significant impact on BI by enabling a more intuitive, easier, and more automated experience for data analysis. Most importantly, generative AI understands natural language, thus allowing non-technical users to ask questions in their natural language and receive personalized dashboards, relevant visualizations, or even analytical summaries immediately. Further, it automates complex queries (SQL or otherwise) and report generation, allowing lower reliance on technical teams. Incorporating generative AI to BI systems means companies are realizing greater responsiveness, accuracy in decision-making, and democratized access to data intelligence.

4.5.3 Generative AI Types

1. Text-Based Generative AI

- **Examples:** GPT-4, Mistral, and Llama.
- **Used for:** Producing reports, summaries, and natural responses.
- **Strength:** Natural language comprehension.
- **Weakness:** Potential hallucinations when not anchored in data.

2. Visual-Based Generative AI

- **Examples:** DALL-E and Stable Diffusion.
- **Used for:** Producing images/diagrams from prompts.
- **Strength:** Ability to generate complex visuals.

- **Weakness:** Not not meant for structured data (e.g., SQL tables).

3. Multimodal Generative AI

- **Examples:** Gemini and GPT-4V.
- **Used for:** Producing combinations of text and visuals (e.g., discussing a graph).
- **Strength:** Natural interaction with heterogeneous data.
- **Weakness:** Complexity of implementation.

4. Code-Based Generative AI

- **Examples:** GitHub Copilot and CodeLlama.
- **Used for:** Generating SQL queries, producing analysis scripts, etc.
- **Strength:** Automation of technical tasks.
- **Weakness:** Possibility of producing syntactically or logically incorrect code.

4.5.4 Selected Type

The system architecture consists of a well-structured multimodal model integrating several constituents performing in harmony to produce effective results. The generative part of the model, the Mistral-Saba-24b, has two key tasks: first, it enables the system to properly interpret and understand natural language user inputs in an extremely intuitive and flexible way; second, it produces sophisticated textual analyses of the input data, producing rich narrative conclusions that are suitable for decision-making. Counterbalancing this generative aspect is a deterministic control layer that ensures the creation of visualizations that are strictly constricted to the underlying data to provide accuracy and reliability. It also ensures compliance with authorized business rules, playing an important validating role by preventing misclassification or inappropriate outputs. Together, these two layers constitute a strong framework that combines the imagination of generative AI with the precision of deterministic control.

4.6 Large Language Model

A large language model (LLM) is a type of artificial intelligence algorithm that is designed to produce and work with human-like text depending on large datasets and advanced deep learning techniques. LLMs are trained on huge amounts of text data, which enables them to understand context, summarize content, produce coherent and contextually relevant material, and even predict the next words or sentences in a sequence. LLMs are a subcategory of generative AI with a particular focus on producing text-based outputs. As a

result of this, the terms "large language model" and "generative AI" are closely linked with one another but not precisely interchangeable because generative AI also encompasses models that produce images, audio, video, and other forms of data. The most notable architectures used for LLMs are transformer-based architectures such as GPT (Generative Pre-trained Transformer) and BERT (Bidirectional Encoder Representations from Transformers). These architectures have revolutionized the natural language processing field with use cases in chatbots, automated content creation, machine translation, and sentiment analysis. The ability to generate human-like language with high accuracy has made them an indispensable part of the majority of AI-driven systems currently in use.

4.6.1 LLM Models Comparison

This section gives a comparative analysis of a number of Language Models (LMs) with the aim of determining their individual strengths and weaknesses as they pertain to the particular requirements of our project. The aim of this exercise is not to realize each model discussed but to carry out focused experiments that will enable us to select the most appropriate model to integrate into our system. This step is required to determine whether the chosen model meets our particular functional requirements and performance criteria. By carefully choosing the most appropriate language model, we aim to guarantee the overall effectiveness, reliability, and appropriateness of AI-enabled functionality developed within the project, and hence the system's ability to deliver correct and efficient results.

Table 4.4: Comparison of Language Models.

Model	Provider	Advantages	Limitations
Mistral (Mistral-saba-24b)	Mistral (via Groq)	- Open source, high performance, low latency (especially when combined with Groq) - Good cost to performance ratio - Multilingual	- Less fine-tuned than GPT-4 for rare edge cases
GPT-4	OpenAI	- Best-in-class contextual understanding - Generated text is great – and so is code - Very robust	- Paid (higher cost) - Proprietary
Llama 2 (70B)	Meta	- Good handling of natural language - Open source - Large open source community	- Slightly less capable than GPT-4 for pure code generation

4.6.2 Selected model

We selected **the Mistral (Mistral-saba-24b)** large language model, which we were able to access through the Groq API. There were several important aspects to my decision. Mistral being open source allows me to know how it works, so it (to a degree) preserves some transparency and potential flexibility as we build. And it performs well on natural language understanding and code generation scenarios, especially when considering SQL generation for queries and long text analyses. Mistral also has strong multilingual capabilities, which is important for processing queries in French. There will be a low-latency response for my integration in Groq, which will allow for a smooth, interactive experience for the user. Relative to proprietary solutions like GPT-4, it has a great balance of quality, speed, and cost. The combination of these factors make Mistral a great option for scalable, business-facing applications related to business intelligence.

4.6.3 Working Mechanism of the Mistral-Saba-24b Model

The Mistral-Saba-24b is built upon a highly advanced transformer architecture that has been specifically developed to maximize large-scale natural language processing. When it receives a query from a user via the interface, the model first breaks down the text into semantic units called tokens, allowing for an extensive examination of the input. With its training on massive datasets, Mistral then goes ahead to understand the general meaning of the query, identifying the user's purpose and key factors necessary for correct comprehension. The system applies preprocessing reasoning to categorize the query, routing it to appropriate SQL queries or relevant processing procedures. Based on this, the model constructs a structured text response that may comprise sophisticated data analysis, personalized suggestions, or instructions for designing interactive visualizations. In the interest of ensuring a fast and effective user experience, a caching mechanism holds the result of repetitive queries so that unnecessary API calls are avoided and response rates are improved. Mistral also capitalizes on the context enabled by data stored in a star schema so that it can return consistent and relevant responses, for example, suggesting actions based on trends observed, thereby enhancing the usability and relevance of the insights being served.

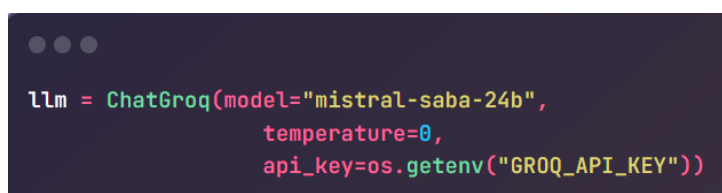
4.7 Integration of Groq API

Groq API is a high-performance interface specifically designed for delivering blazing-fast inference for big language models (LLMs). In our project of building interactive and dynamic dashboards from natural language requests, the choice of using Groq API was driven by its better performance and extremely low latency. These characteristics are important in providing real-time responsiveness, which is essential in providing a fluid

and seamless user experience in the chatbot interface. Answering in real time to user queries provides an interactive experience where users can ask complex questions and receive instant visual analytics without such delays. Developing on the Groq API makes it possible to handle high data volumes in a seamless way and high-level AI computations, so it is possible to provide visual insights near real-time. This responsiveness not just makes the business intelligence platform more useful but also brings a higher level of user satisfaction by minimizing waiting and enabling faster decision-making through real-time information. Overall, the integration of Groq API is among the most significant features of offering a next-gen, interactive BI solution addressing the level of expectation for performance by today's users.

4.7.1 API Key and Authentication

Having chosen Groq as our vendor for our large language model (LLM) needs, the integration of their API into the application is built with strong and efficient authentication. The system begins by implementing the use of a single authentication key for an individual, which regulates that solely identified access to the Groq API is provided. In order to integrate it to be used, the ChatGroq class is invoked within the system, with various required parameters being set. These include the authentication key itself, the model to use (in this case, the mistral-saba-24b model), and a temperature parameter set to 0 to make the responses deterministic and reproducible from the model. The authentication key is loaded securely from a .env config file using the `os.getenv("GROQ_API_KEY")` function that prevents any exposure of sensitive credentials in the application's source code. This solution conforms to best practices in security and key management in software development. In summary, this configuration, figure 4.2, provides a secure and reliable connection to Groq's robust LLM functionality to allow seamless creation of responsive and accurate natural language responses for the application.



```
llm = ChatGroq(model="mistral-saba-24b",  
               temperature=0,  
               api_key=os.getenv("GROQ_API_KEY"))
```

Figure 4.2: Setting up Groq LLM with API key in project.

4.7.2 Groq Cloud Usage

The chatbot operates by directly interacting with Groq Cloud, where it sends the users' questions in natural language. These questions are then submitted and executed on the Large Language Model (LLM) stored on Groq Cloud, specifically the Mistral-SABA-24B model in our case. This intelligent model reads and understands the questions of the

users to deliver right and contextually relevant responses. Such responses can be in the form of SQL queries to retrieve data or detailed instructions for constructing interactive visualizations, which are then passed back to the chatbot interface for visualization to the user. Groq Cloud offers a full-featured real-time monitoring dashboard, as illustrated in figure 4.3, providing graphical displays of API consumption and resource utilization. This dashboard shows critical metrics like the quantity of requests run within specified time intervals and tokens used by the Mistral-SABA-24B model. To ensure optimization of API use, prevention of quota exhaustion allocated, and provision of continuous availability and best quality of the service in our project scenario, access to such detailed monitoring tools is indispensable. By implementing this dashboard, we will be able to monitor active usage of Groq Cloud usage limits and implement dynamic rate-limiting rules for our API calls based on real-time consumption volumes, thereby guaranteeing an effective and smooth user experience.

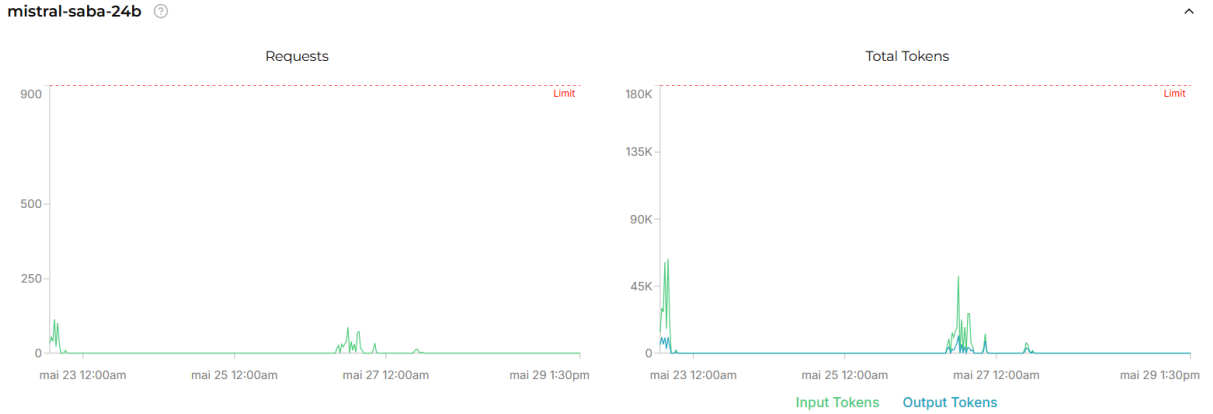


Figure 4.3: API activity and resource consumption.

4.8 Architecture diagram of the intelligent agent

This vertical architecture diagram, figure 4.4, depicts the **AI-based transportation analytics system** that processes natural language queries through a multi-layered pipeline before serving them up as actionable insights. Centralized is a **hybrid query processor** that combines pre-defined SQL templates with queries generated from LLM, all rigorously proved against the database schema. After users ask questions via the Streamlit interface, the system initially leverages a Mistral-powered validator to determine whether the query is relevant to SEBN-TN2's transportation operations (routes, staff, buses, etc.) and secondarily conducts spelling correction automatically. The engine subsequently attempts to find matches for the question from cached questions or, if not possible, generates dynamically optimized SQL using constrained LLM prompting. Executed queries undergo various tests like syntax checking and column checking before reaching the MySQL database via SQLAlchemy. Processed results are subjected to smart visualization selection (Plotly

charts for trends, tables for raw data, indicators for KPIs) and final enrichment through Mistral’s analytical commentary. The system maintains tight-knit integration amongst components without hallucinations through schema-aware prompt engineering and validation gates so that each user query passes through from natural language input to validated business insight through this structured vertical workflow. Edge cases like fuzzy category matching and fallback visualizations are managed by special modules, and the architecture remains robust for experts but accessible for casual users.

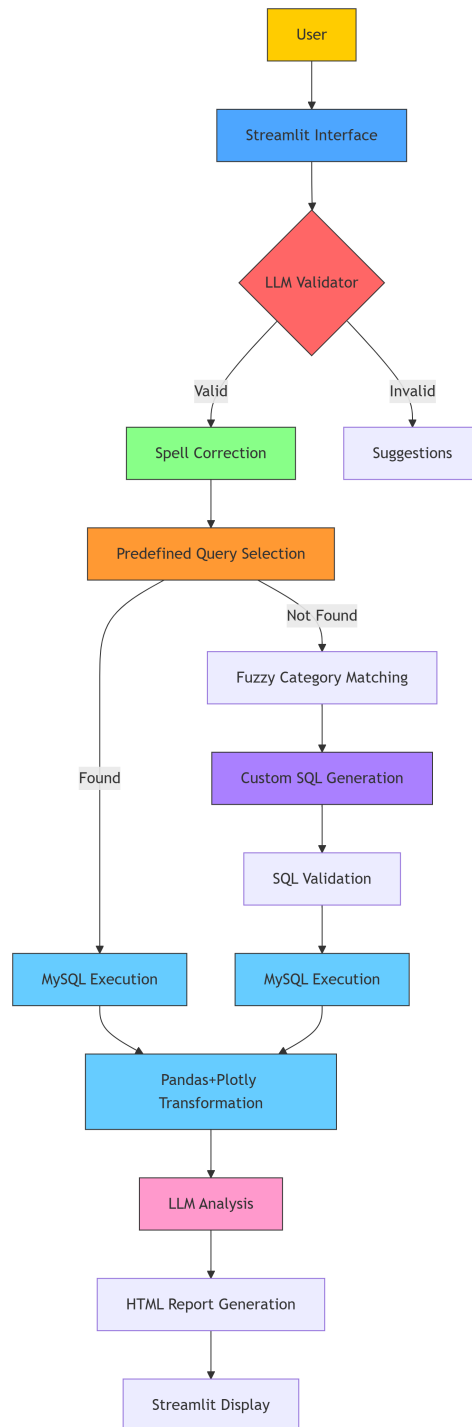


Figure 4.4: Architecture diagram of the intelligent agent.

4.9 Realization

Once authenticated, the user is directed to the main interface of the dashboard generation platform as shown in the Figure 4.5. The interface is minimalist, clean, and intuitive. On the left-hand side, a separate pane displays the history of all the previous analysis requests so that the users can conveniently look at earlier queries and results.

In the center of the screen, an interaction area takes center stage. This is where all the output generated — SQL queries, findings, and interactive visualizations — is displayed dynamically. It is the main output section for the chatbot's response.

At the top of the interface, there is a text entry field for the user to input natural language questions regarding the business data. This prompt is forwarded to the chatbot for processing using Groq Cloud and the appropriate response is thus displayed in the middle section. The layout offers a seamless user experience by presenting all the necessary elements in sight and within reach.

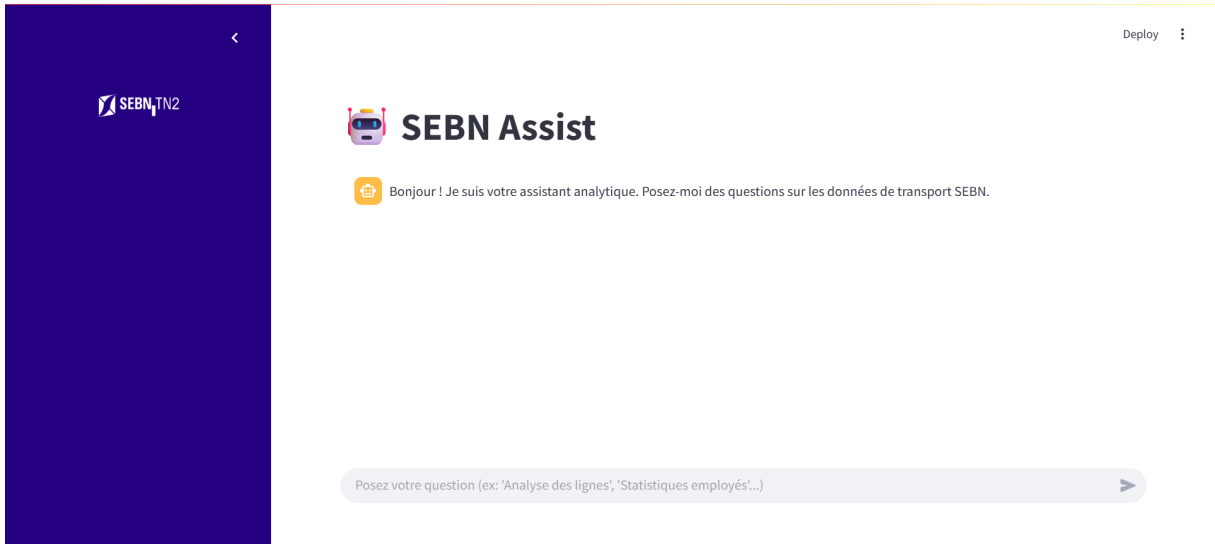


Figure 4.5: Main Interface of the Dashboard Generation Platform.

Figure 4.6 illustrates the result of the natural language query provided by the user, in this case "Analyse des employés". The figure illustrates how the system interprets the input given by the user and displays the respective analysis in a structured and compact manner. It illustrates the capability of the chatbot to interpret the question asked in natural language and offer appropriate insights in the form of graphical representation and summaries, facilitating the user to understand employee-related data effectively.



Figure 4.6: Natural language query submitted by the user: “Analyse des employés”.

When this request was executed, the chatbot created an interactive dashboard automatically made up of four informative visualizations shown in the figure 4.7 and 4.8. They include two bar graphs reporting the number of employees by shift and bus line, a scatter plot indicating the total employee contribution (cotisation) to transport cost by shift, and a line graph illustrating the yearly trend in worker numbers from 2022 to the second month of 2025. The implicit selection of each of the charts with respect to the underlying data structure was both relevant and understandable.

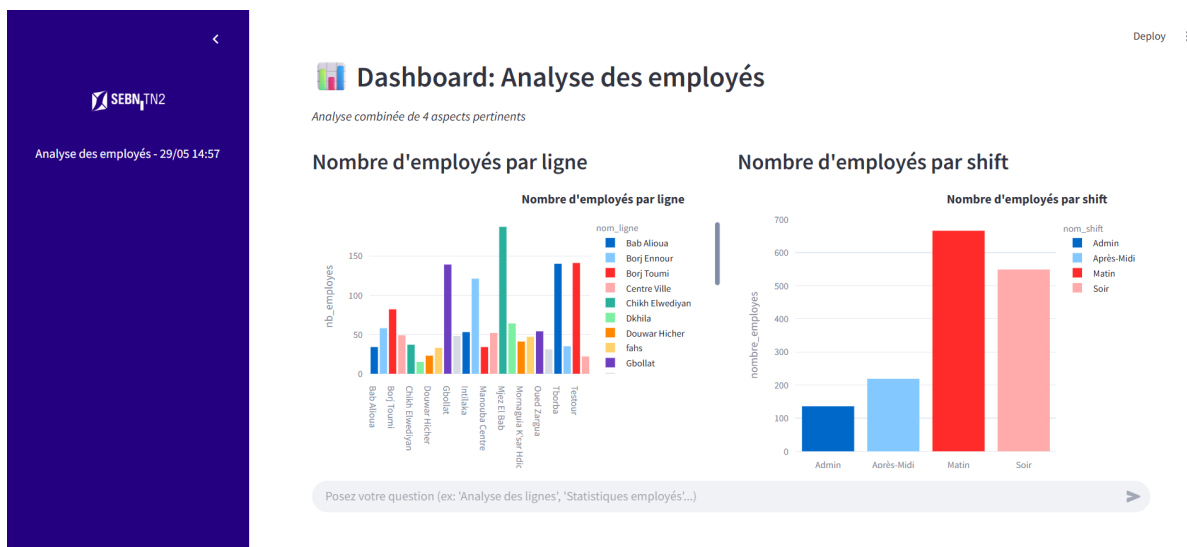


Figure 4.7: Analytical visualization of the number of employees per shift and per bus line.

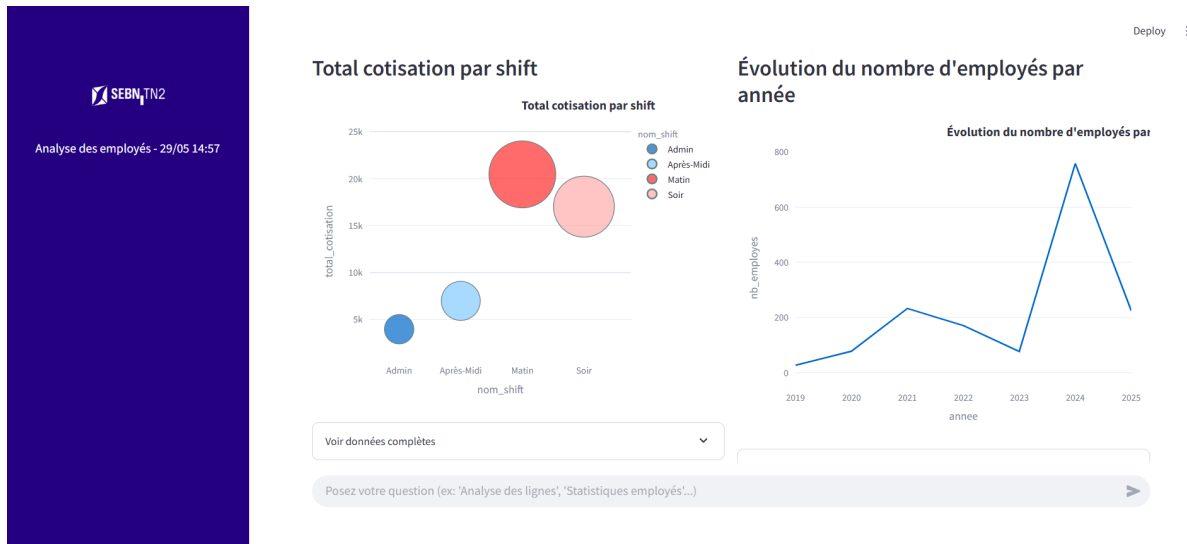


Figure 4.8: Analytical visualization of the total employee contribution and the yearly trend in worker numbers.

Below the graphics, there is a special section aimed at presenting automatic analysis and strategic recommendations. The intention behind this feature is to assist decision-makers by making it easier to comprehend emerging trends and identify possible opportunities or avenues of action. In addition, to provide more usability and facilitate more use of the insights, there is a button provided which enables users to download the entire PDF report to a local machine. This report can be easily circulated to stakeholders or retained for reference. This functionality is depicted in figure 4.9.

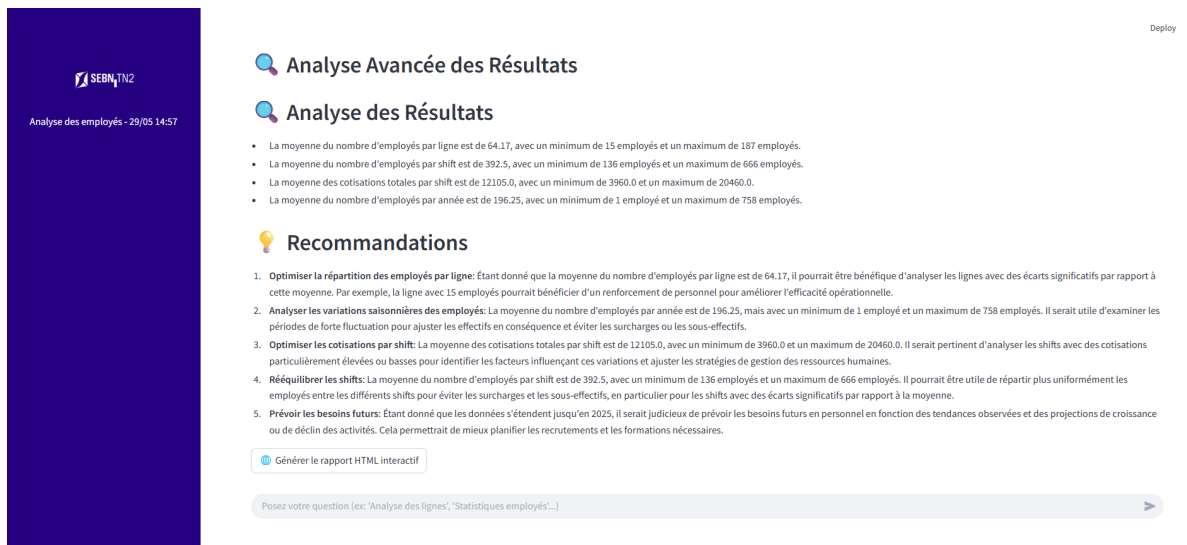


Figure 4.9: Automatic analysis and strategic recommendation.

This is a demonstration of what generative AI is capable of in the field of business intelligence. By translating natural language queries into actionable, data-driven findings in a few seconds, the system significantly eases the process of analytics. Besides speeding

analysis, the chatbot lowers hurdles to advanced BI features, making them accessible to many more individuals. This enables even non-technologists to merely engage with advanced data and make well-informed, data-driven decisions with confidence. All of this is accomplished through a simple, smart, and natural conversational interface that simplifies interaction and empowers users at every level of the organization.

4.10 Conclusion

In conclusion, this sprint highlights the implementation of more advanced artificial intelligence techniques, combining natural language understanding with dynamic data visualization and automated analysis to combine the system to provide users with a smarter, easier, and more efficient method of extracting insights from their data. This marks a significant advancement in delivering modern, user-friendly, and intelligent analytics for the organization. In the next sprint, focus will be laid in developing the employee management platform wherein core functionalities such as employee data registration, bus allocation, attendance tracking, and real-time monitoring would be implemented to lay down the foundation of a comprehensive and intelligent transport management system.

CHAPTER 5

SPRINT 3:TRANSPORT MANAGEMENT

5.1 Introduction

This chapter aims to clarify the progression of Sprint 3 implementation. It is divided into three sections: the first section contains the backlog and analysis, including the global use case diagram for the sprint, along with the detailed use cases diagrams; the second section focuses on the design phase, and the third section looks at the implementation phase using the developed user interfaces.

5.2 Sprint 3 Backlog

Sprint 3 backlog outlines the entire set of prioritized user stories and features to be developed during this phase. It covers essential functionalities essential for handling the system, including login for users, viewing dashboards, employee and bus administration, and real-time tracking. It is a formal plan to guide development efforts and align with project goals and user needs.

Table 5.1: Sprint 3 Backlog.

ID	Functionality	US ID	User Story	Duration in days
1	Authentication	1	As a user, I want to authenticate myself.	1
2	Manage users	2.1	As an HR Manager, I want to view the users list.	1
		2.2	As an HR Manager, I want to add a user.	1
		2.3	As an HR Manager, I want to delete a user.	1

Table 5.1: Sprint 3 Backlog (Continued).

ID	Functionality	US ID	User Story	Priority
3	Manage Employees	3.1	As an HR employee, I want to view the employees list.	1
		3.2	As an HR employee, I want to add an employee.	1
		3.3	As an HR employee, I want to modify an employee.	1
		3.4	As an HR employee, I want to delete an employee.	1
4	Manage Buses	4.1	As an HR employee, I want to view the Buses list.	1
		4.2	As an HR employee, I want to add a Bus.	1
		4.3	As an HR employee, I want to modify a Bus.	1
		4.4	As an HR employee, I want to delete a Bus.	1
5	Manage Lines	5.1	As an HR employee, I want to view the Lines list.	1
		5.2	As an HR employee, I want to add a Line.	1
		5.3	As an HR employee, I want to modify a Line.	1
		5.4	As an HR employee, I want to delete a Line.	1
6	Manage Stops	6.1	As an HR employee, I want to view the Stops list.	1
		6.2	As an HR employee, I want to add a Stop.	1
		6.3	As an HR employee, I want to modify a Stop.	1
		6.4	As an HR employee, I want to delete a Stop.	1
7	View complaints	7	As an HR employee, I want to read complaints.	1
8	Display Fill Rate	8	As an employee, I want to view the fill rate for each bus line and shift.	1
9	Access to dashboards	9	As an HR employee, I want to see a well-structured Dashboard.	1
10	Manage Reservations	10.1	As an HR employee, I want to view the reservations list.	1
		10.2	As an HR employee, I want to accept a reservation.	1
		10.3	As an HR employee, I want to refuse a reservation.	1

Table 5.1: Sprint 3 Backlog (Continued).

ID	Functionality	US ID	User Story	Priority
11	Manage changing requests	11.1	As an HR employee, I want to view the changing requests list.	1
		11.2	As an HR employee, I want to accept a changing requests.	1
		11.3	As an HR employee, I want to refuse a changing requests.	1

5.3 Sprint 3 Analysis

In this section we present a general analysis of the work done in Sprint 3. We begin by presenting the general use case diagram to provide a graphical system interaction overview. We then proceed to a detailed explanation of each of the significant use cases, explaining their function, actors, and specific workflow. Particular attention will be given to important features such as user authentication, highlighting how different components of the system interact to ensure secure and proper functioning. This analysis is aimed at defining the functional scope and guide subsequent development phases.

5.3.1 Sprint 3 Use Case Diagram

Figure 5.1 illustrates the overall use case diagram for Sprint 3

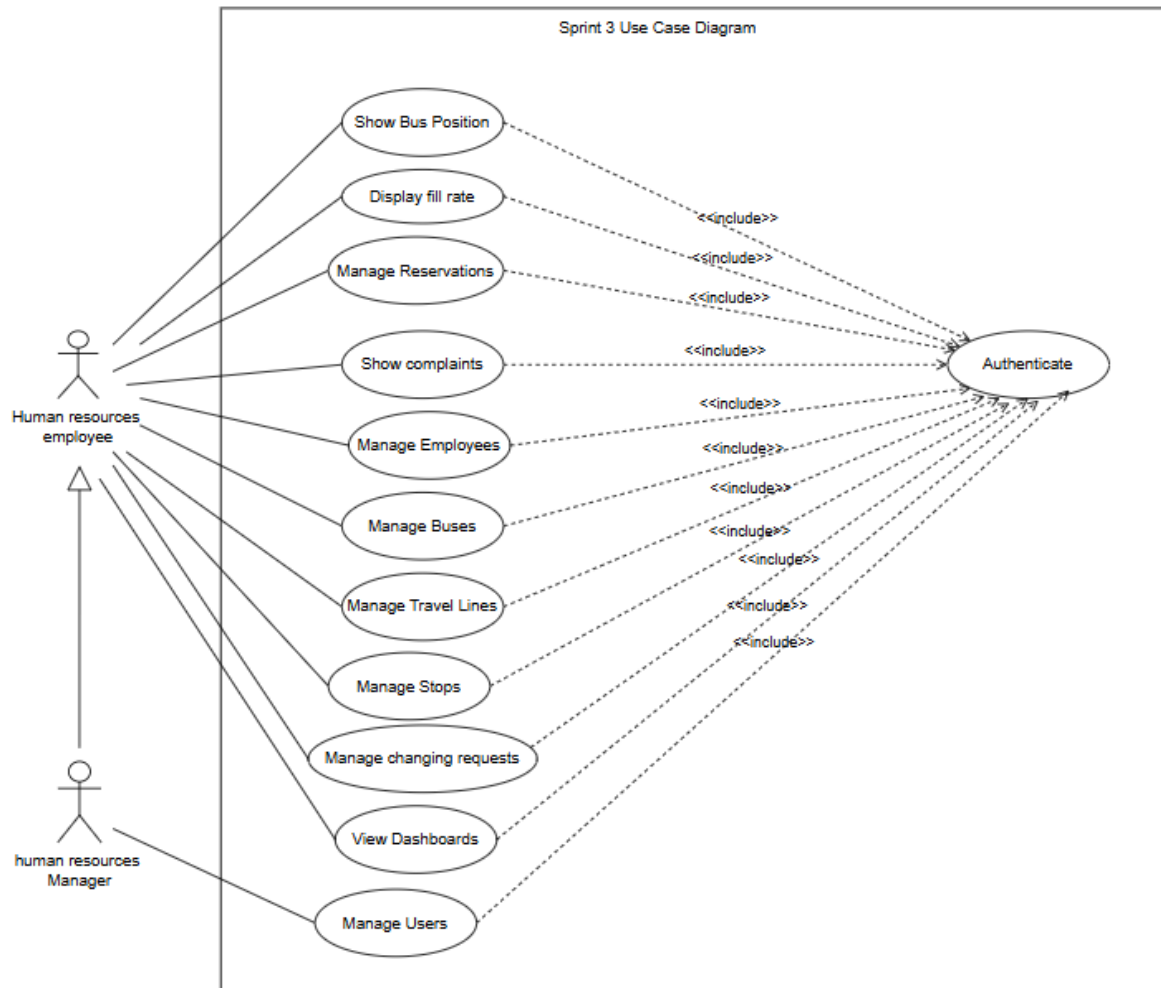


Figure 5.1: Sprint 3 Use Case Diagram.

5.3.2 Detailed use case study

The detailed use case study is meant to provide a better understanding of how the system has to behave under various scenarios. It captures the interaction among the users and the system for each specific feature, such that all the functional requirements become clearly defined and evident. It guides both the design and development processes by indicating expected behavior, user actions, and system responses.

5.3.2.1 Use case diagram for the “Authenticate” functionality

This use case diagram, figure 5.2, presents the authentication process for a human resources employee. It shows that the employee must first enter their email address, followed by their password, which are the necessary information to verify their identity. These actions trigger the Authenticate use case, during which the system validates the information provided to ensure secure access. This mechanism guarantees that only authorized human resources employees can log in, thus protecting the sensitive information and the

general security of the system. The diagram outlines the most important steps of the authentication as well as the user-system interaction within the process.

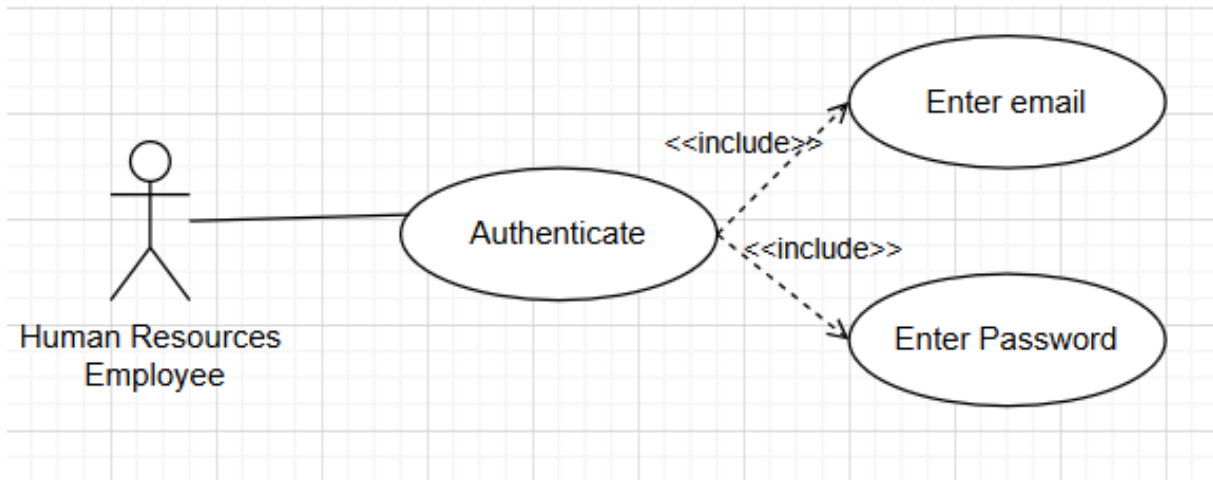


Figure 5.2: Use case diagram for the “Authenticate” functionality.

5.3.2.2 Use case diagram for the “Display Fill Rate” functionality

This use case diagram, figure 5.3, illustrates the process of displaying the fill rate of buses that are used to transport workers. The situation begins with the login of an Human Resources employee for safe access. After authentication, the user may choose a specific shift, a date, and the direction of travel (for instance, to or from work). Based on these selections, the system carries out the "Show passengers" task to query and display the quantity of employees that commuted by bus based on the specified criteria. Lastly, the system calculates and displays the fill rate, enabling HR to track bus occupancy and schedule transport more efficiently.

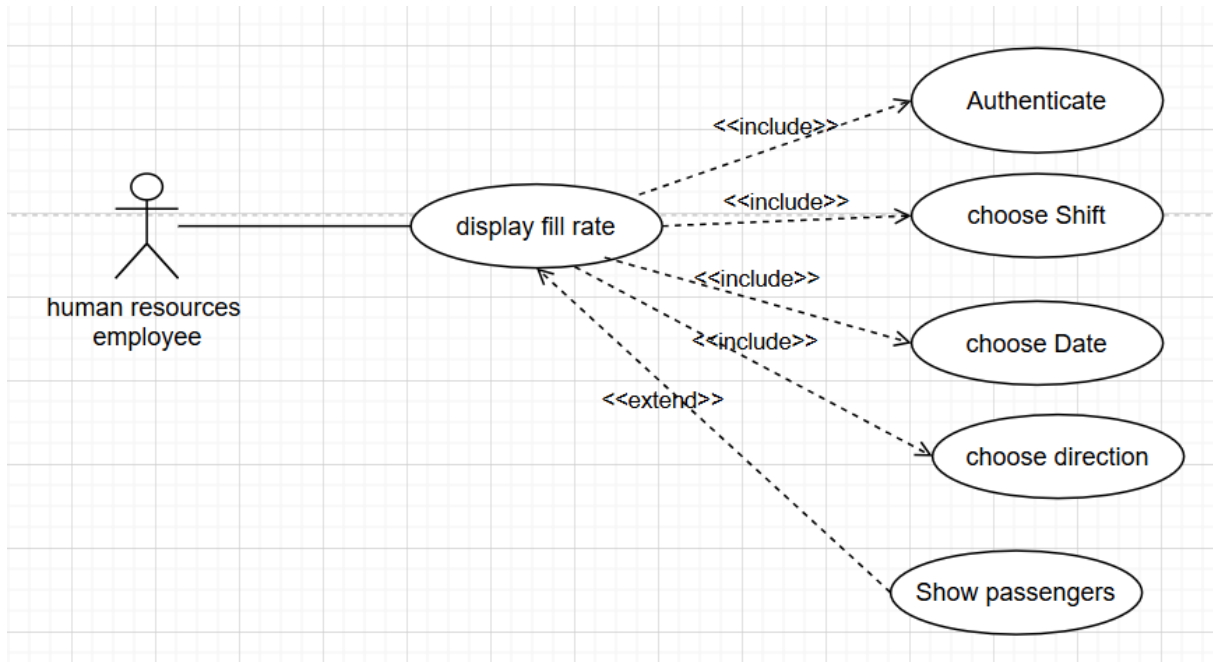


Figure 5.3: Use case diagram for the “Display Fill Rate” functionality.

5.3.2.3 Use case diagram for the “Manage employees” functionality

This use case diagram, figure 5.4, emphasizes the employee management functionalities offered to the Human Resources users of the system. On successful authentication, HR personnel can look at the employee list, providing a summary of all staff records. They can also insert new employees, modify existing employee information, and delete employee records as needed. These operations are the basic management operations through which HR can maintain valid and updated employee data. The diagram shows the HR actor use case and the main system operations within successful employee management.

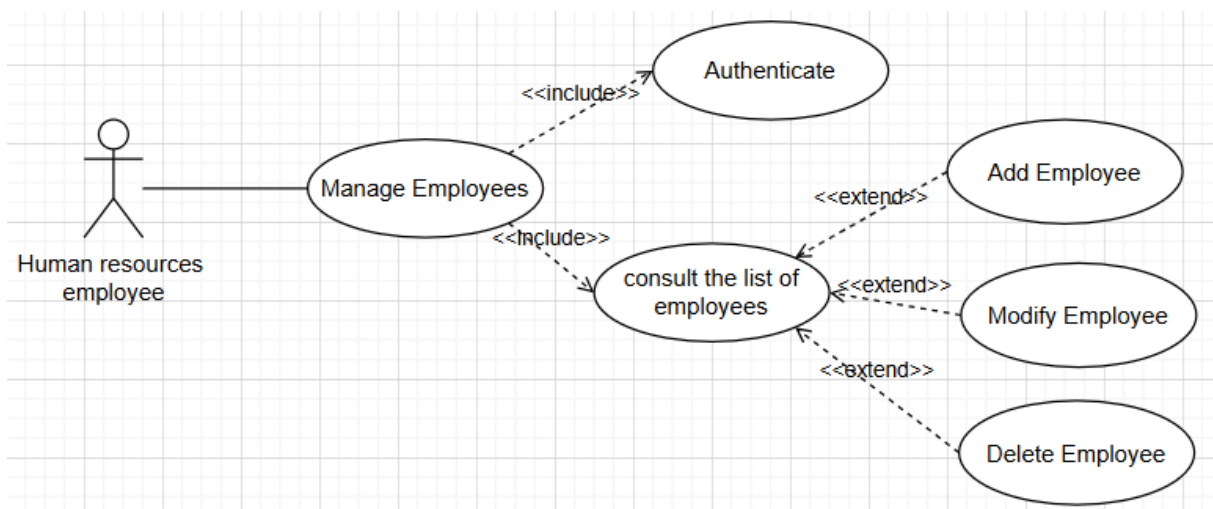


Figure 5.4: Use case diagram for the “Manage employees” functionality.

The use cases ‘Manage Lines’, ‘Manage Stops’, and ‘Manage Buses’ have the

same functional logic and format as the 'Manage Employees' use case. Each use case allows the HR employee to perform **add**, **edit**, and **delete** function on the corresponding entities (lines, stops, buses) via a defined interface, all with the same pre-conditions, post-conditions and user interaction scenarios.

5.3.3 Text descriptions of use case scenarios

Text descriptions of use case scenarios provide a sequential account of how a user interacts with the system to accomplish a goal. The table 5.2 represents the text descriptions of the use case "Authenticate", The table 5.3 represents the text descriptions of the use case "Display Fill Rate" and finally the table 5.4 represents the text descriptions of the use case "Add employee".

5.3.3.1 Text description of use case "Authenticate"

Providing a textual description of the "Authenticate", table 5.2, use case has certain advantages in terms of understanding and communicating the important steps in user authentication.

Table 5.2: Text description of use case "Authenticate".

Use Case	Authenticate
Main Actor	HR employee
Precondition	The user has an active account in the system
Postcondition	The user has access to features associated with their profile
Nominal Scenario	1. The user is in the login page. 2. The system shows the authentication form. 3. The user enters his username and password 4. The user clicks "Login". 5. The system checks the entered data. 6. The system sends the user to Home page
Alternative scenario	The user submits incorrect data or missing data: 1. The system shows an error message. 2. resume to step 2 of the nominal scenario.

5.3.3.2 Text description of use case "Display Fill Rate"

Assigning the "Display Fill Rate" use case a text description ,table 5.3, yields a clear, structured view of how the system is helpful to HR staff in acquiring access to some operating data.

Table 5.3: Text description of use case "Display Fill Rate".

Use Case	Display Fill Rate
Main Actor	HR employee
Precondition	the HR employee is authenticated
Postcondition	The passenger count per line is presented
Nominal Scenario	1. The HR employee chooses the shift, the date and the direction. 2. The system shows the number of employees per line.

5.3.3.3 Text description of use case "Add employee"

The text description of the "Add Employee" use case, table 5.4, gives a concise but comprehensive description of the process through which HR employees are able to add new staff records into the system.

Table 5.4: Text description of use case "Add employee".

Use Case	Add employee
Main Actor	HR employee
Precondition	the HR employee is authenticated
Postcondition	Employee added
Nominal Scenario	1. The HR employee accesses the employee list interface. 2. The HR employee clicks on the add button. 3. the system displays the add form. 4. The HR employee fill out the form and click the add button. 5. The system validates the data. 6. The system saves The changes and displays a success message.
Alternative scenario	HR employee enters missing data or an existing employee ID: 1. The system shows an error message. 2. resume the step 3 of the nominal scenario.

5.4 Conception

The section is dedicated to the stage of project conception, which is the critical phase in the transformation of functional requirements into a clearly defined and formulated technical vision. It serves as a link between analysis and development, making the system organized and in accordance with users' requirements. Here we will unveil the important design specifics that describe the way the system has to be structured and how the various parts of the system will interact. Here we minimize development-time mistakes as well as make the application simpler to scale and maintain in the future.

5.4.1 Sequence Diagrams

the sequence diagrams gives a dynamic view by showing the interaction between the elements of a system over time for specific functionalities. They describe how objects communicate with one another through sending messages to achieve user-driven operations such as authentication or staff administration.

5.4.1.1 Authenticate sequence diagram

This sequence diagram, figure 5.5, illustrates the authentication of a user who is an HR employee logging into the system. The process begins when the HR employee enters their credentials and initiates the login action. The system first checks whether all the fields are completed. If any field is blank, it immediately displays an error message that all the fields must be completed and halts the login process. If validation passes, the system then tries to verify the user's credentials by searching for the user in the database. If the user is not found, a message duly informs them of invalid credentials or a nonexistent account. If the user is found the system marks the user as authenticated successfully and checks this state before redirecting the user to the Home Page, granting access to the application's features. Overall, the diagram effectively explains successful and failed login attempts with emphasis on error handling, data validation, and secure authentication.

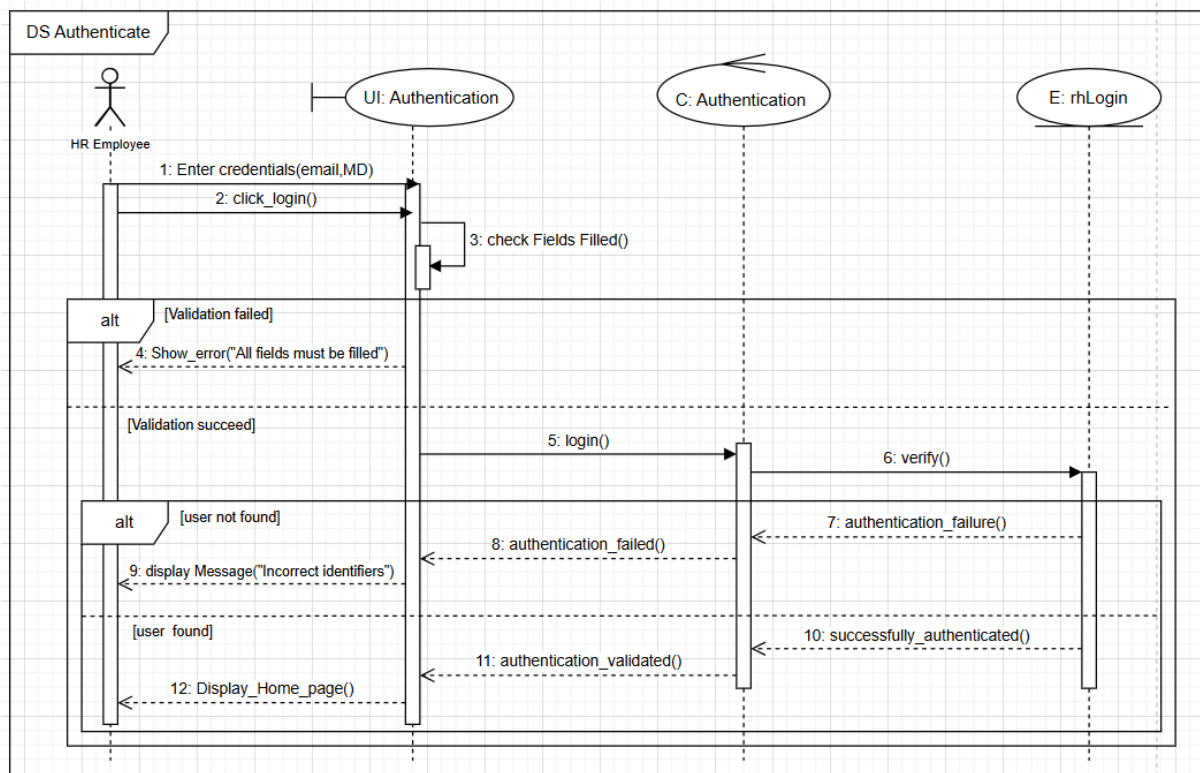


Figure 5.5: Sequence diagram of the “Authenticate” functionality.

5.4.1.2 Display Fill Rate sequence diagram

This sequence diagram, figure 5.6, illustrates the "Fill Rate" consultation process by which an employee of Human Resources can consult employee transportation fill rates by line. The process begins when the HR employee interacts with the system by selecting a specific shift and specifying the direction of travel (inbound or outbound). Once these choices are made, the system starts to calculate the fill rate of each of the transport lines. For each line, the system collects the list of workers scanned, constructs this data into a

fill rate list, and ultimately presents the results graphically to the HR user. Should the HR employee want to examine more specific details, they may click on a specific line, and the system will present a detailed view for the scanned employees of the line. This sequence neatly supports both a summary view of transport occupancy and the ability to drill down to detailed line-level data, enabling the HR personnel to better manage and optimize transport usage.

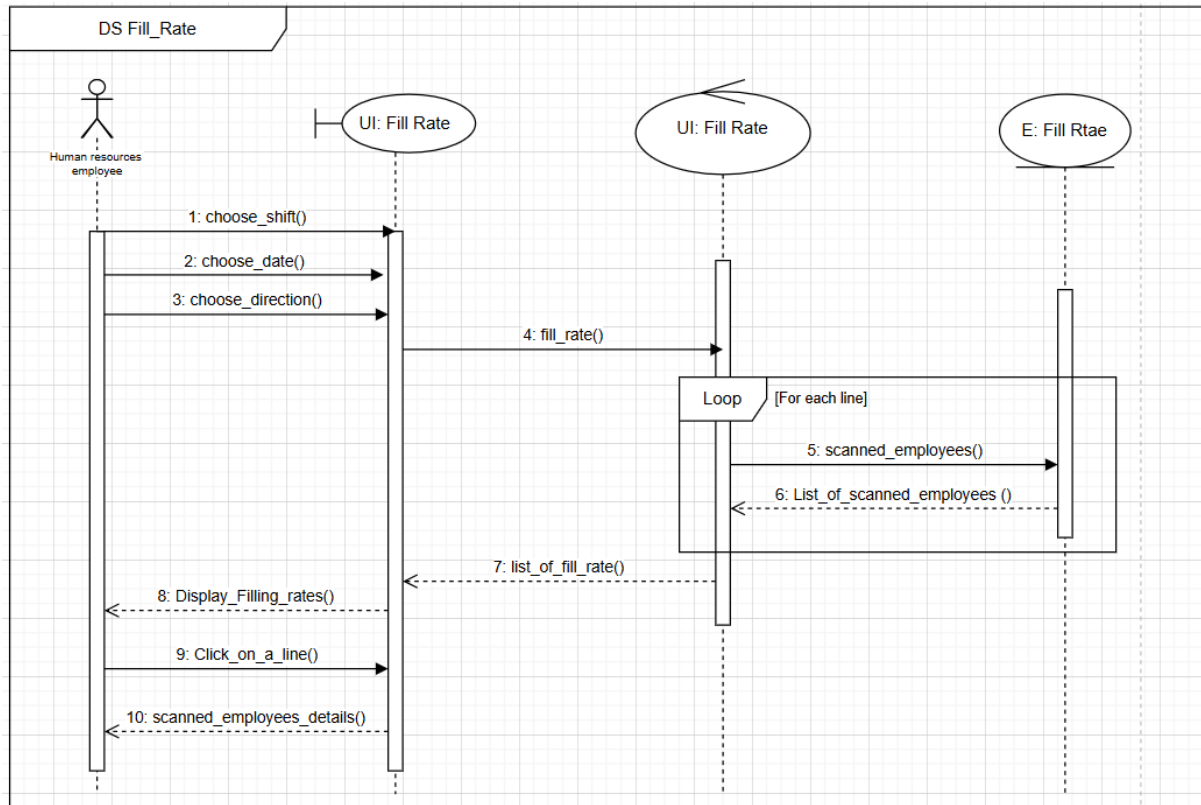


Figure 5.6: Sequence diagram of the “Display Fill Rate” functionality.

5.4.1.3 Add employee sequence diagram

This sequence diagram, figure 5.7, illustrates the Add Employee process in the system, showing how a Human Resource (HR) user inputs new employee details with validation and duplication prevention. The process starts when the HR user inputs the employee details. The system performs input validation to ensure that all mandatory fields are completed. If any field is left blank, the system displays an error message requesting the user to fill in the blank fields: Once the form is filled in correctly, the system checks whether an employee with the same matricule (employee ID) does not already exist in the database. If there is a duplicate, the system blocks the addition and shows an appropriate error message indicating that the employee already exists. If no duplication is found, the system adds the new employee, verifies the successful insertion, and then displays a success message to the user. Finally, the employee list is refreshed to display the newly inserted employee. This flow illustrates the value of input validation, duplicate prevention, and

meaningful user feedback in maintaining data integrity and improving the overall user experience of adding an employee.

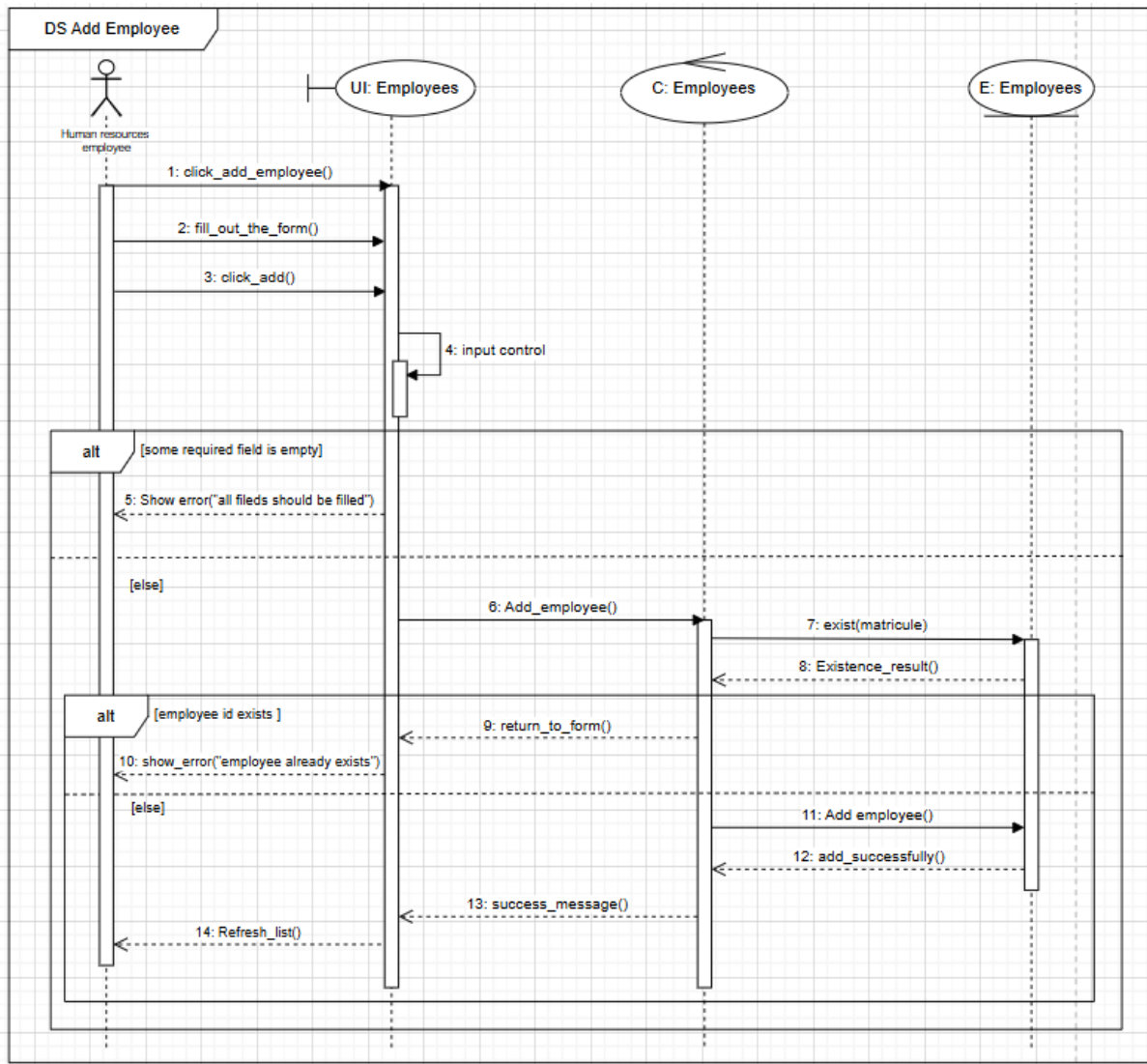


Figure 5.7: Sequence diagram of the “Add employee” functionality.

5.4.2 Class Diagram

The class diagram provides a static representation of the system by showing the prominent classes, their methods, variables, and how they interact with each other. It serves as a pattern for object-oriented application design and facilitates organizing code. Figure 5.8 illustrates the class diagram of Sprint 3:

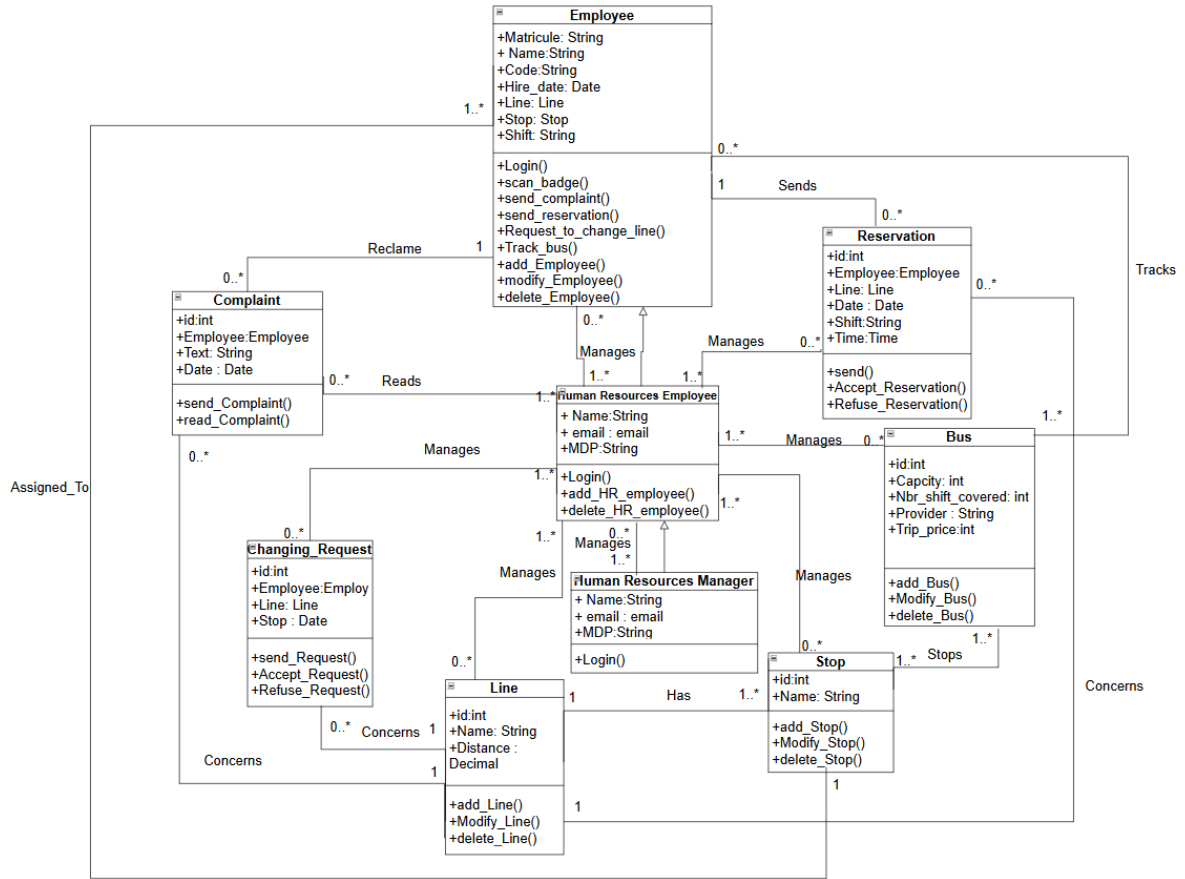


Figure 5.8: Class Diagram of Sprint 3.

5.5 Realization

This section is dedicated to present some of the main interfaces of the platform. It serves to illustrate the visual aspect of the system and provides an overview of the work carried out during development.

5.5.0.1 Authentication interface

The user interface of the application begins with the Login Page, figure 5.9, consisting of two text fields to input the email and password, and a "Send" button to invoke authentication.

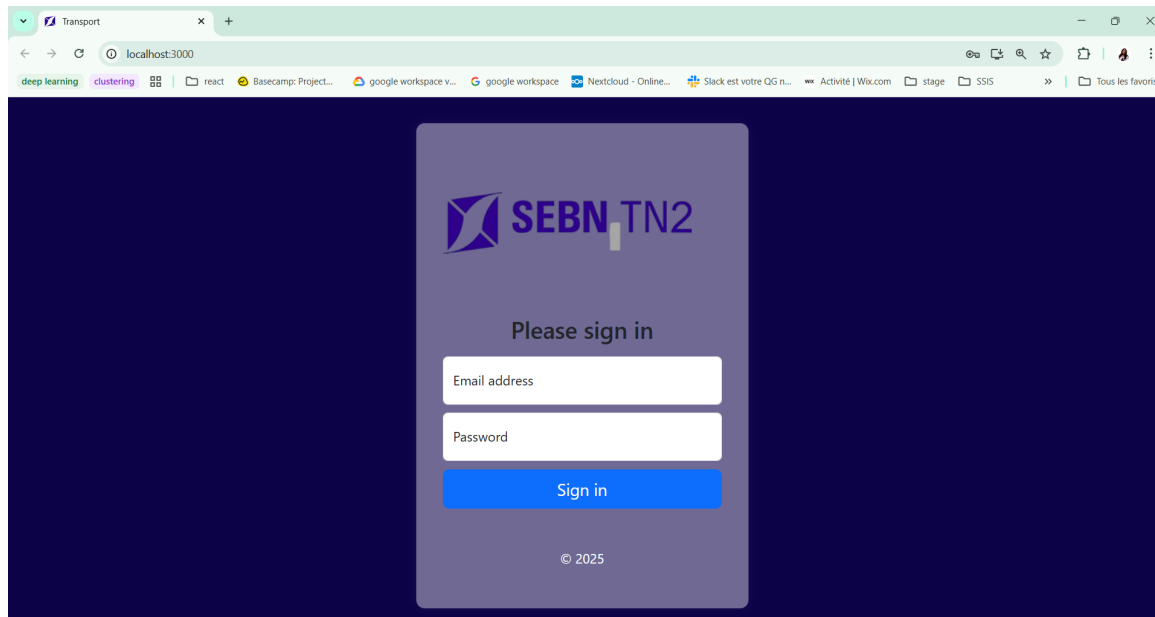


Figure 5.9: the HR employee authentication interface.

5.5.0.2 Home interface

Upon successful authentication, the user is routed to the Home Page, figure 5.10. The page is made simple and easy to go through. The user's name appears at the top left, while a navigation sidebar on the left gives direct access to different modules. The date and time appear at the top right to keep the user informed. The middle part displays an overview of transport data in the form of easy-to-glance cards. Under these cards are two buttons — one is to the Fill Rate page, figure 5.11, and the other to the Tracking module.

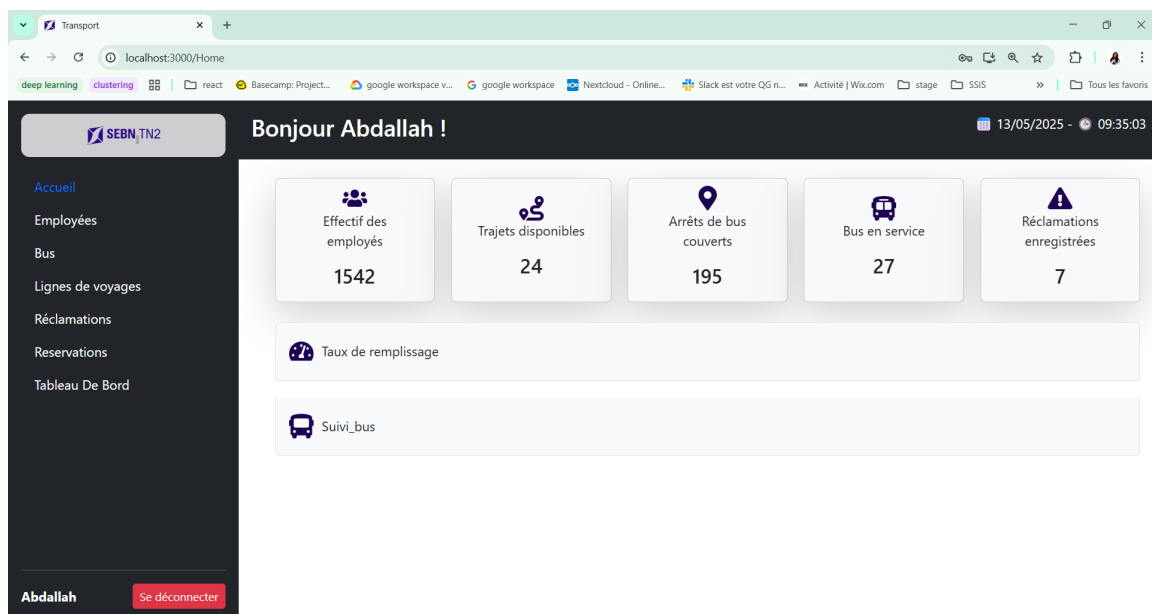


Figure 5.10: The Home interface of the web platform.

5.5.0.3 Fill rate interface

The Fill Rate interface, figure 5.11, displays the bus lines as cards. After the HR user selects the options they require (e.g., shift, direction, date), the system shows how many employees per line. When the user clicks a button on a particular line card (Figure 5), a detail view is displayed where the list of employees who used the respective bus is indicated.

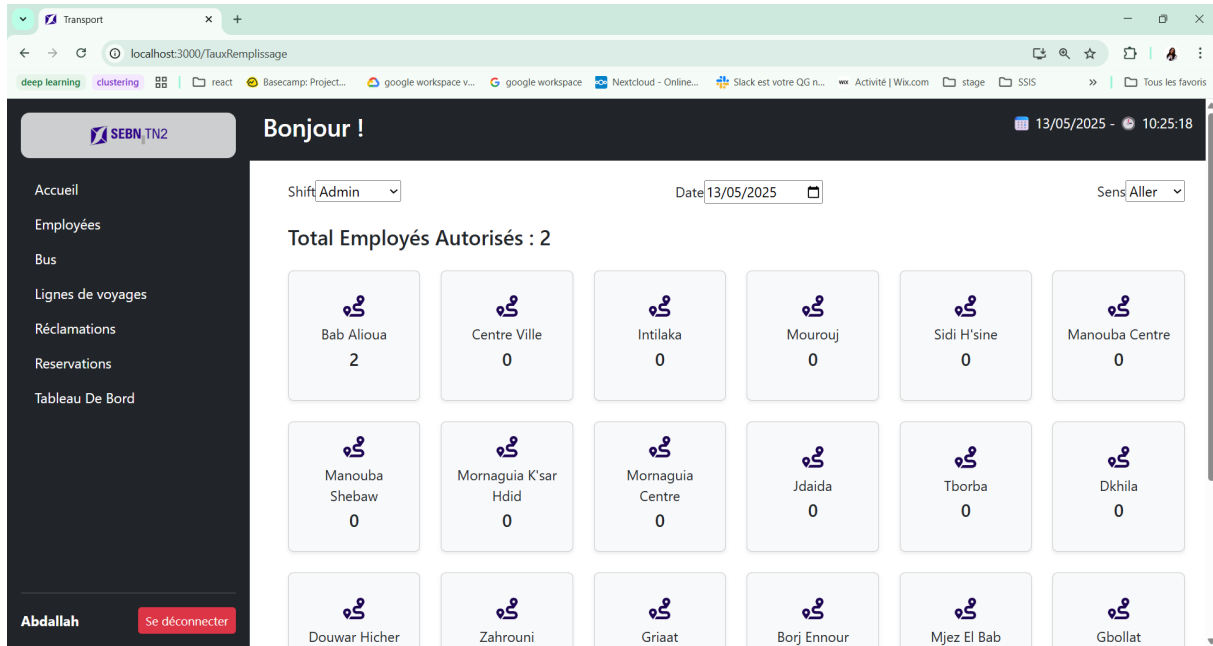


Figure 5.11: Fill rate interface.

When the user clicks a button on a particular line card, figure 5.12, a detail view is displayed where the list of employees who used the respective bus is indicated.

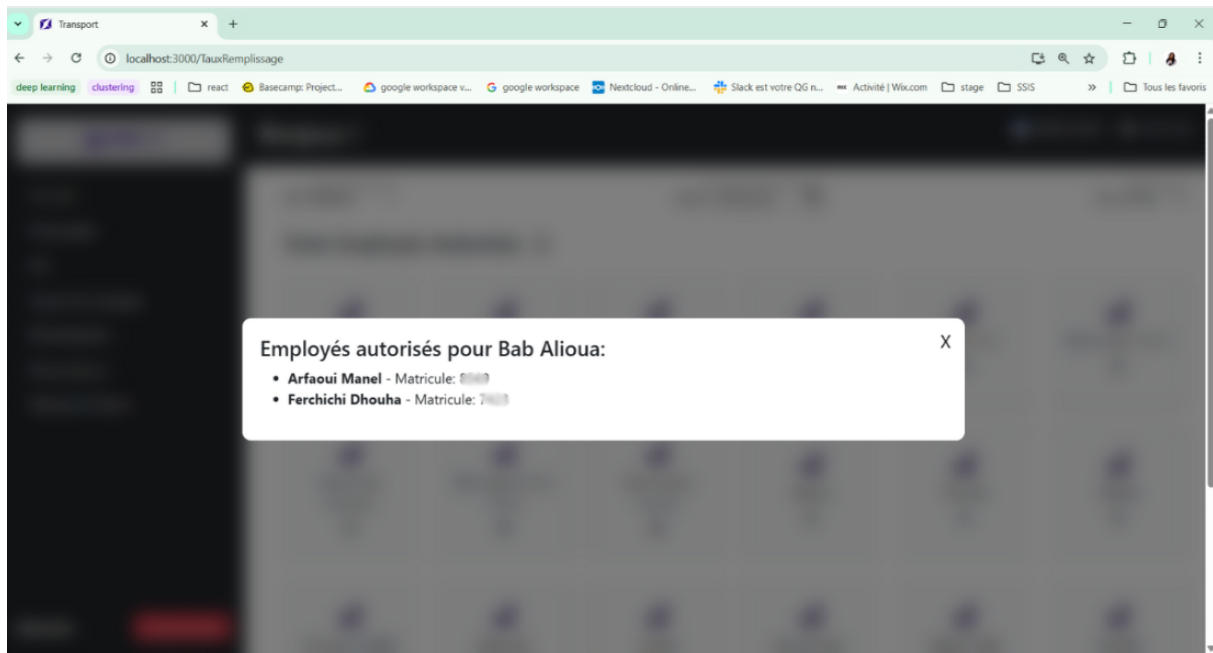


Figure 5.12: Modal of the scanned employees.

5.5.0.4 Employees interface

Figure 5.13 illustrates the Employees interface.

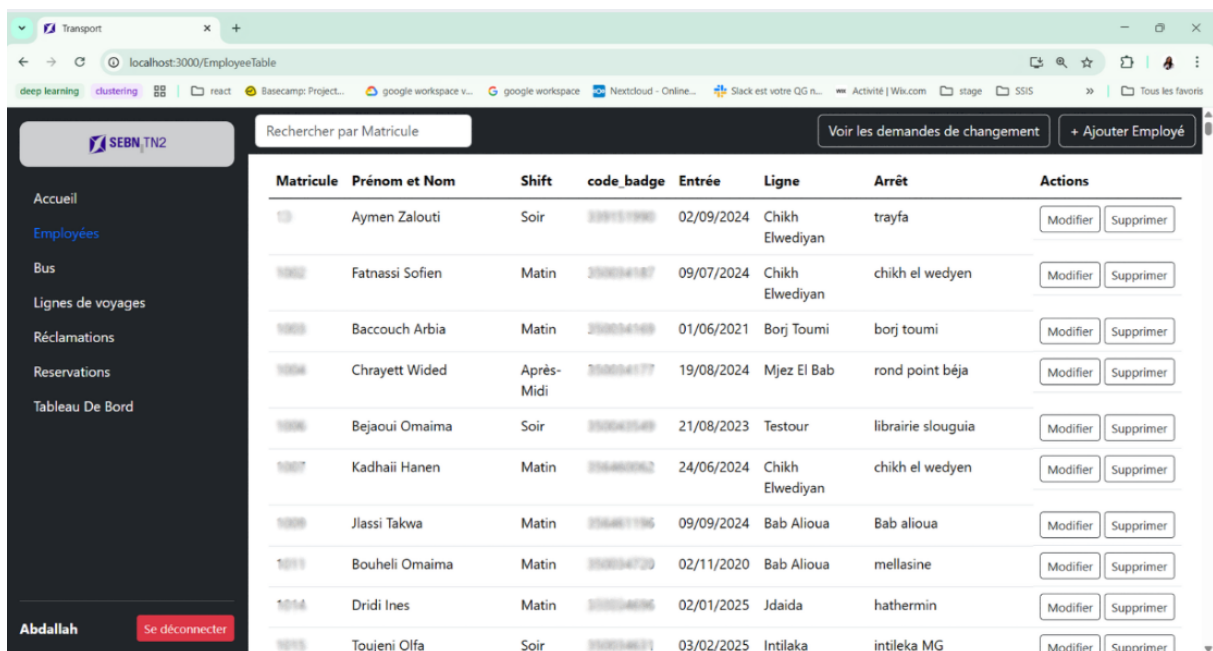


Figure 5.13: Manage employees interface.

There is a "Add Employee" button in the top right corner. When clicked, it will show a form, figure 5.14, where the employee details can be manually entered by the HR user. The form has data fields for employee details and is designed to enable quick and accurate data entry.

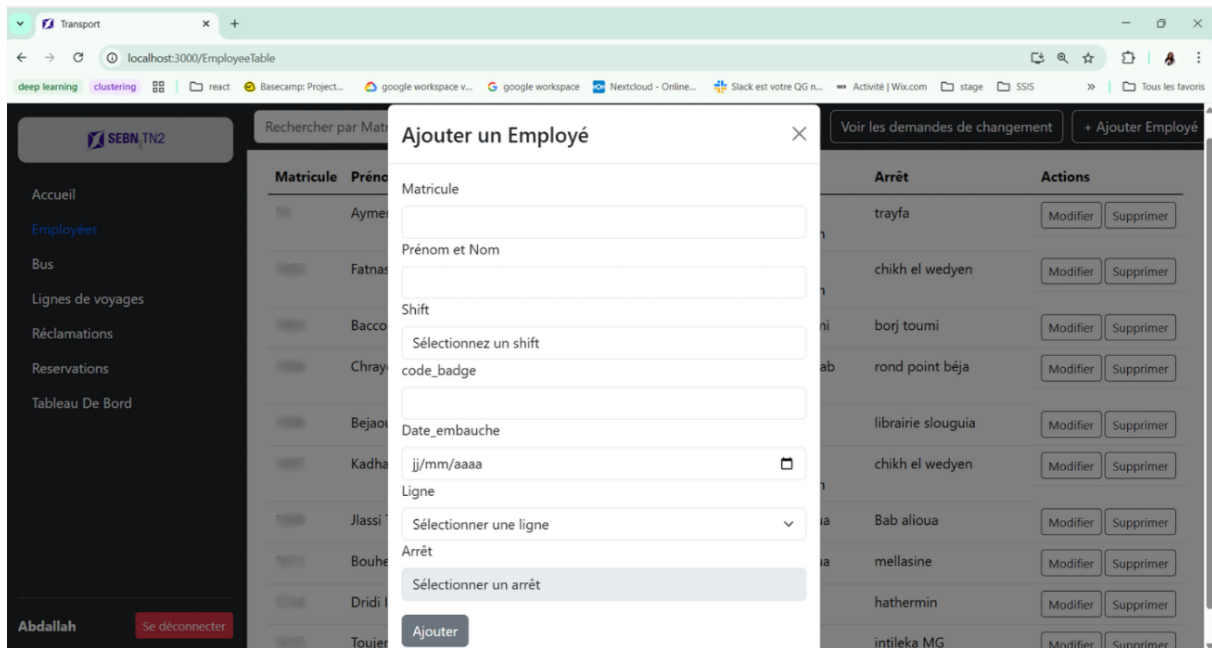


Figure 5.14: Add employee form.

5.6 Conclusion

During this sprint, we managed to develop the Human Resources platform through which the HR team can manage all the aspects related to employee transportation. The platform includes core features such as employee management, fill rate monitoring, and access to transportation dashboards that provide a centralized and user-friendly interface for effective operational management. We move to Sprint 4, which is one of the most important parts of the system. This sprint is involved in developing the badge scanning interfaces and the mobile application for the employees, both of which play a very crucial role in real-time data capture and simple day-to-day usage by the employees.

CHAPTER 6

SPRINT 4: REAL-TIME EMPLOYEE AUTHENTICATION AND TRANSPORT REQUEST SYSTEM

6.1 Introduction

This chapter describes the results of Sprint 4, in terms of a backlog and analysis, design and implementation of the operational functionality created through user interfaces in this sprint

6.2 Sprint 4 Backlog

The Sprint 4 backlog introduces functionalities that enhance the employee's interaction with the system, with features on the user side taking top priority. These include badge scanning, complaints sending, bus seat booking, and even real-time bus tracking location. The user stories are prioritized in a way that important functionalities are delivered early to ensure a seamless and responsive experience for the employees while catering to their operational needs in the transportation system.

Table 6.1: Sprint 4 Backlog.

ID	Functionality	US ID	User Story	Duration in days
1	Authentication	1	As a user, I want to authenticate myself.	1
2	Scan Badge	2	As an employee, I want to scan my badge.	14
3	Show Bus Position	3	As a user, I want to track buses in real time.	2
4	Send complaints	4.1	As an employee, I want to send a complaint.	1
		4.2	As an employee, I want to see my complaints history.	1

Table 6.1: Sprint 4 Backlog (continued).

ID	Functionality	US ID	User Story	Duration in days
5	Request to change line	5	As an employee, I want to request to change my line.	1
6	Reserve a Seat	6.1	As an employee, I want to reserve a bus.	1
		5.2	As an employee, I want to see my reservations history.	1

6.3 Sprint 4 Analysis

This section displays the analysis completed during Sprint 4, which pertains to the features implemented within this sprint. It begins with a global use case diagram that provides an overview of the system's interactions. A detailed look at the most significant use cases is then given to describe their behavior and logic. Finally, textual descriptions of the various use case situations are provided to specify how the system must act under different circumstances in order to provide a clear and structured understanding of the requirements before implementation.

6.3.1 Sprint 4 Use Case Diagram

Figure 6.1 illustrates the overall use case diagram for Sprint 4

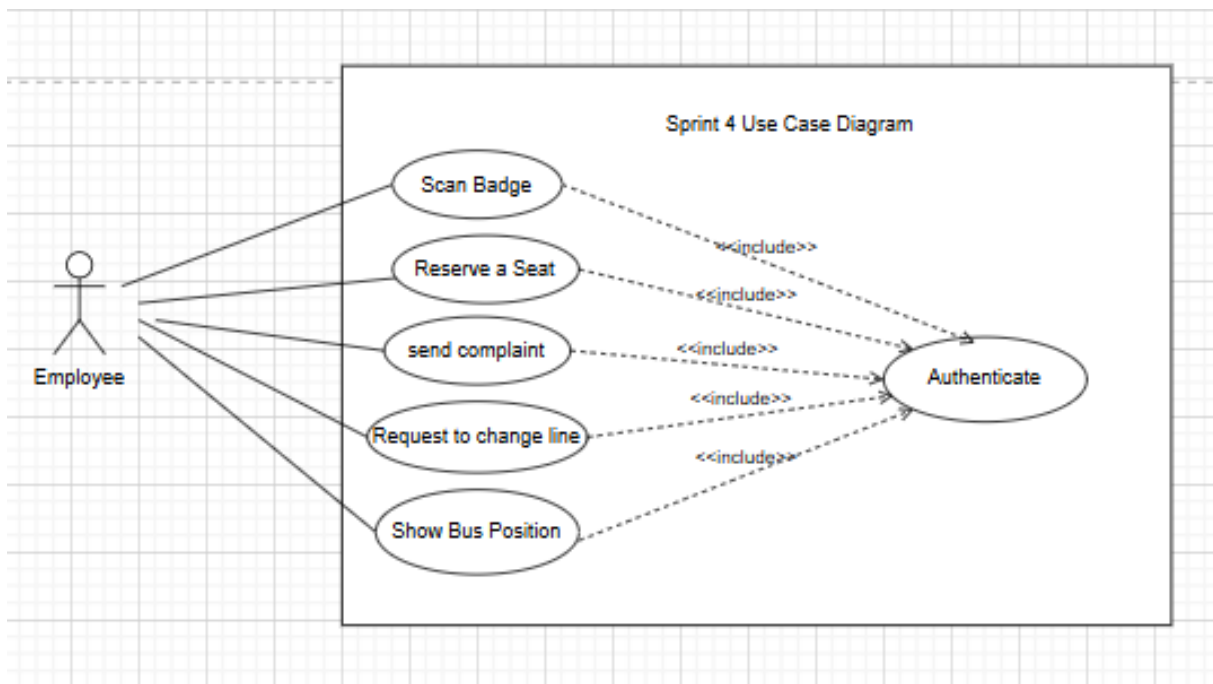


Figure 6.1: Sprint 4 Use Case Diagram.

6.3.2 Detailed use case study

This section provides a careful analysis of the prominent use cases realized in Sprint 4. Each use case is addressed individually to determine the actors, interaction sequence, and the emergent outcomes. This scrutiny is needed to develop an understanding of the system’s behavior and to guide the subsequent design and implementation phases.

6.3.2.1 Use case diagram for the “Scan Badge” functionality

This use case, figure 6.2, explains the flow through which an employee scans his/her badge to authenticate and obtain access to the platform.

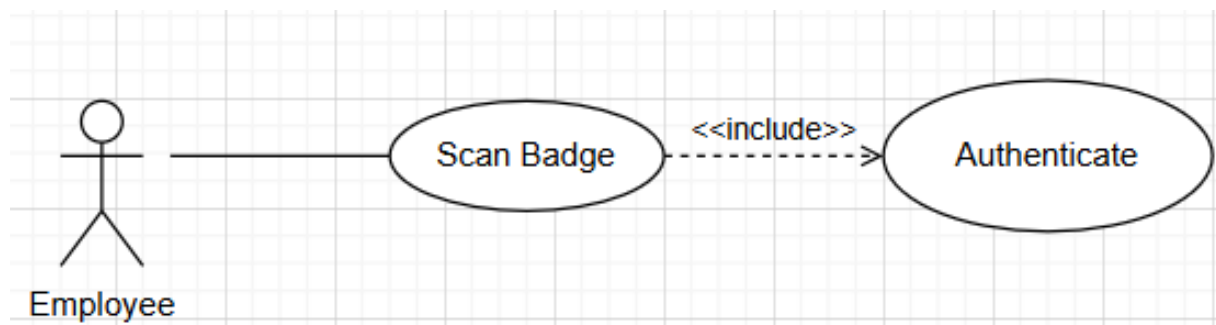


Figure 6.2: Use case diagram for the “Scan Badge” functionality.

6.3.2.2 Use case diagram for the “Send complaint” functionality

This diagram, figure 6.3, illustrates the use case through which an authenticated employee is capable of sending a complaint and viewing his/her complaint history.

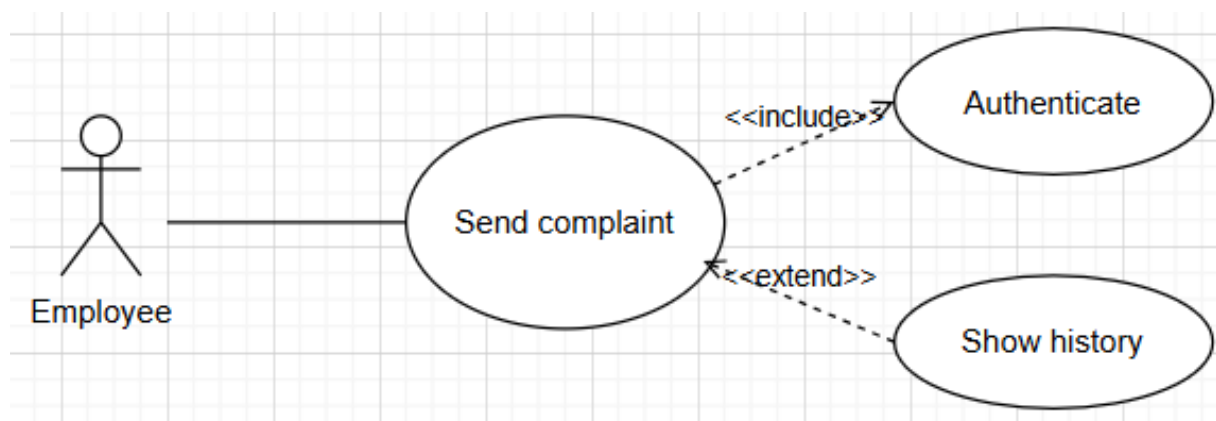


Figure 6.3: Use case diagram for the “send complaint” functionality.

6.3.2.3 Use case diagram for the “Show bus position” functionality

This use case, figure 6.4, explains how an authenticated employee can view the location of the bus in real-time.

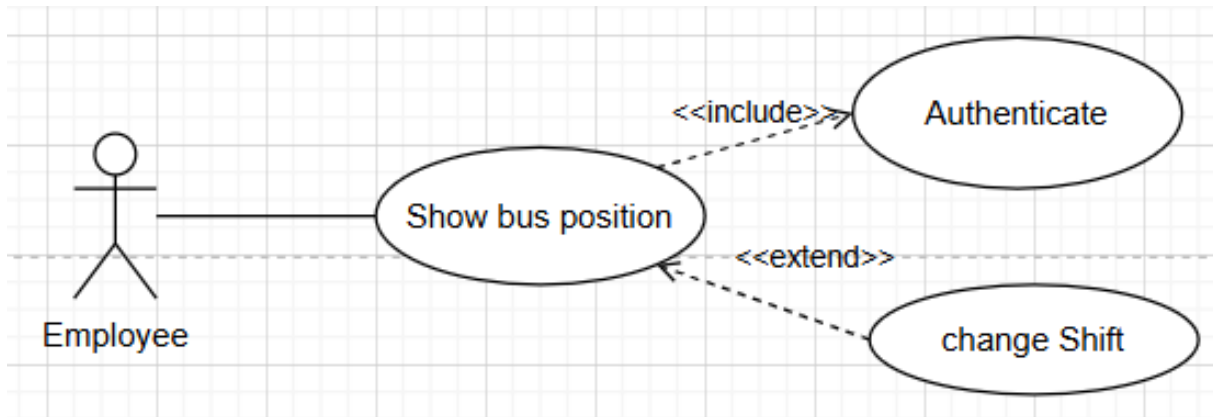


Figure 6.4: Use case diagram for the “Show bus position” functionality.

6.3.3 Text descriptions of use case scenarios

This part provides textual explanations of the various use case scenarios. These descriptions are meant to illustrate, step by step, how each function should operate in real-case situations. They are a helpful complement to the diagrams in that they provide a clearer view of user input and system output.

6.3.3.1 Text description of use case "Scan Badge"

Table 6.2 details the step-by-step interaction between the user and the system during the badge scanning process, ensuring accurate tracking and validation.

Table 6.2: Text description of use case "Scan Badge".

Use Case	Scan Badge
Main Actor	Employee
Precondition	Employee not authorized to use the bus
Postcondition	Employee authorized to use the bus
Nominal Scenario	1. The employee scans their badge. 2. The system authorizes access to the bus. 3. The employee is checked in.
Alternative scenario	The employee does not belong to this bus and the system does not authorize access

6.3.3.2 Text description of use case "Send complaint"

Table 6.3 explains the detailed steps involved when a user submits a complaint, highlighting the system's response to ensure effective issue management.

Table 6.3: Text description of use case "Send complaint".

Use Case	Send complaint
Main Actor	Employee
Precondition	The employee is authenticated
Postcondition	complaint sent
Nominal Scenario	1. the employer enters his complaint and clicks on the send button. 2. The system shows success message . 3. The system saves the complaint.
Alternative scenario	The system displays an error message if the employee sends an empty complaint

6.3.3.3 Text description of use case "Show bus position"

Table 6.4 details the sequence of actions and system responses involved in displaying the real-time location of a bus, providing clarity on how this feature operates to assist users effectively.

Table 6.4: Text description of use case "Show bus position".

Use Case	Show bus position
Main Actor	Employee
Precondition	Employee is authenticated
Postcondition	The bus is tracked
Nominal Scenario	1. The Employee chooses the shift. 2. The system shows the history of bus positions.
Alternative scenario	The system shows no position available if the bus is not running.

6.4 Conception

This section focuses on our system design phase, during which we define the structural and behavioral foundations of the application. The aim is to specify exactly how individual system components communicate and how data is structured and processed. To this end, we present two useful modeling tools: the sequence diagram and the class diagram.

6.4.1 Sequence Diagrams

The sequence diagram shows process interactions in order of time. The diagram depicts the objects and processes utilized and messages passed in sequence that are required to realize the functionality.

6.4.1.1 Scan Badge sequence diagram

The sequence diagram in figure 6.5 illustrates the badge scanning process in the system to activate an employee to use the transport service by scanning his/her badge. It starts when the employee scans his/her badge and the system authenticates it. It initially checks whether the badge has already been scanned before. If it did, the system denies by displaying a message the badge has been scanned previously, denies access, sends an alert, and updates the unauthorized scan list. In case it is a new scan, the system then verifies whether the employee is authorized to use the transportation. If the employee is not authorized, access is denied and the unauthorized list is updated. But in the event that the authorized employee, the system gives access, lists on the approved list, and records the scan. This system keeps only approved staff members able to use the transport service, prevents the repetition of scans, and maintains a record of both approved and unapproved accesses.

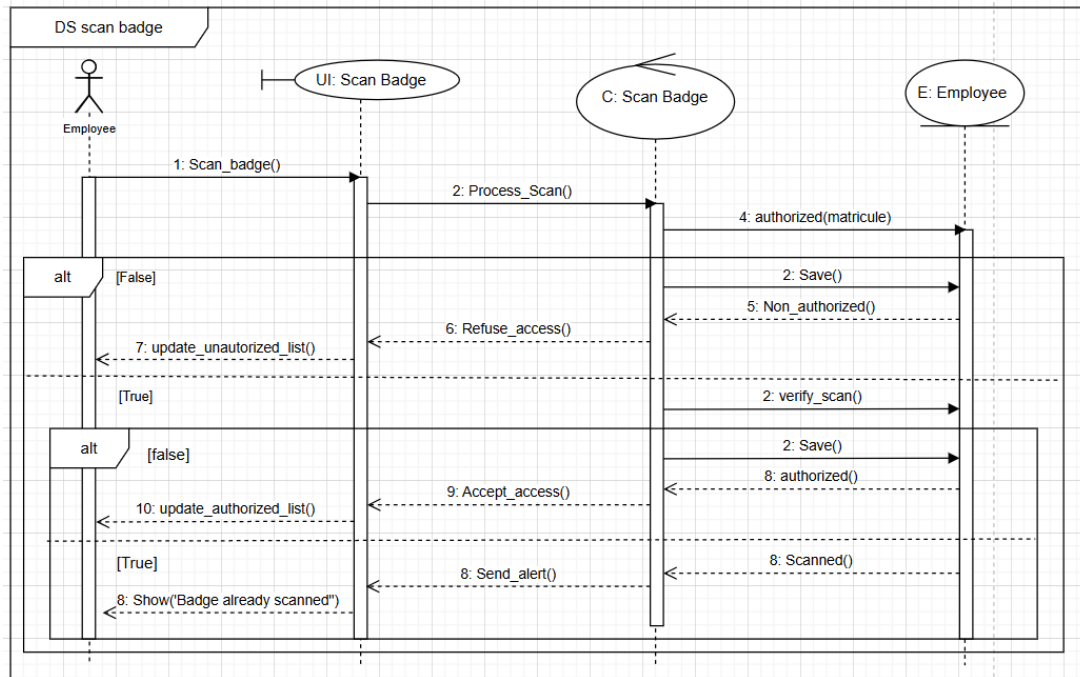


Figure 6.5: Sequence diagram of the “Scan Badge” functionality.

6.4.1.2 Send complaint sequence diagram

The sequence diagram illustrated in figure 6.6 indicates the flow of submitting complaints, where an employee is capable of submitting a transport-related complaint through the system. The process begins when the staff member fills out the complaint form and clicks on the send option. The system then verifies the input to avoid leaving the complaint field blank. If the field is blank, the system halts the process and displays an error message asking the user to provide the required information. When the input is correct, the system continues to send the complaint and store it successfully in the database. When sent successfully, the system notifies the user through a success message stating that the complaint has been sent. This process ensures correct validation of the input, gives clear user feedback, and handles data optimally to make a secure and friendly experience in sending complaints.

This method ensures proper input validation, effective feedback, and healthy data management, resulting in a smooth and user-friendly complaining process.

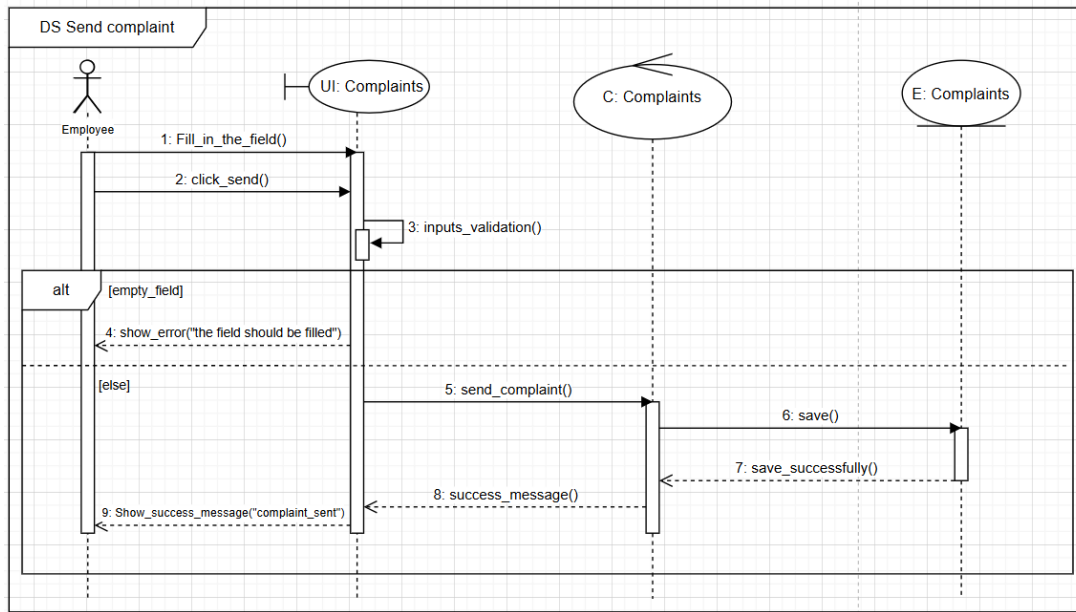


Figure 6.6: Sequence diagram of the “Send complaint” functionality.

6.4.1.3 Show bus position sequence diagram

The sequence diagram shown in figure 6.7 illustrates the process for tracking bus positions in the system. The process begins when the employee requests tracking. The system attempts to obtain badge scan records in order to examine if there is any data for tracking or not. In case no scan records are available, the system informs the user that there are no position details and further processing is discontinued. However, if there are scan records, the system processes relevant scan data. There is no position information returned if there is no specific scan available corresponding to the request. If there is a matching scan, the system constructs and returns the requested bus positions and tracking history. The action ensures employees receive accurate and timely information regarding bus locations, enhancing real-time tracking while facilitating transparency and reliability in managing employee transport.

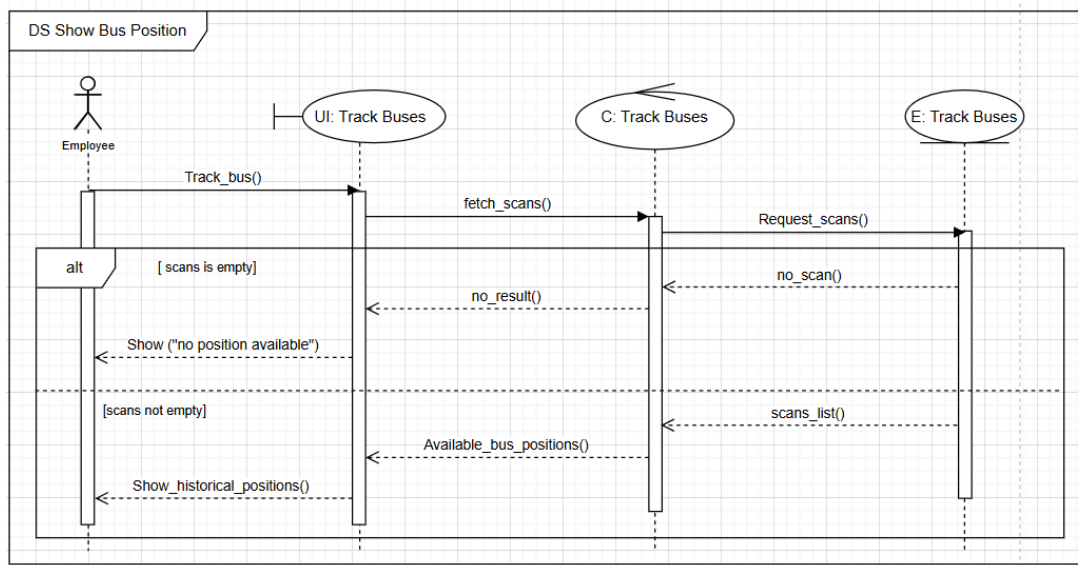


Figure 6.7: Sequence diagram of the “Show bus position” functionality.

6.4.2 Class Diagram

The class diagram is the building block of object-oriented modeling. It is utilized for overall conceptual modeling of the application framework, and for detailed modeling, translating the models into programming code. Figure 6.8 illustrates the class diagram of sprint 4.

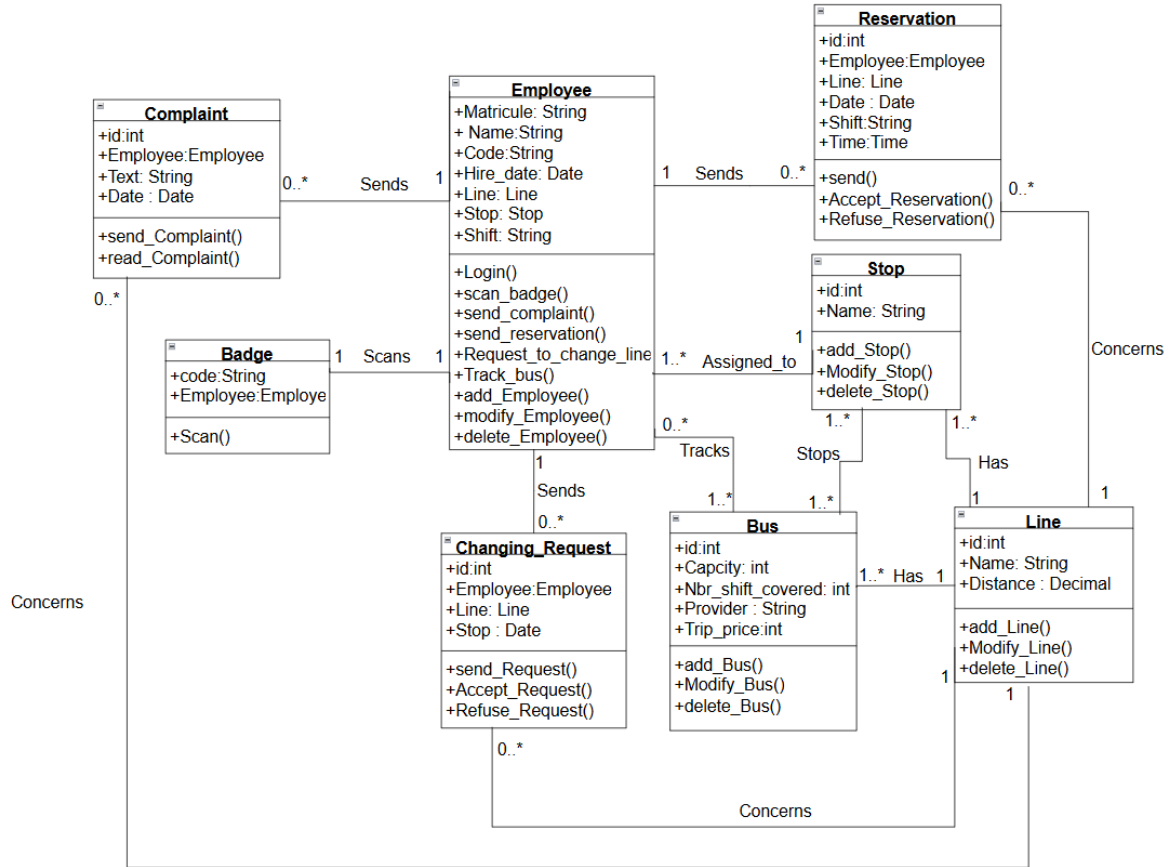


Figure 6.8: Class Diagram of sprint 4.

6.5 Realization

In this section, we present a few of the user interfaces developed during Sprint 4. These interfaces show the practical application of the functionalities hitherto explored, illustrating how the platform supports employees on an everyday level.

6.5.1 Scan badge Interface:

The "Badge Scan" interface is intended to enable real-time employee access verification for company transportation services. It is used most frequently at a bus stop, at the beginning or end of a shift, to scan whether employees are allowed to board the bus. The page is centered with the company logo at the top for the sake of brand consistency. Directly below is a discreet title "Badge Scan" to declare the purpose of the interface. A panel of information is shown to show the transport information like the selected bus line, the corresponding work shift, and the travel direction (outward or return). These values are highlighted for visibility. Two main lists are also present on the interface: one showing authorized individuals and the other showing unauthorized individuals. When the badge is scanned by an employee, the system immediately processes the data from the badge

and checks whether the person is scheduled for the current shift, on the desired line, and in the correct direction. The interface responds by presenting a brief notification message stating the result of the scan, including the name of the employee and other related information. When the bus is authorized for the employee to board, the interface displays a green success message of notification along with the employee name and status as shown in the figure 6.9. Meanwhile, the interface background color is altered to a light green tint in order to give a quick visual cue to the employee as well as the operator. The employee name is added to the "Authorized Persons" list, which helps to keep an updated attendance record for the current trip.



Figure 6.9: Scan Result Interface – Authorized Badge.

In case the employee is rejected, the interface shows a red error message containing the person's name and the rejection reason, figure 6.10. The page background color is also briefly altered to light red to create visual feedback indicating that access is being denied. The employee's name is recorded in the "Unauthorized Persons" list so the operator can take appropriate action or inquire further if needed.



The interface is titled "Scan des Badges" and is set against a red background. It features a white box for "Informations de Transport" containing fields for "Ligne: Bab Alioua", "Shift: Après-Midi", and "Sens: Aller". Below this, the text "Ferchichi Dhouha - Centre Ville" is displayed in orange. The section "Personnes autorisées" contains a green box with "Arfaoui Manel - Hopital Militaire". The section "Personnes non autorisées" contains a pink box with "Ferchichi Dhouha - Centre Ville".

Scan des Badges

Informations de Transport

Ligne: Bab Alioua

Shift: Après-Midi

Sens: Aller

Ferchichi Dhouha - Centre Ville

Personnes autorisées

Arfaoui Manel - Hopital Militaire

Personnes non autorisées

Ferchichi Dhouha - Centre Ville

Figure 6.10: Scan Result Interface – Unauthorized Badge.

In the situation an employee tries to scan their badge more than once, the system recognizes the duplicate scan and displays a warning message indicating that the badge has already been scanned.



Figure 6.11: Scan Result Interface – Warning message.

All the alert messages and background color changes disappear automatically after five seconds. This keeps the interface free from unwanted messages and ready for the next scan, allowing easy processing of a large number of employees in less time. Its objective is to streamline the employee boarding process, verify that only legitimate users of company transport access it, and provide for clear, real-time feedback to drivers. It improves command of operations and worker safety through precise and up-to-date tracking of company bus usage.

6.5.2 Send complaints Interface

The "Send complaints" user interface, figure 6.12, allows employees to report complaints or ideas from within the app. This is a simple and informative user interface that encourages users to express their concerns while promoting polite communication. Towards the top of the screen, a customized AppBar with the title of "Réclamations" and an icon button is displayed, which may be tapped by users in order to view their past complaints. The AppBar is styled with the dark blue background and white icons to provide visual continuity throughout the app. Once the user has entered their message, they can submit it by clicking the "Submit" button, sending an invocation to the backend server. When

the complaint has been successfully sent and received by the server and is valid, a success message is shown in a SnackBar, and the input field is cleared. If the submission failed due to some reason, such as a server or validation error, an appropriate error message is shown to inform the user. This feedback process keeps the user at all times informed about the result of his action.

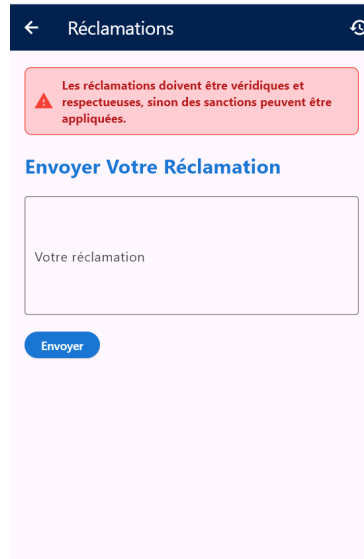


Figure 6.12: Send complaints Interface.

The interface supports a history view option for submitted grievances. On tapping the history icon in the AppBar, the user initiates a call to fetch all complaints made by them ever. A loading spinner is shown while data is loading to indicate processing. When the complaints are loaded, the primary interface is blurred using a transparent background filter to ensure a modal popup is highlighted on the center of the screen. This mode, constructed in the form of a white padded rounded card, displays a list of past complaints in the order they were submitted as illustrated in the figure 6.13. Each complaints listed is in the form of a card whose color indicates its status: green if the administrator has marked the complaint as "read," and red if it remains unread. Each card displays the complaint text and submission date. This visual facilitates users to directly recognize addressed and pending issues. Underneath the modal, a "Close" button allows a user to close the view of history and return to the main interface.



Figure 6.13: The history of submitted complaints.

6.5.3 Track Bus Interface

The Track Bus interface, figure 6.14, provides users with a real-time overview of the path of a single employee's bus. Upon entering the page, the user is presented with a briefly titled heading and a warning message describing that the tracking data displayed is derived from live data, which at times may be inaccurate or erratic. This warning is intended to manage user expectations for the validity of the data. Under the warning, the interface displays the name of the bus line the employee is assigned to, giving context to the tracking information. The user chooses their shift—morning, afternoon, evening, or administrative—from a dropdown list. The filter reloads the scan data dynamically to show only the most recent records of bus stops relevant to the chosen shift, making the data current and specific. If there is no bus stop data for the selected criteria, a bold and centered "No stops available" message informs the user that there are no stops to display. Where data is present, however, the interface displays each stop in a tidy, card-designed list. Each card shows the name of the bus stop and the exact time the bus was scanned at the stop, with the most recent stop highlighted visually by a subtle blue background. This layout is simple for users to track the bus journey and view at a glance the latest bus movement.

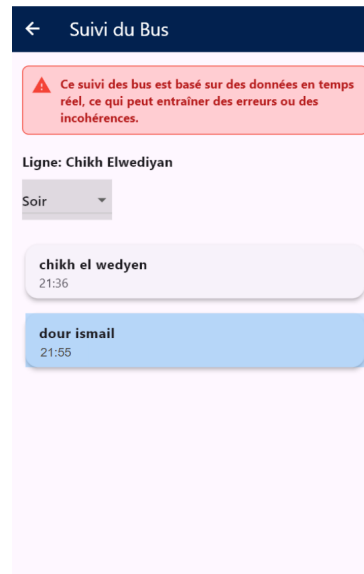


Figure 6.14: Track Bus Interface.

6.6 Conclusion

In conclusion, Sprint 4 was an important step where we integrated all essential functionalities of the application. Having implemented authentication, complaints, reservation, line changing requests, and bus tracking, we have officially finished the project through this sprint. Overall, the project is a complete product with all functions maturing together and ready to go into the real world.

GENERAL CONCLUSION AND PERSPECTIVES

This end-of-studies project represents the culmination of my academic journey in Business Intelligence and provided a valuable opportunity to apply my theoretical knowledge in a professional context. The primary objective was to design and develop an intelligent system to optimize employee transportation within an organization, leveraging modern technological trends such as web development, data analytics, and generative AI.

Throughout the different stages of the project, I explored and integrated various disciplines—from building dynamic dashboards to developing a secure, full-stack web application for managing transport requests. Each sprint contributed to a scalable, user-centered digital solution that addressed a real-world operational challenge, aligned with contemporary trends in digital transformation.

A particularly innovative aspect of this project was the incorporation of generative AI to create personalized dashboards. This feature demonstrated the potential of AI to enhance the flexibility, autonomy, and interactivity of Business Intelligence tools. It deepened my practical understanding of AI's impact on how users interact with and derive insights from data.

Beyond the technical achievements, this experience was also transformative on a personal level. It taught me how to manage priorities, work collaboratively in an agile environment, adapt quickly to new technologies, and deliver functional solutions under time constraints. Most importantly, it reinforced the importance of user-centric design and iterative development.

In conclusion, this internship was both a professional challenge and a rich learning experience. It allowed me to grow technically, personally, and professionally while contributing to a meaningful digital solution. I am confident that the skills, knowledge, and mindset I developed through this project will serve as a strong foundation for the next stages of my academic and professional career.

Looking forward, there are several promising directions to keep enhancing and building on the current system. While the current system is an effective model for employee

transport management, there remain some places where future development can be added. They are including more advanced predictive analytics to predict transport demand and dynamically optimize route planning. In addition, extending the application of AI to natural language processing could enable interaction with the system even more naturally by voice . In addition, further work would also cover mobile application development for greater convenience and direct communication between employees and transport managers. Finally, enhancing security capabilities and scalability will be essential as the system grows to accommodate bigger organizations or multiple sites.

BIBLIOGRAPHY

- [1] Sebn-tn2's logo., 2025. <https://www.keejob.com/offreemploi/companies/18991/sebn-tn2/> (Accessed: 4 February 2025).
- [2] Scrum process., 2025. <https://www.pm-partners.com.au/insights/the-agile-journey-a-scrum-overview/> (Accessed: 4 February 2025).
- [3] Scrum roles., 2025. <https://nulab.com/learn/project-management/quick-guide-scrum-roles-responsibilities/> (Accessed: 4 February 2025).
- [4] Technical architecture of the system., 2025. <https://www.altexsoft.com/blog/complete-guide-to-business-intelligence-and-analytics-strategy-steps-processes-and-implementation/> (Accessed: 29 April 2025).
- [5] Inmon approach definition., 2025. <https://datascientest.com/bill-inmon-biographie> (Accessed: 29 April 2025).
- [6] Inmon approach., 2025. <https://dwgeek.com/various-data-warehouse-design-approaches.html/> (Accessed: 29 April 2025).
- [7] Kimball approach definition., 2025. <https://datascientest.com/methode-de-kimball-tout-savoir> (Accessed: 29 April 2025).
- [8] Kimball approach., 2025. <https://dwgeek.com/various-data-warehouse-design-approaches.html/> (Accessed: 29 April 2025).
- [9] Visual studio code., 2025. <https://visualstudio.microsoft.com/fr/> (Accessed: 20 March 2025).
- [10] Mysql., 2025. <https://fr.wikipedia.org/wiki/MySQL>.

- [11] phpmyadmin., 2025. <https://fr.wikipedia.org/wiki/PhpMyAdmin> (Accessed: 09 April 2025).
- [12] Xampp., 2025. <https://fr.wikipedia.org/wiki/XAMPP> (Accessed: 09 April 2025).
- [13] Talend., 2025. <https://www.synaltic.fr/blog/partenaires/talend#:~:text=Talend%20est%20l'un%20des,rapidement%20vers%20les%20syst%C3%A8mes%20cibles>. (Accessed: 09 April 2025).
- [14] Talend's logo., 2025. <https://www.audaxis.com/news/sortie-de-talend-v5-3/> (Accessed: 09 April 2025).
- [15] Power bi's logo., 2025. <https://www.jeveuxetredata scientist.fr/logo-power-bi/> (Accessed: 09 April 2025).
- [16] Draw.io's logo., 2025. <https://medium.com/@hasaliedirisinghe/user-guide-to-draw-io-6a1a4d7b5d33> (Accessed: 20 March 2025).
- [17] Overleaf's logo., 2025. <https://fr.wikipedia.org/wiki/Overleaf> (Accessed: 09 April 2025).
- [18] Explanation of the etl process., 2025. <https://medium.com/analytics-vidhya/understanding-etl-pipeline-76718d299a08> (Accessed: 29 April 2025).

Optimizing employee transportation System via Business Intelligence and Generative-AI

RAPPORT DE SIMILITUDE

3%

INDICE DE SIMILITUDE

1%

SOURCES INTERNET

1%

PUBLICATIONS

1%

COPIES DE L'ÉTUDIANT

SOURCES PRINCIPALES

1	da Silva, Daniel Filipe Rasteiro. "Implementation of the Fireloc Application", Universidade de Coimbra (Portugal), 2024 Publication	<1 %
2	Submitted to Arab Open University Copie de l'étudiant	<1 %
3	Submitted to Southern New Hampshire University - Continuing Education Copie de l'étudiant	<1 %
4	www.coursehero.com Source Internet	<1 %
5	ntnuopen.ntnu.no Source Internet	<1 %
6	Keke Gai, Jing Yu, Liehuang Zhu. "Introduction to Cybersecurity in the Internet of Things", CRC Press, 2024 Publication	<1 %
7	Thomas B. Hilburn, Massood Towhidnejad. "Software Engineering Practice - A Case Study Approach", CRC Press, 2020 Publication	<1 %
8	ijhis.pubmedia.id Source Internet	<1 %
9	Submitted to Kuala Lumpur Infrastructure University College Copie de l'étudiant	<1 %

Abstract

As companies seek to prioritize their optimization of operational processes through business intelligence, automated and artificial intelligence systems and web technologies, the modernization and digital transformation of the current landscape has inspired my graduation project at SEBN TN2. This project proposes the design and development of an intelligent system for employee transportation optimization. The central objective was to generate a comprehensive and scalable solution that includes interactive dashboards for urgent insights, customizable reporting tools using generative AI and a web application for managing and requesting employee transportation safely and securely.

The different project phases addressed multiple disciplines; including data visualization techniques for data-driven decisions, integration of generative AI to create a personalized and responsive user experience, and the design and development of a web application to meet the needs of Human resources and employees. The project therefore identified an innovated digital solution to a practical and operational realities to also, apply streamlined methods, agility, and user-centered design.

keywords: Employee Transportation, Business Intelligence, Generative AI, Data Visualization, Web Application.